

Center Midfields

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Player counts

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In this project, there are a total of 182 center midfielders considered, with

- 46 center midfielders in 12 teams in the English FA Women's Super League
- 43 center midfielders in 12 teams in the German Frauen Bundesliga
- 57 center midfielders in 16 teams in the Spanish Liga F
- 36 center midfielders in 10 teams in the Italian Serie A Women

Results

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Results for 2 clusters

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Breakdown of results by league

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league	FA Women's Super League	Frauen Bundesliga	Liga F	Serie A Women
kmeans_2_cluster				
0	23	25	32	21
1	23	18	25	15

The split is rather more biased towards cluster 0, except for England Women's Super League where the split is equal.

Breakdown of results by features

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- Cluster 0 shows more positiveness towards forward pass percentage, pass lengths, average shot distance, fouls won while defending, and pass more than carry when it comes to progression.
- On the other hand, positiveness in cluster 1 can be seen in positional metrics higher up the pitch, as they carry more than pass as a method of progression, while attempting more through balls and passes into the box.

Therefore, based on these results,

- Cluster 0 center midfielders can be seen as deep midfielders
- Cluster 1 center midfielders can be seen as advanced midfielders

Breakdown of results by players

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Results for 3 clusters

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Breakdown of results by league

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league	FA Women's Super League	Frauen Bundesliga	Liga F	Serie A Women
kmeans_3_cluster				
0	14	7	15	7
1	11	17	16	12
2	21	19	26	17

Comparing results for 2 clusters against 3 clusters

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kmeans_3_cluster	0	1	2
kmeans_2_cluster			
0	0	18	83
1	43	38	0

It seems like $k = 3$ cluster 1 is a shared zone between the two $k = 2$ clusters. Interestingly, no players from $k = 2$ cluster 0 landed in $k = 3$ cluster 0, while no players from $k = 2$ cluster 1 landed in $k = 3$ cluster 2.

Breakdown of results by features

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- Cluster 0 midfielders passed more backward and shorter, while attempted more through balls in passes into opposition box. They carried forward more, and were active at higher areas up the pitch, including receiving and applying pressures and winning the ball.
- While for cluster 1, these midfielders attempted more passes forward and low and progressive into the final third, while receiving locations vary more. Passing is their main progression method.
- Cluster 2 midfielders, on the other hand, made longer passes and further shots, and were active at lower zones, especially shown in blocks in own box.

Hence, the following conclusions can be made:

- Similar to $k = 2$ cluster 0, $k = 3$ cluster 2 center midfielders can be seen as deep pivot midfielders
- Similar to $k = 2$ cluster 1, $k = 3$ cluster 0 center midfielders can be seen as advanced carrying midfielders
- Cluster 1 can be seen as progressive passing midfielders.

Further clustering of each type of midfielders (Defensive/Center/Attacking) will be done based on the results of $k = 3$ clusters.

Appendix

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Radar plots for 2 clusters center midfielders' averages

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Passing — kmeans_2_cluster comparison (raw)

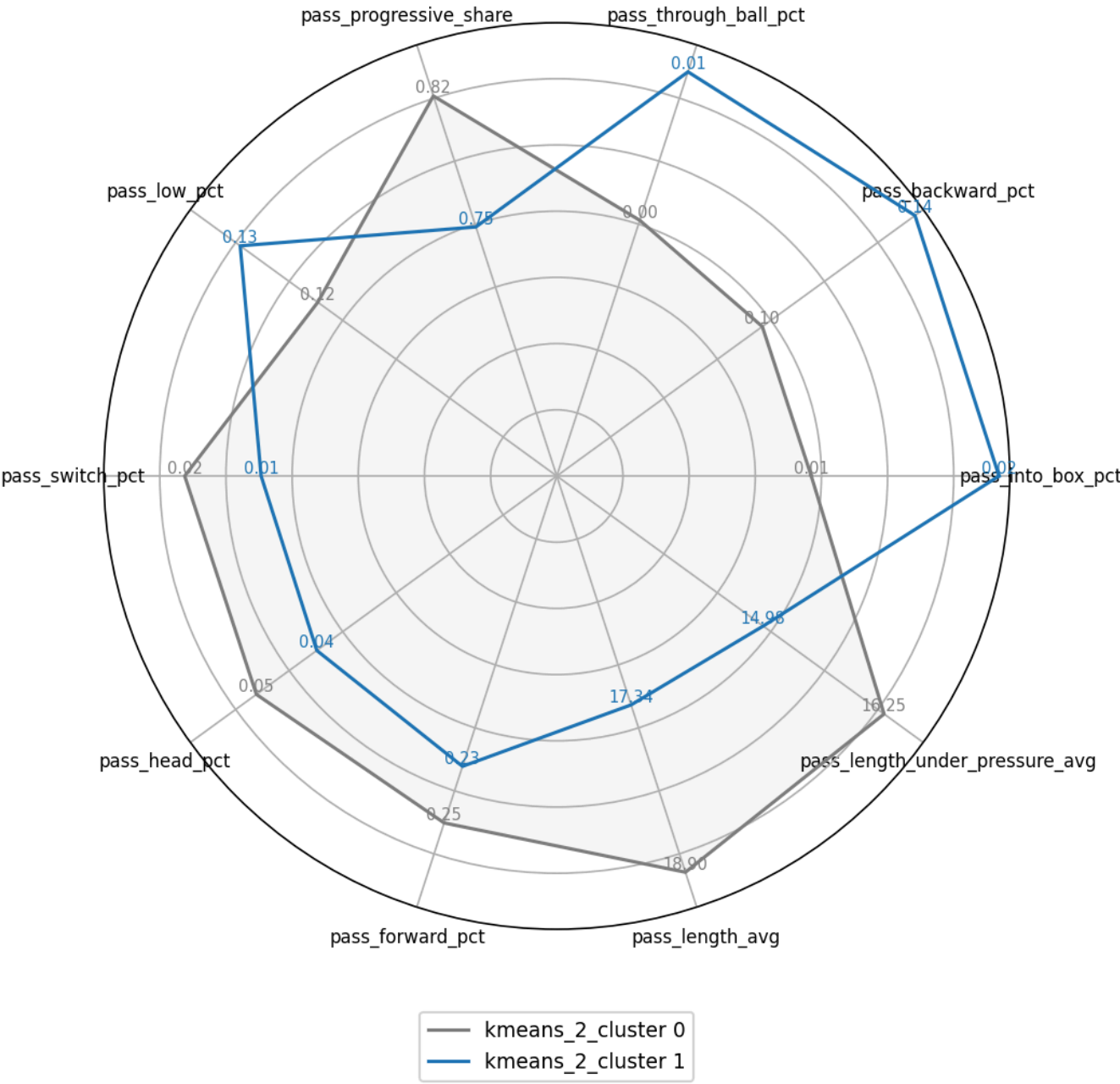


Figure 1: Center Midfields Passing — Raw Statistics Comparison, 2 Clusters

Passing — kmeans_2_cluster comparison (adj)

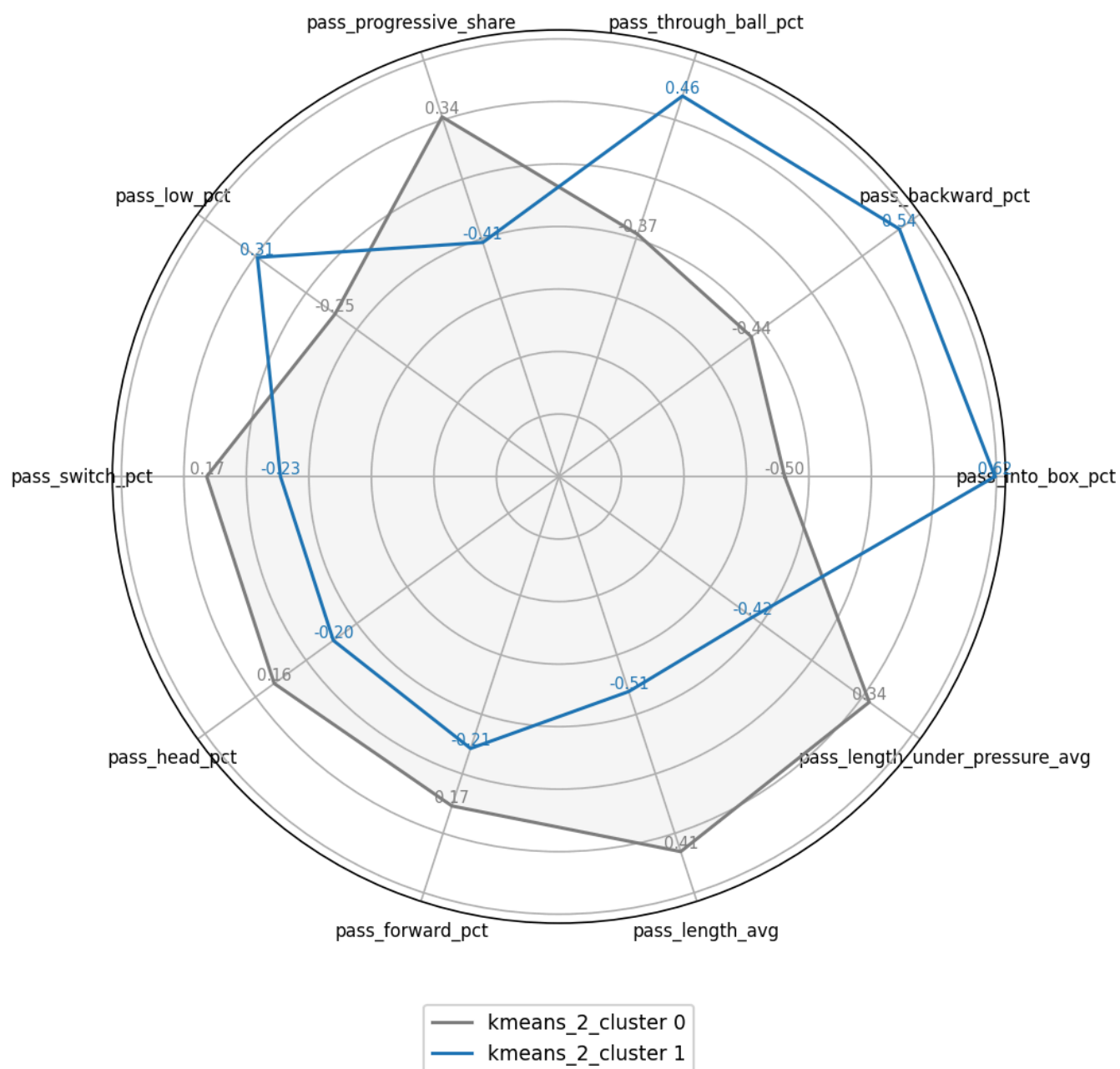


Figure 2: Center Midfields Passing — Adjusted Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (raw)

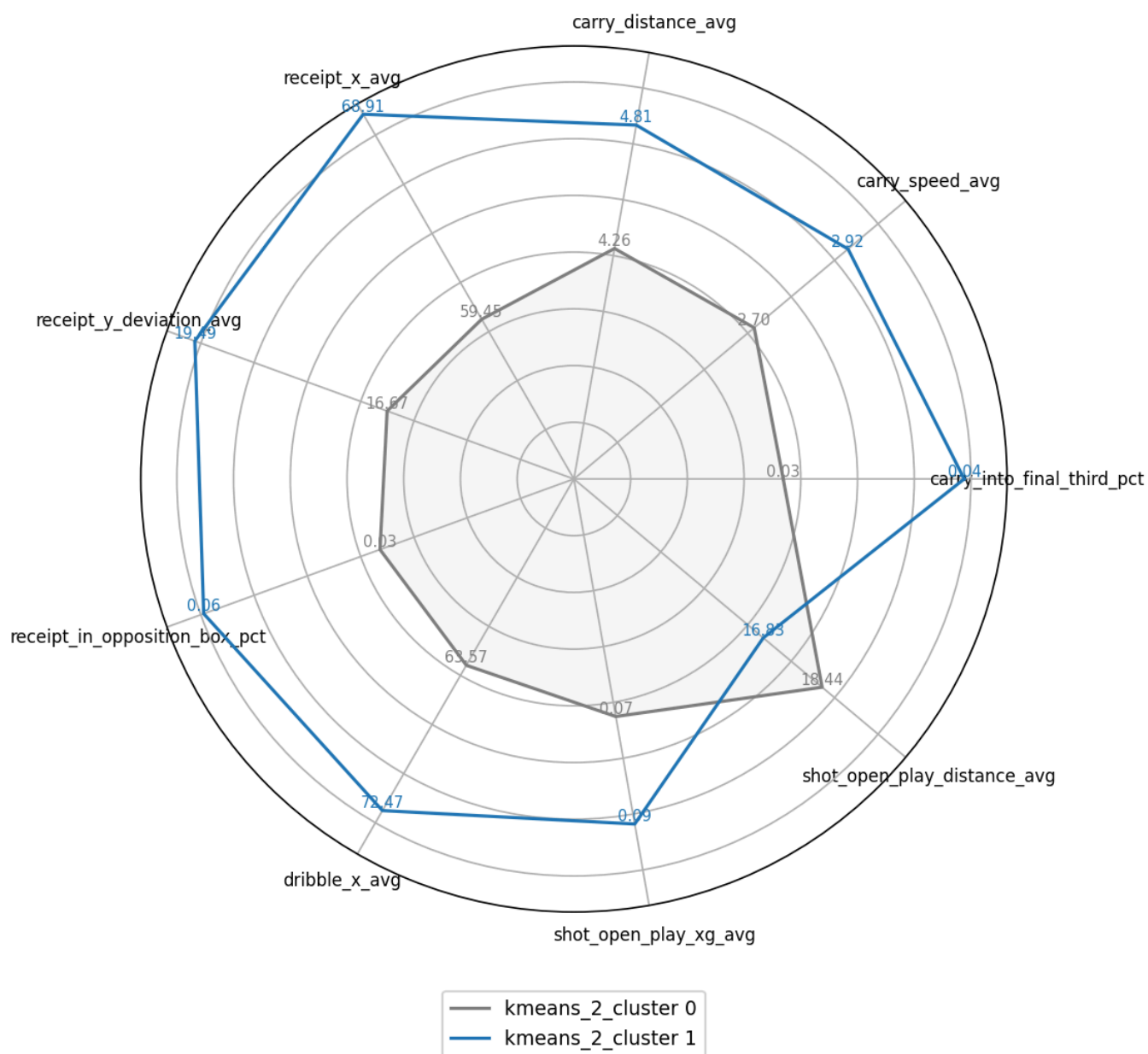


Figure 3: Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (adj)

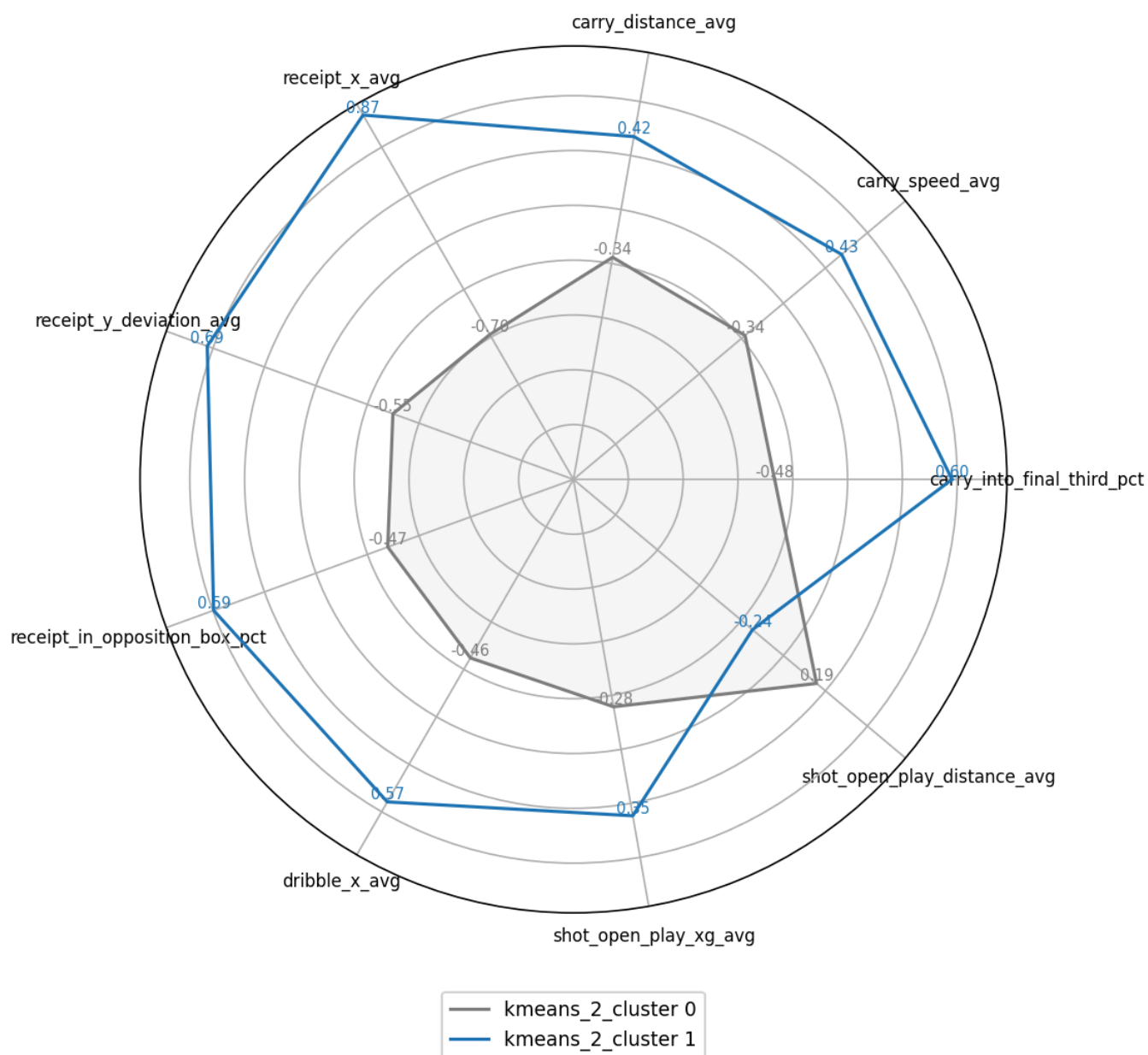


Figure 4: Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (raw)

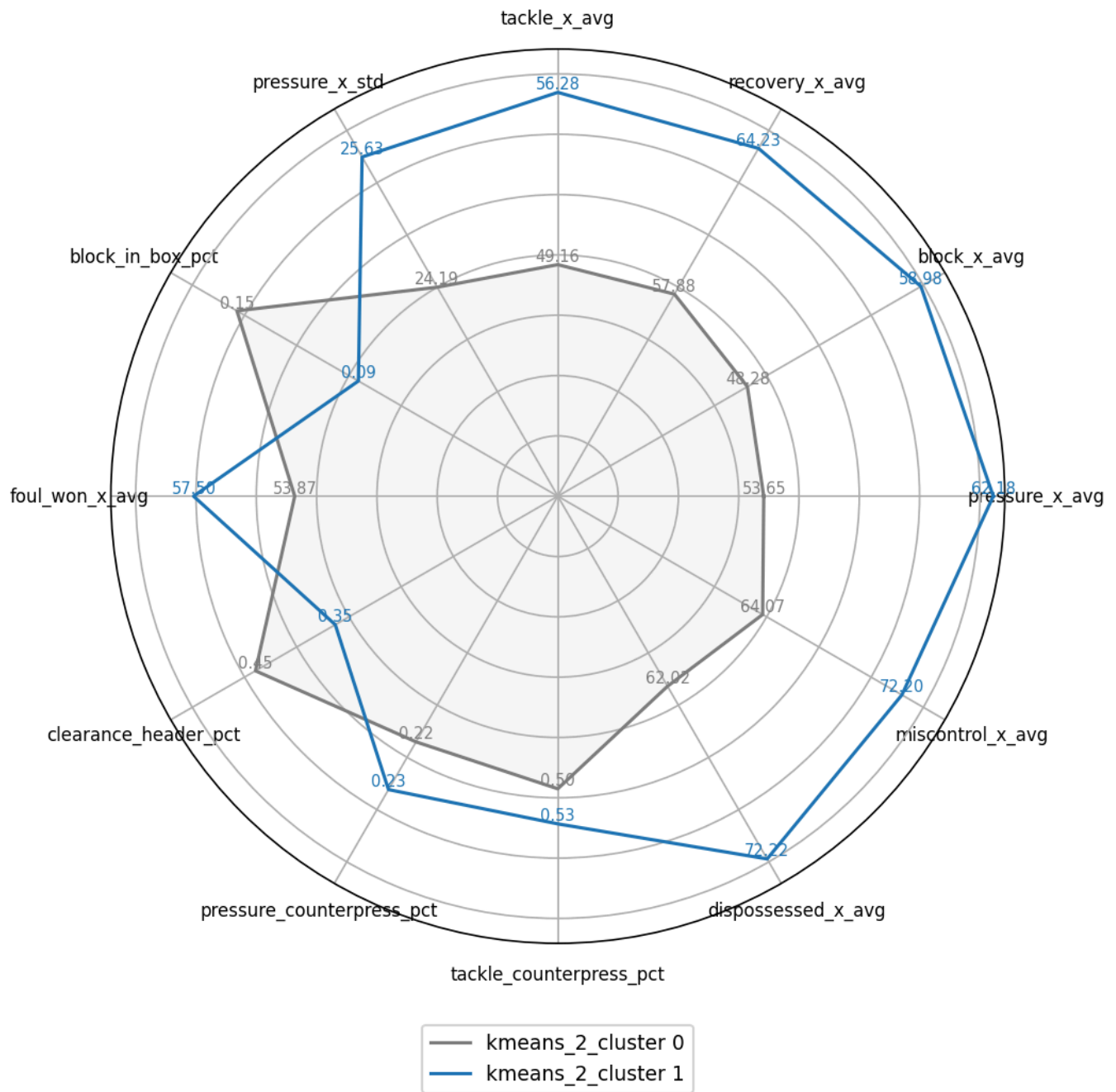


Figure 5: Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (adj)

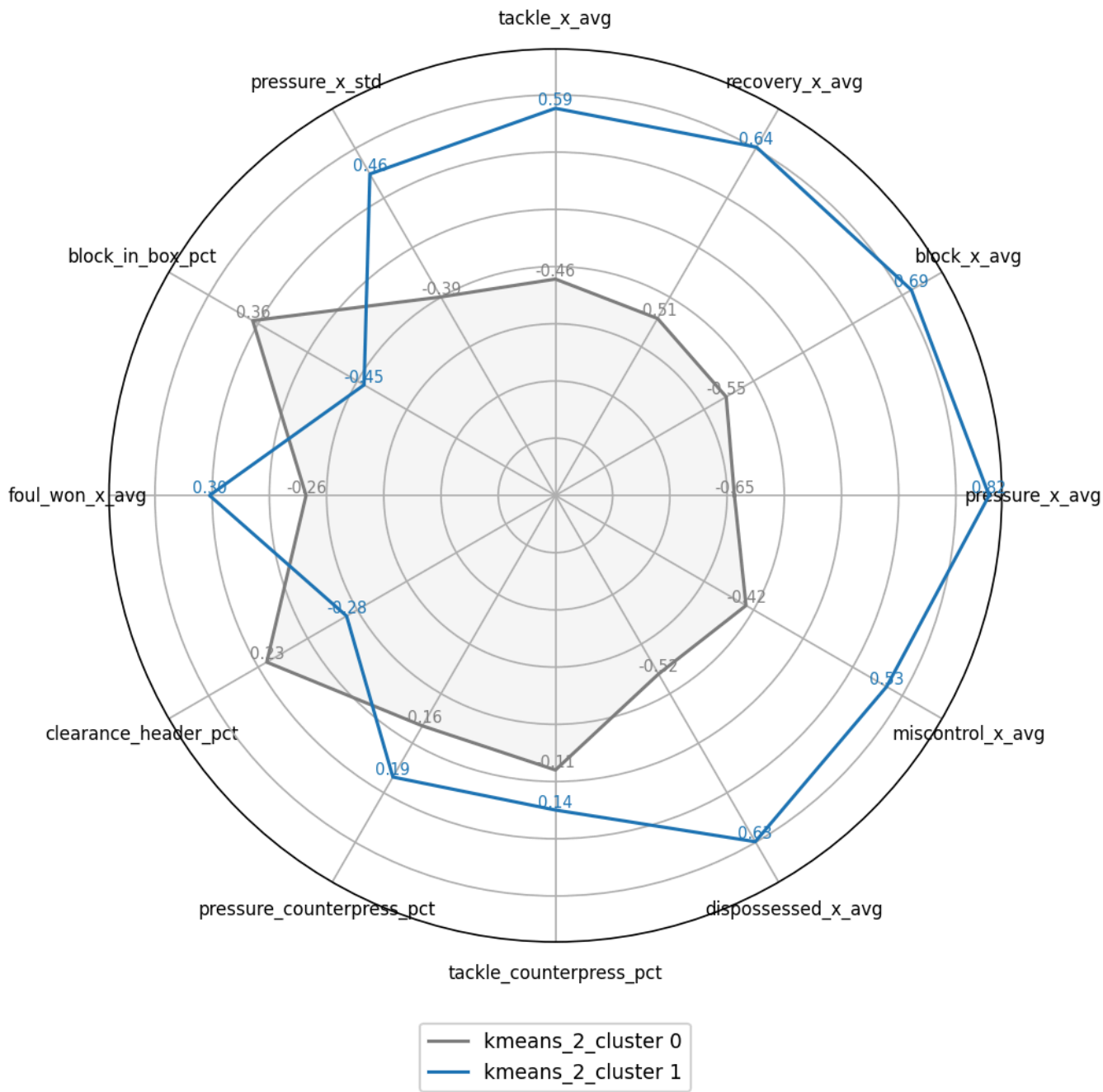


Figure 6: Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

Passing — kmeans_2_cluster comparison (FA Women's Super League, raw)

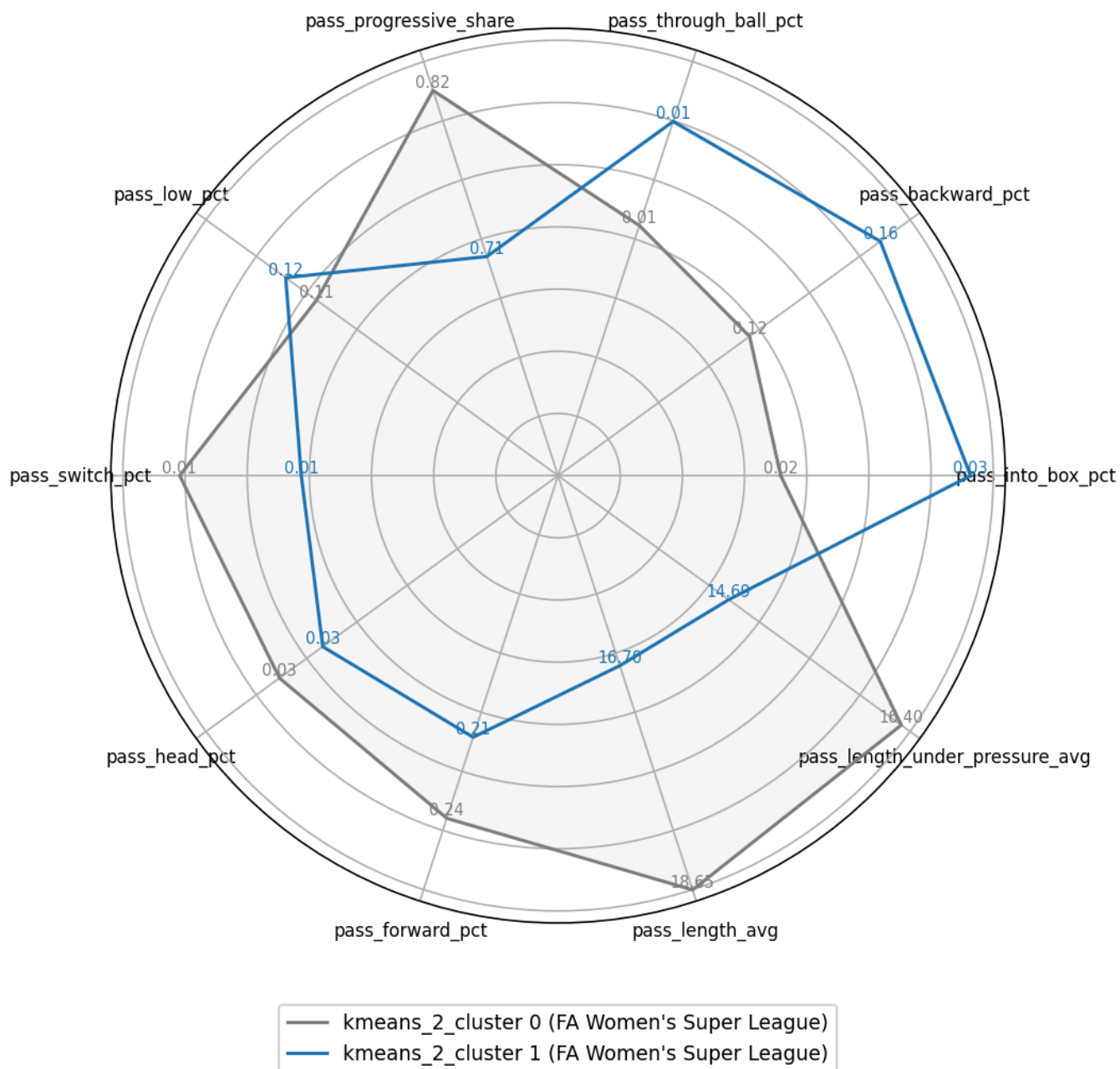


Figure 7: England Center Midfields Passing — Raw Statistics Comparison, 2 Clusters

Passing — kmeans_2_cluster comparison (FA Women's Super League, adj)

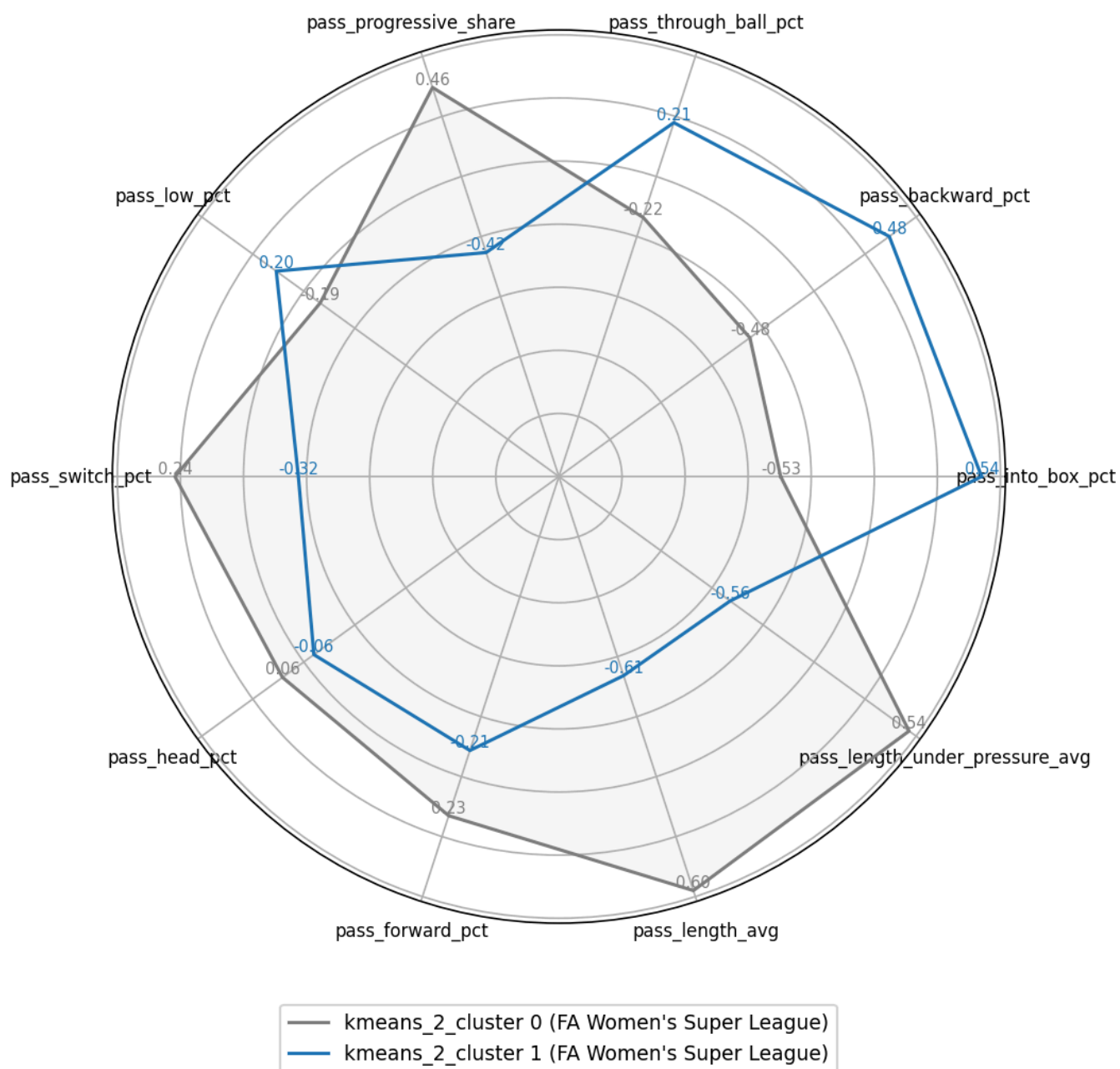


Figure 8: England Center Midfields Passing — Adjusted Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (FA Women's Super League, raw)

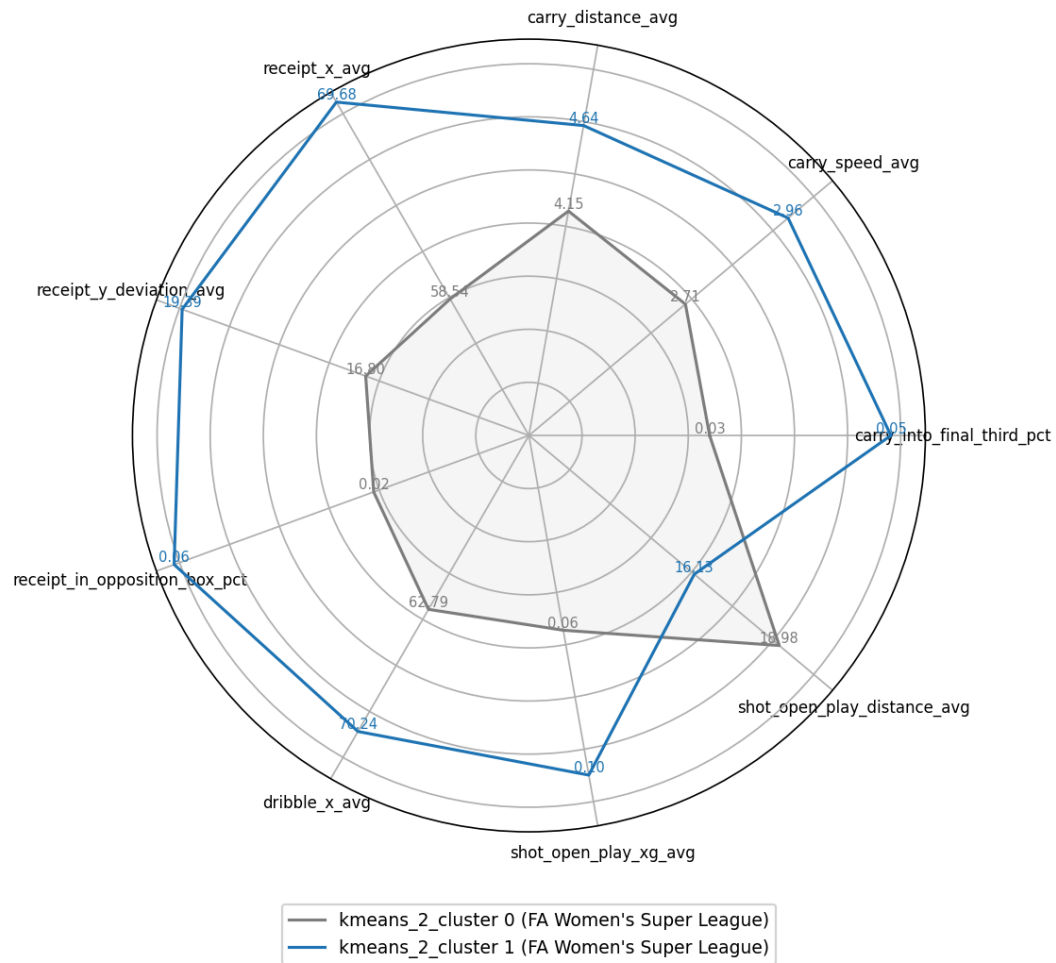


Figure 9: England Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (FA Women's Super League, adj)

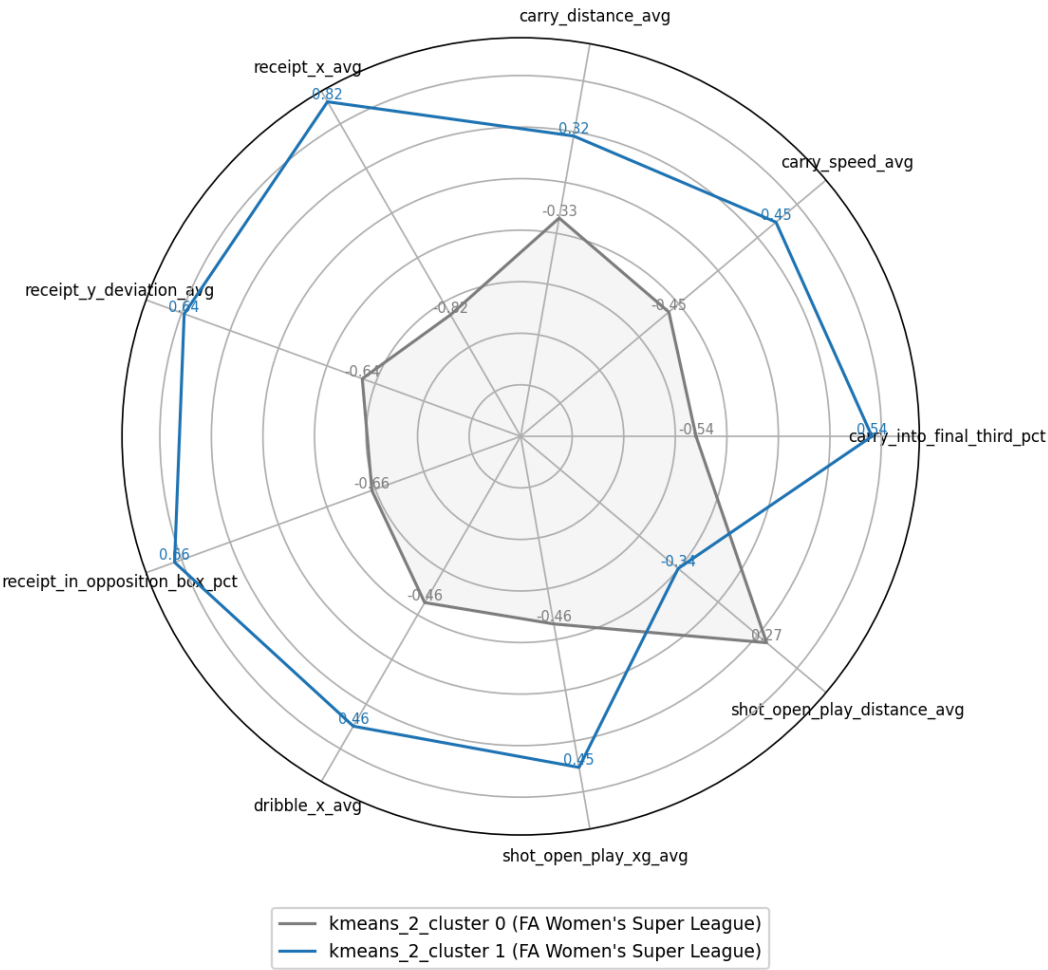


Figure 10: England Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (FA Women's Super League, raw)

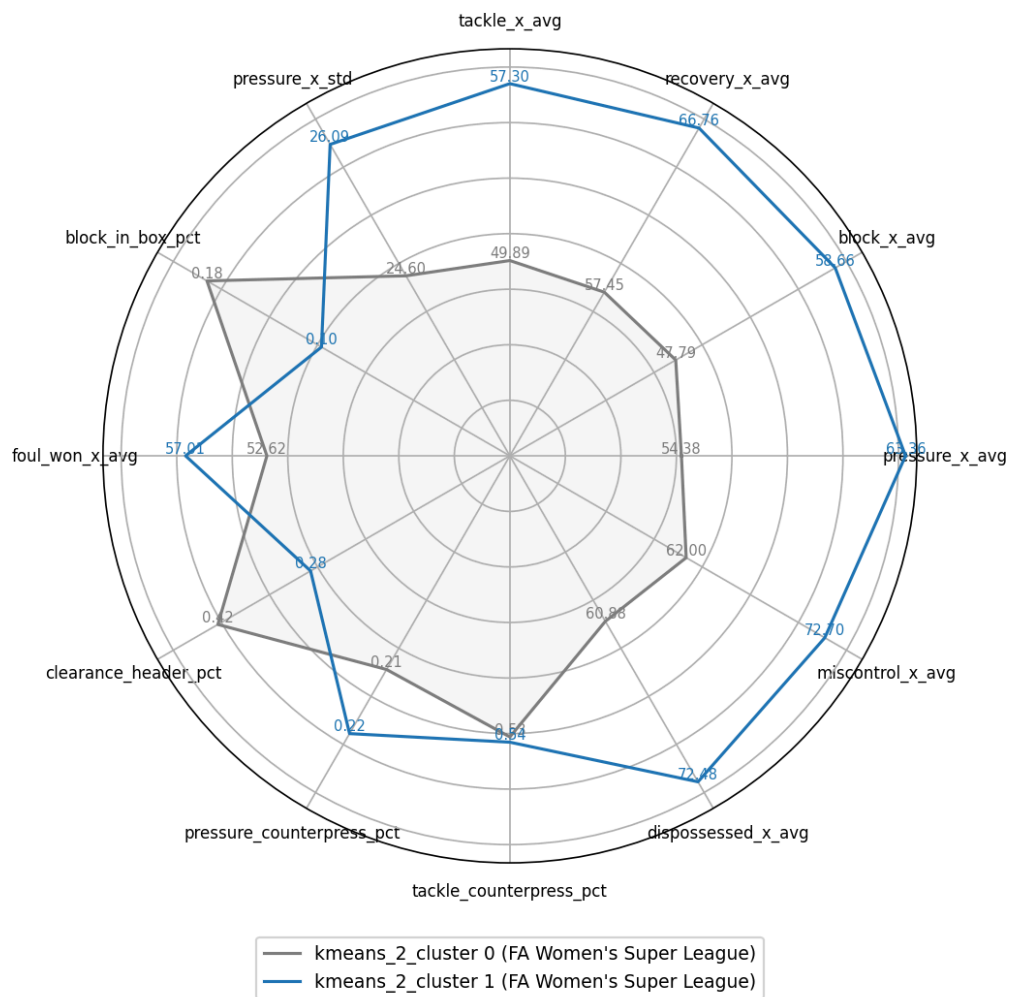


Figure 11: England Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (FA Women's Super League, adj)

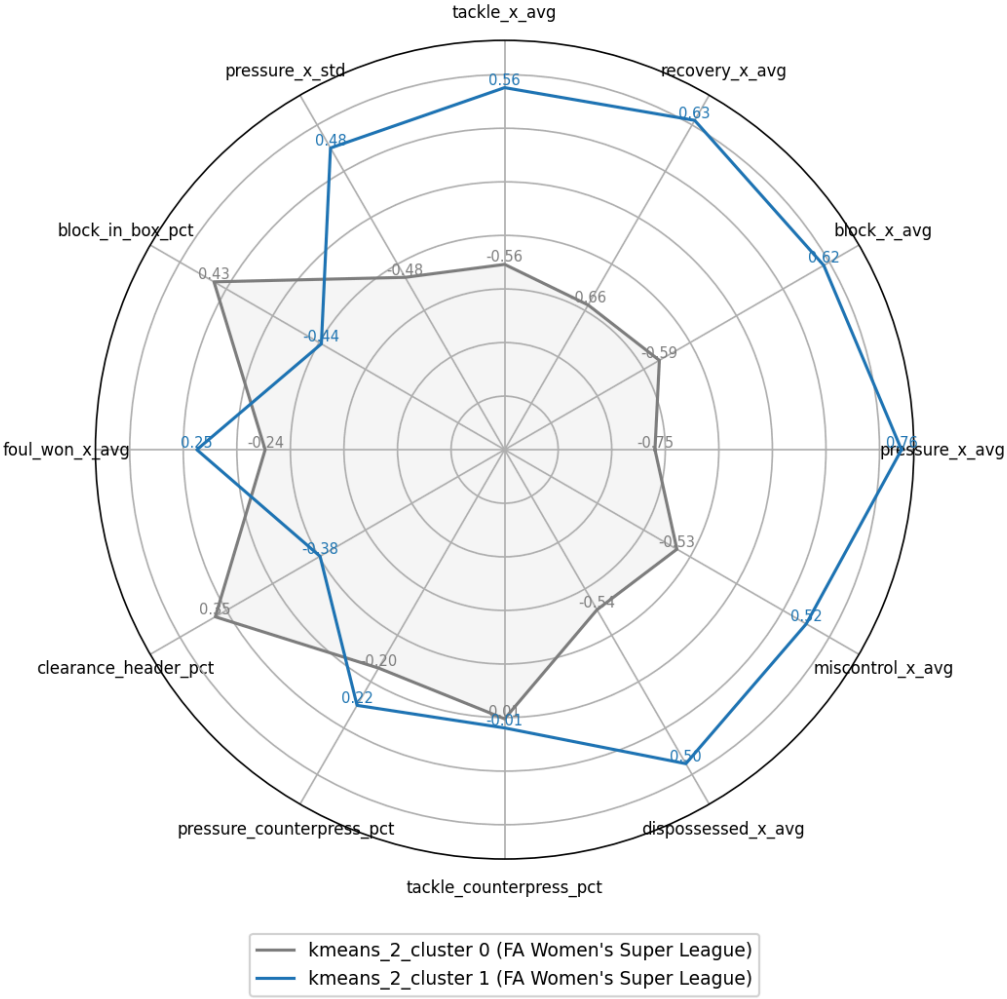


Figure 12: England Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

Frauen Bundesliga

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Passing — kmeans_2_cluster comparison (Frauen Bundesliga, raw)

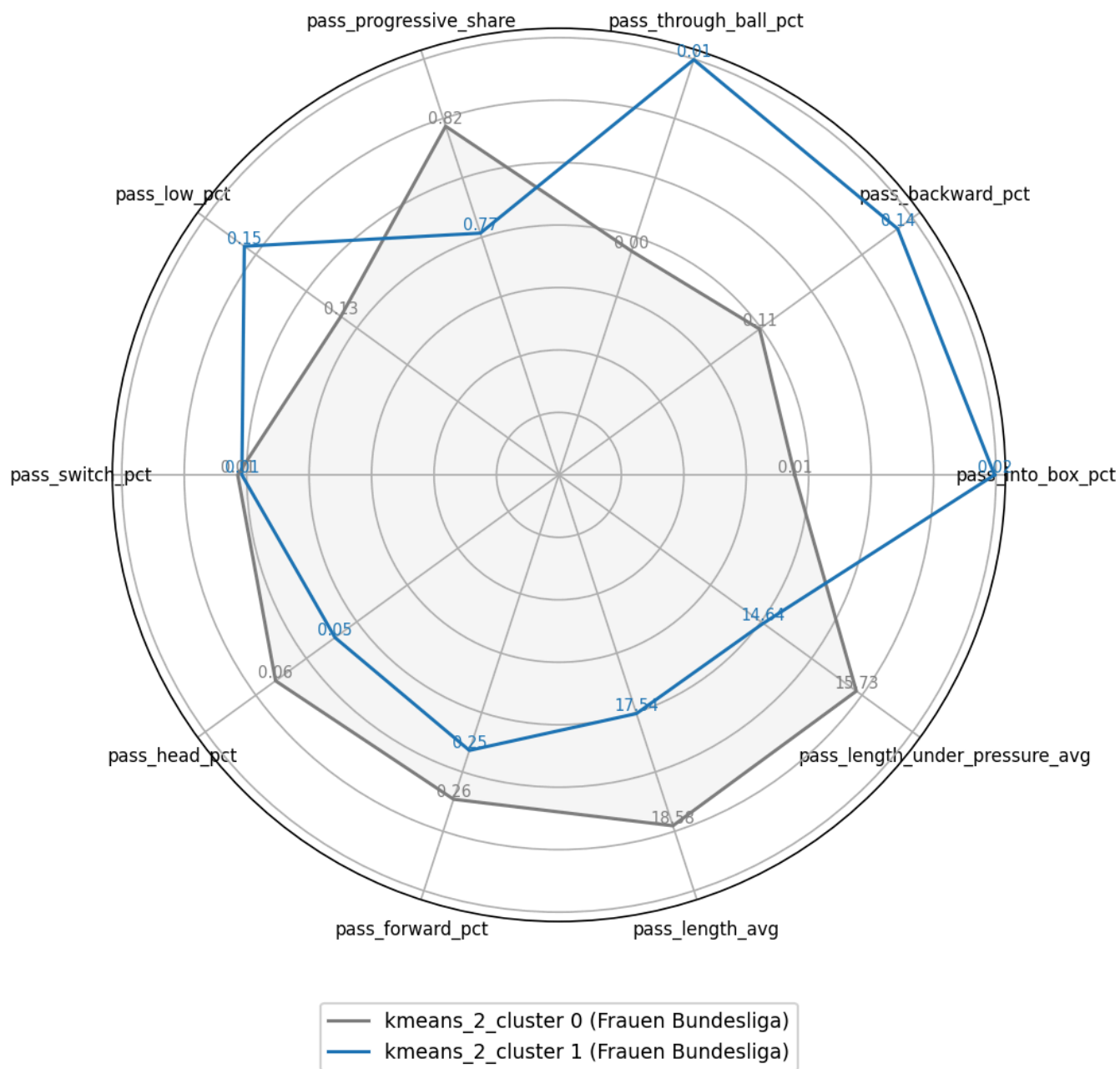


Figure 13: Germany Center Midfields Passing — Raw Statistics Comparison, 2 Clusters

Passing — kmeans_2_cluster comparison (Frauen Bundesliga, adj)

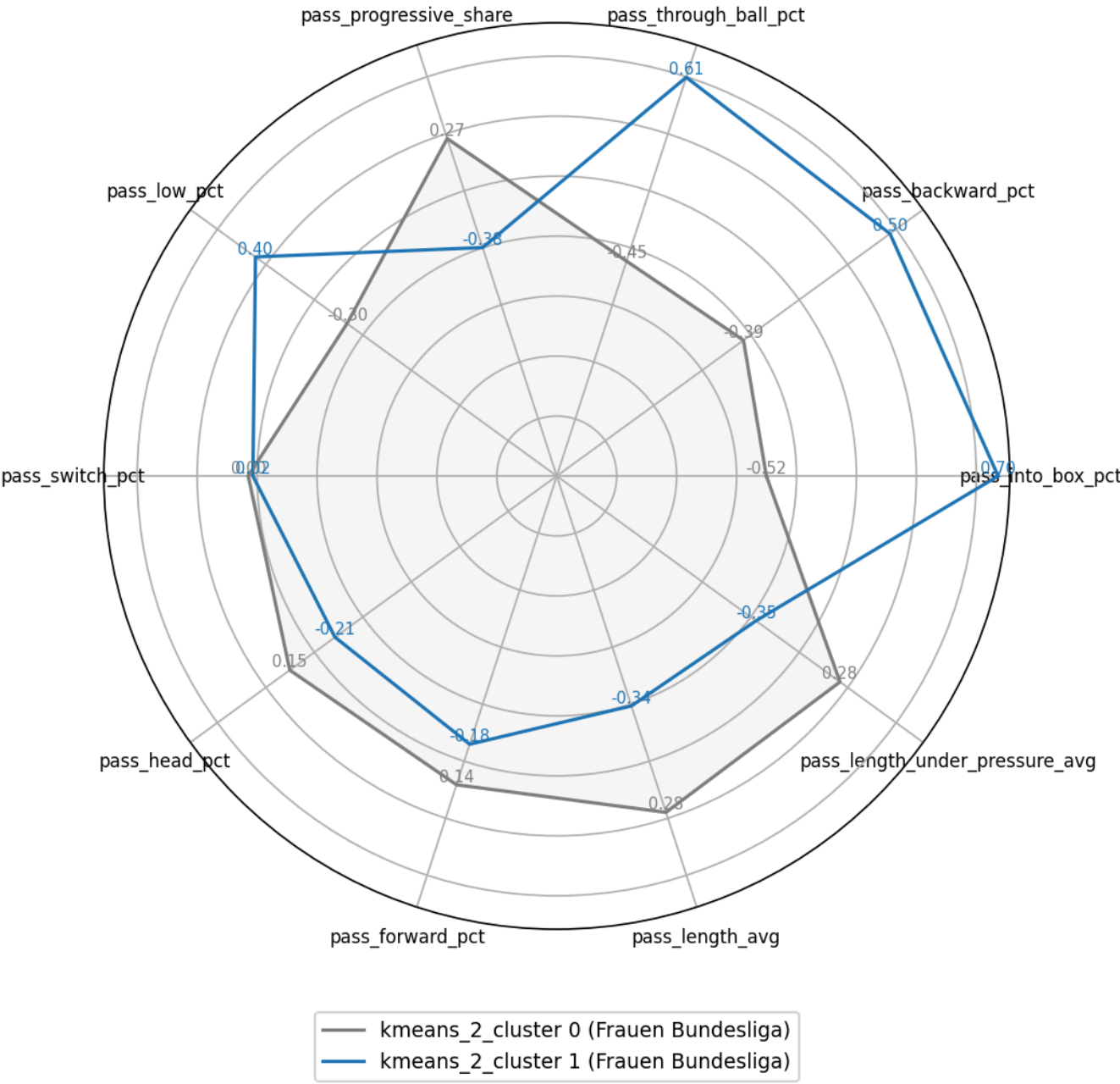


Figure 14: Germany Center Midfields Passing — Adjusted Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (Frauen Bundesliga, raw)

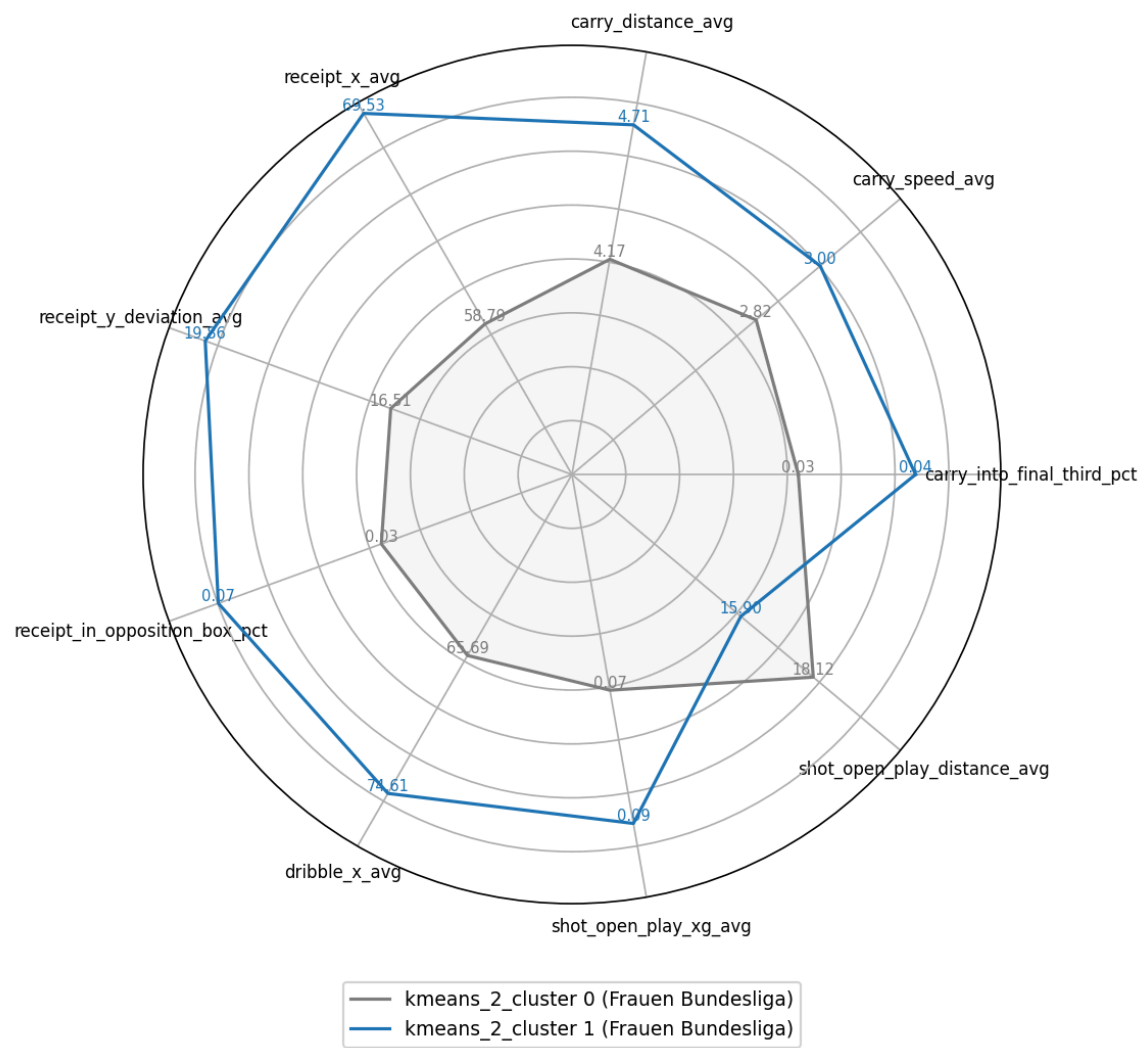


Figure 15: Germany Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (Frauen Bundesliga, adj)

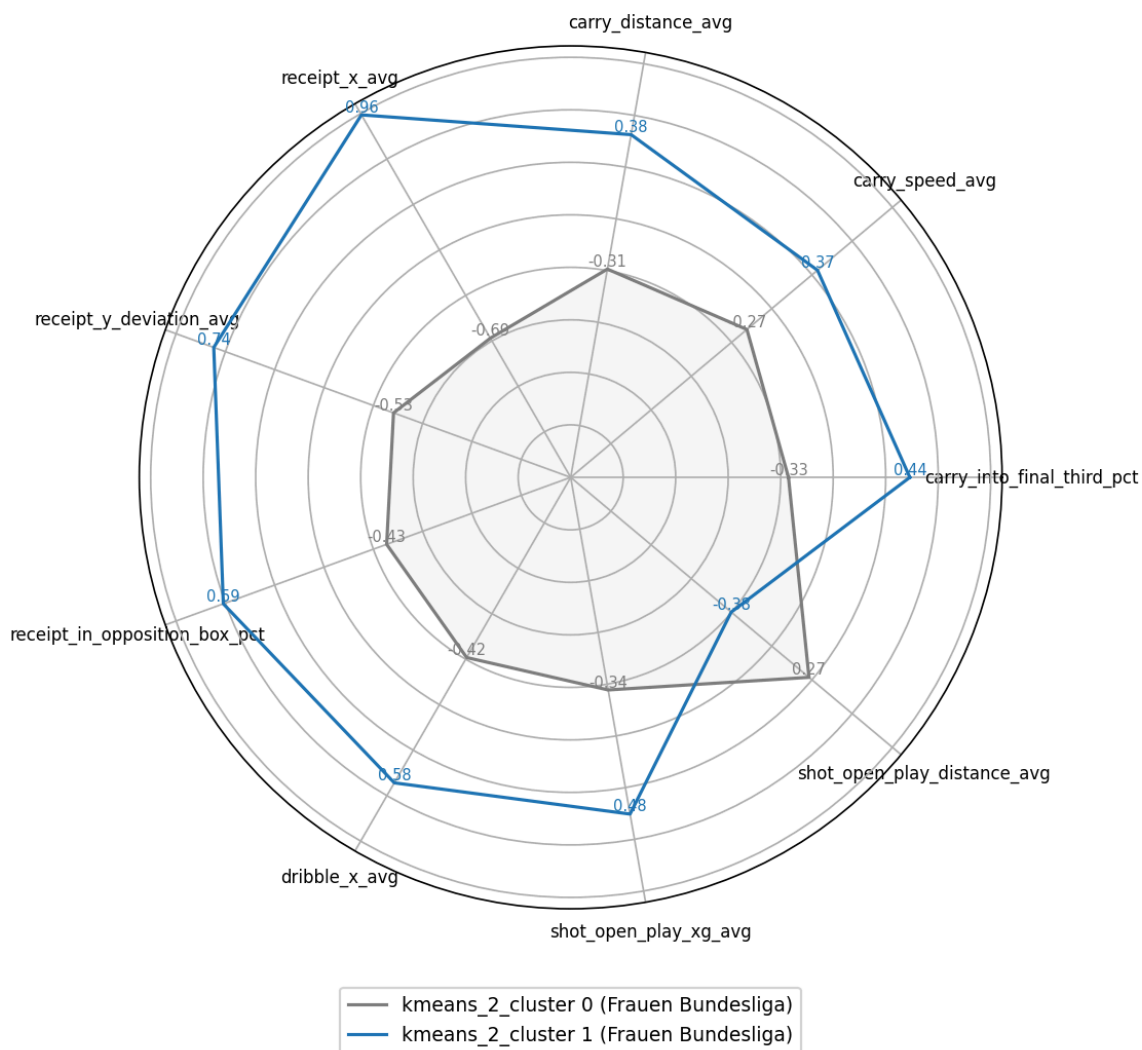


Figure 16: Germany Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (Frauen Bundesliga, raw)

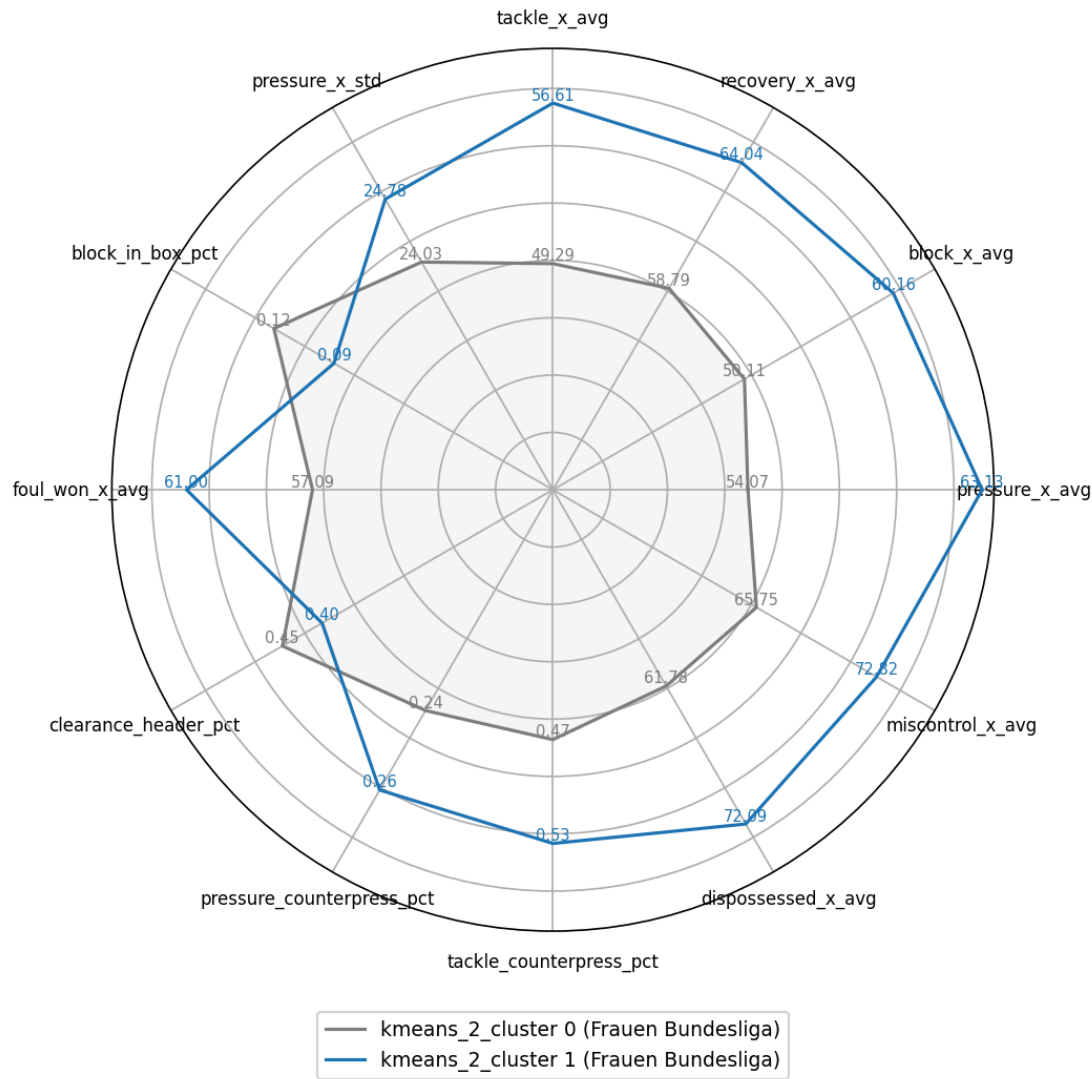


Figure 17: Germany Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (Frauen Bundesliga, adj)

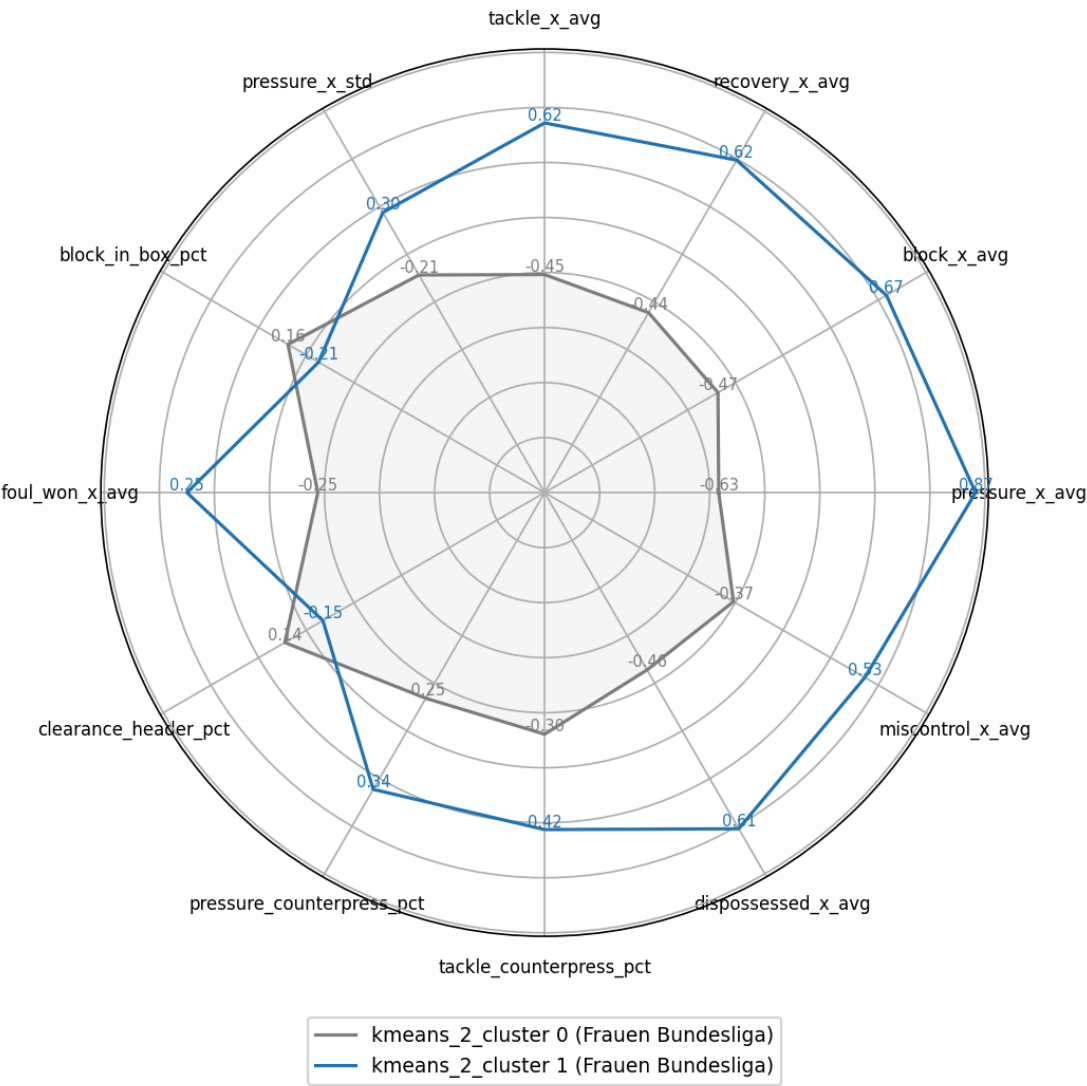


Figure 18: Germany Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

Liga F

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Passing — kmeans_2_cluster comparison (Liga F, raw)

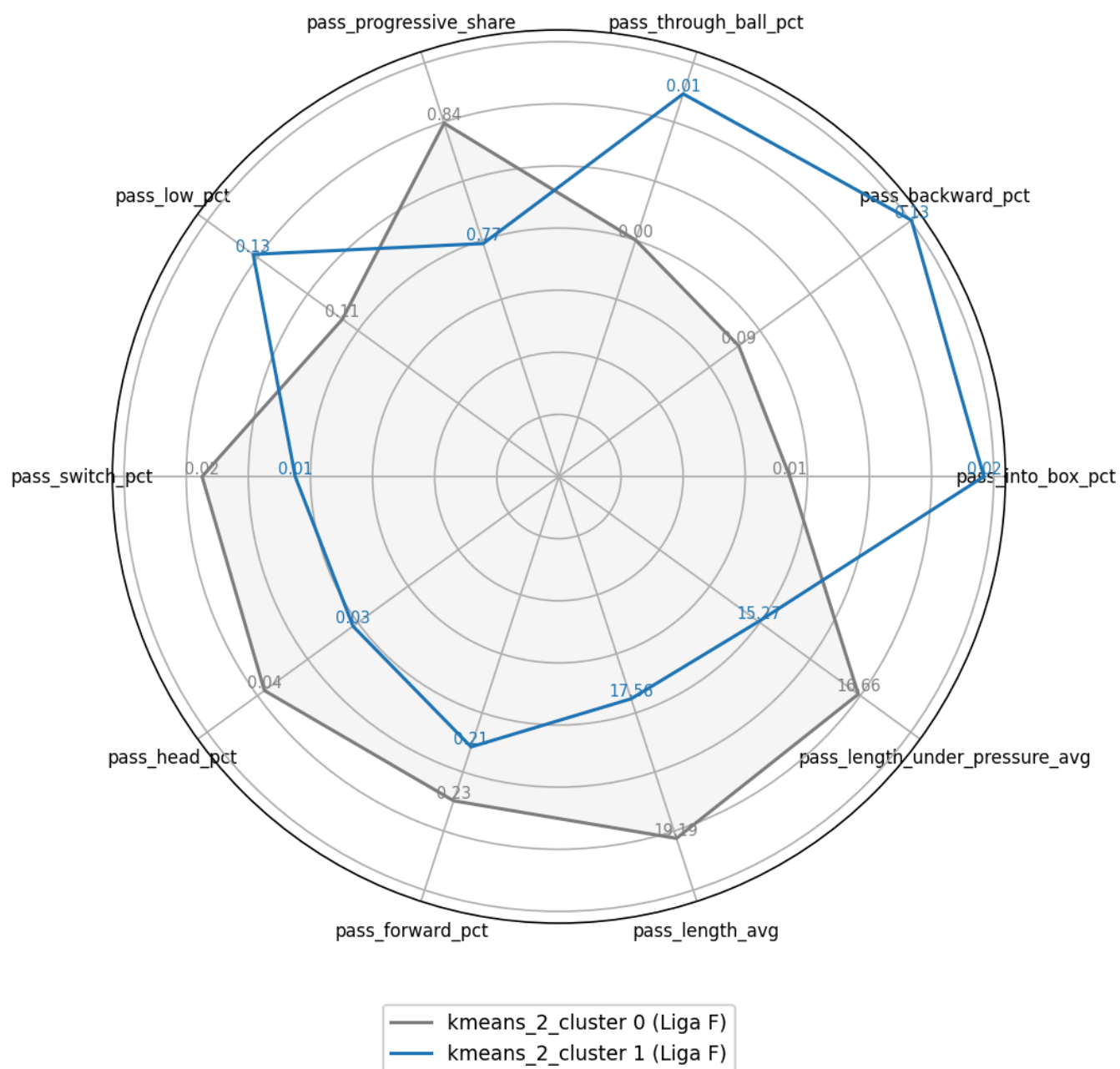


Figure 19: Spain Center Midfields Passing — Raw Statistics Comparison, 2 Clusters

Passing — kmeans_2_cluster comparison (Liga F, adj)

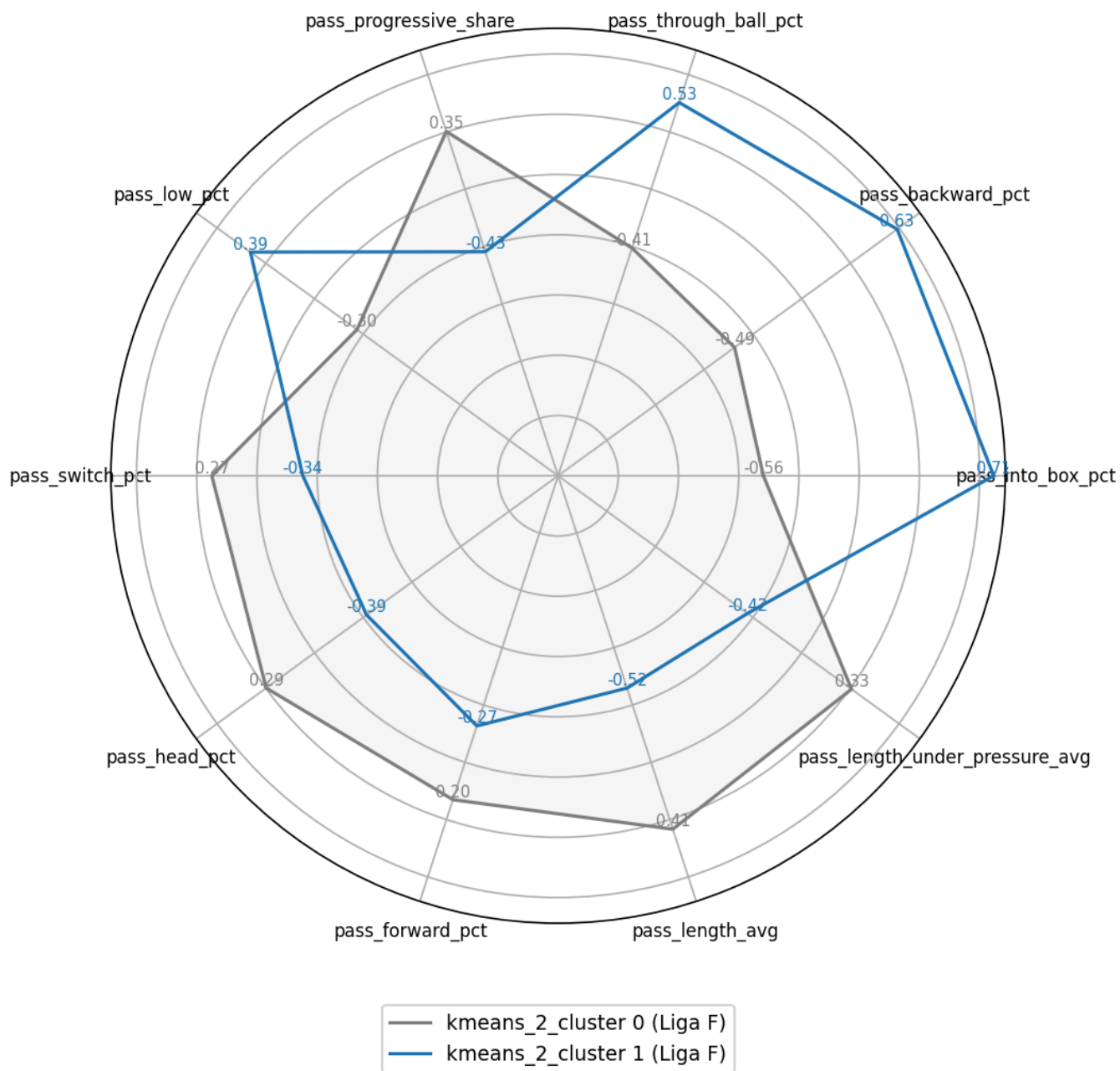


Figure 20: Spain Center Midfields Passing — Adjusted Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (Liga F, raw)

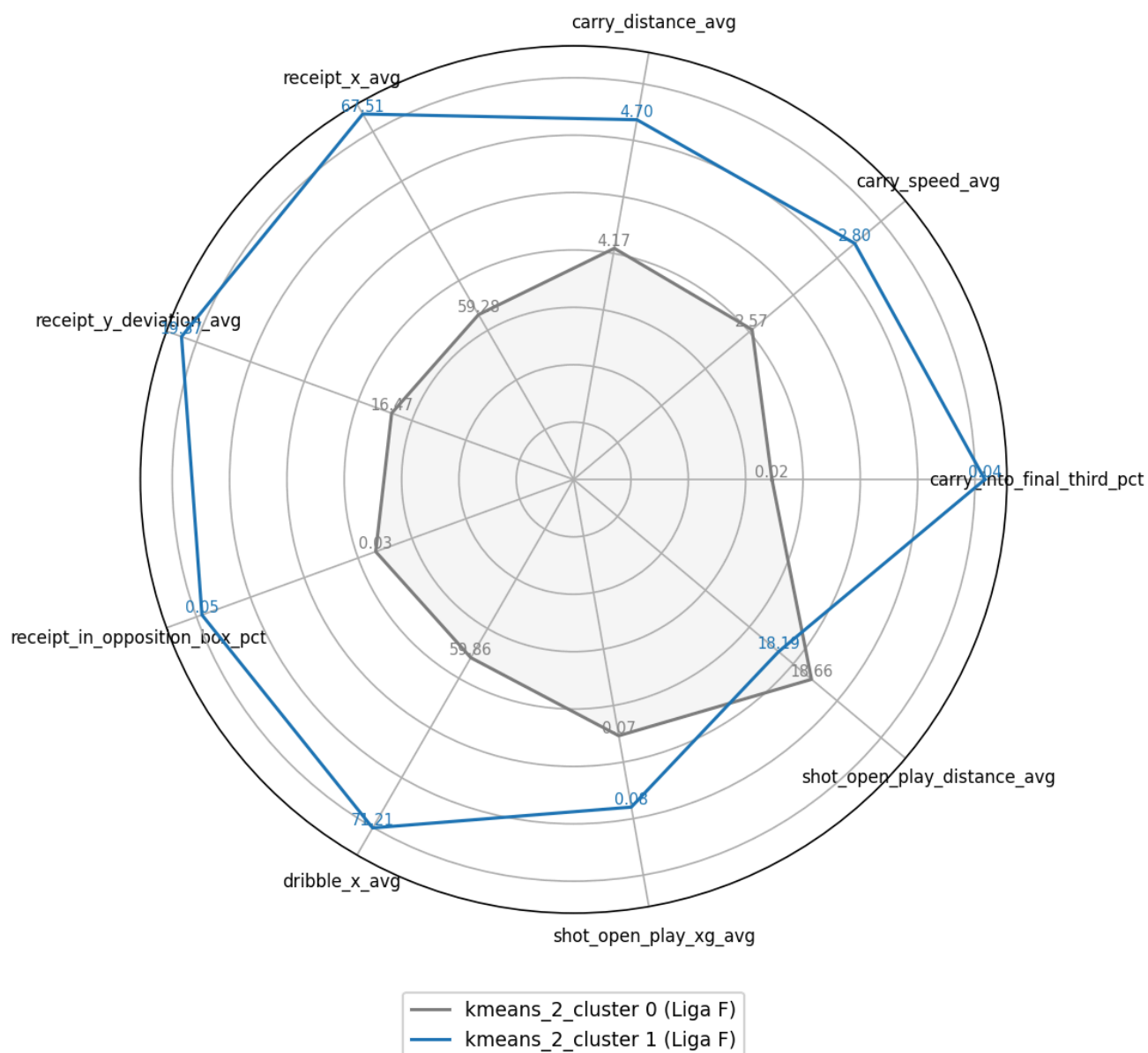


Figure 21: Spain Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (Liga F, adj)

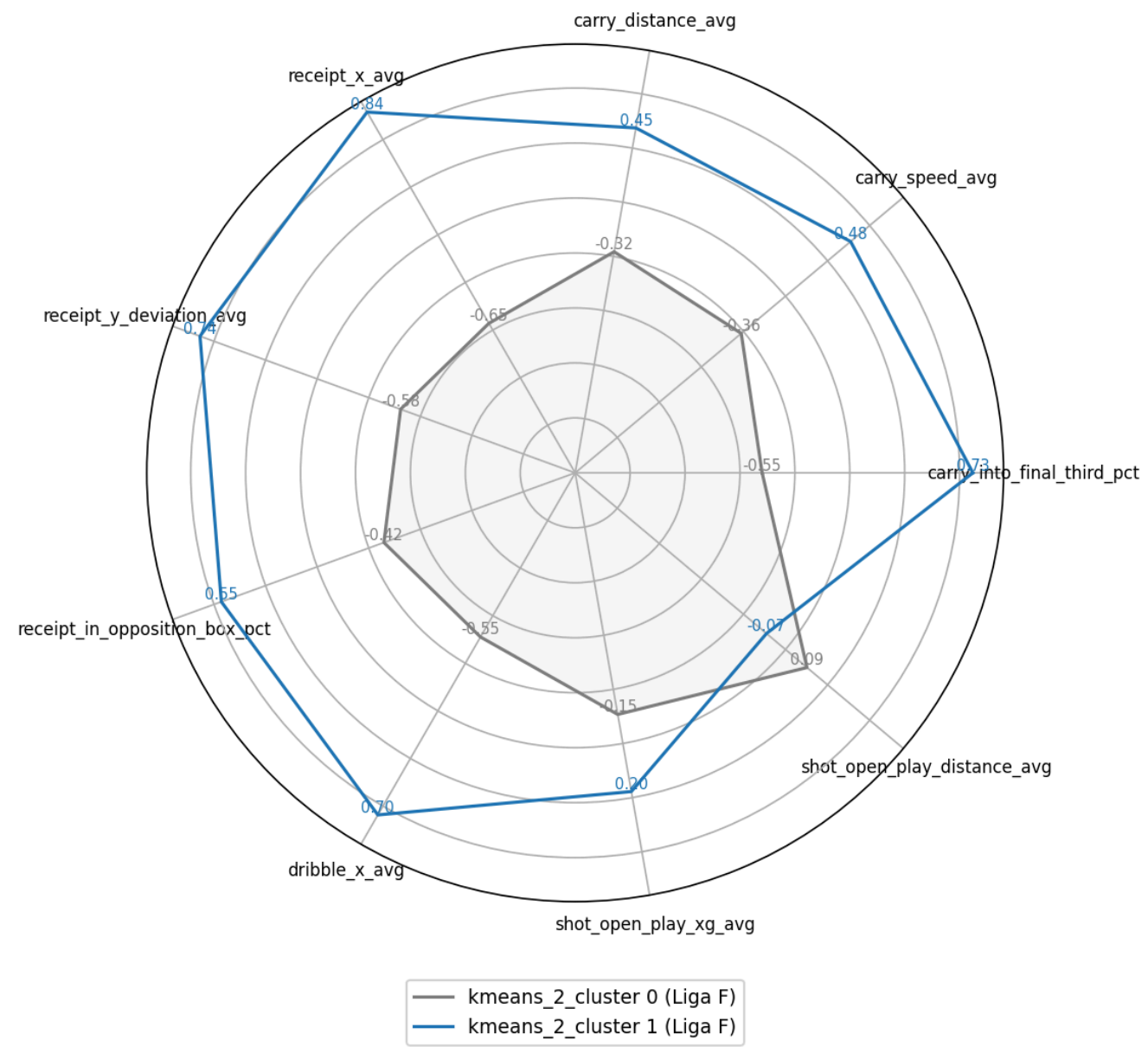


Figure 22: Spain Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (Liga F, raw)

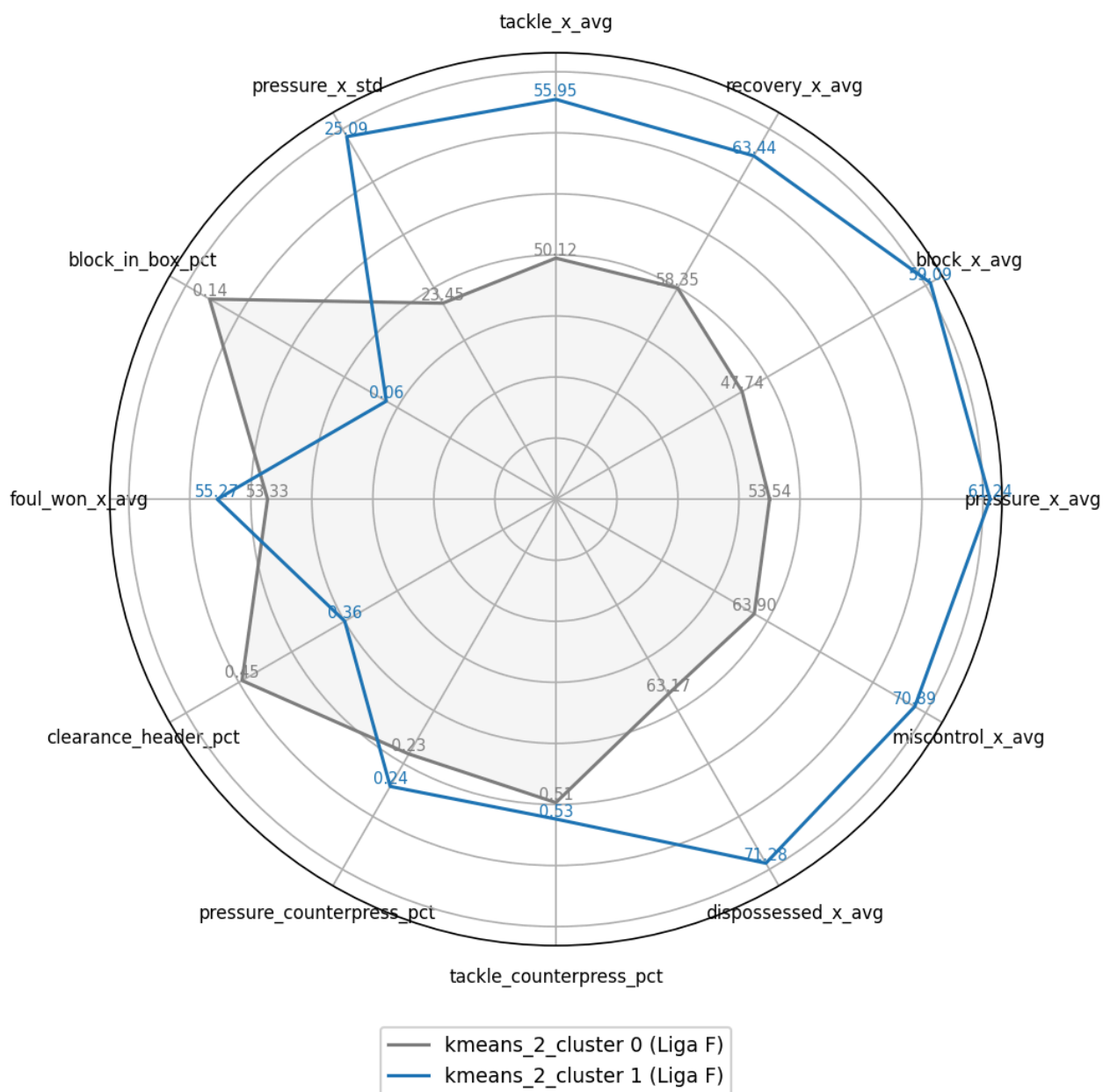


Figure 23: Spain Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (Liga F, adj)

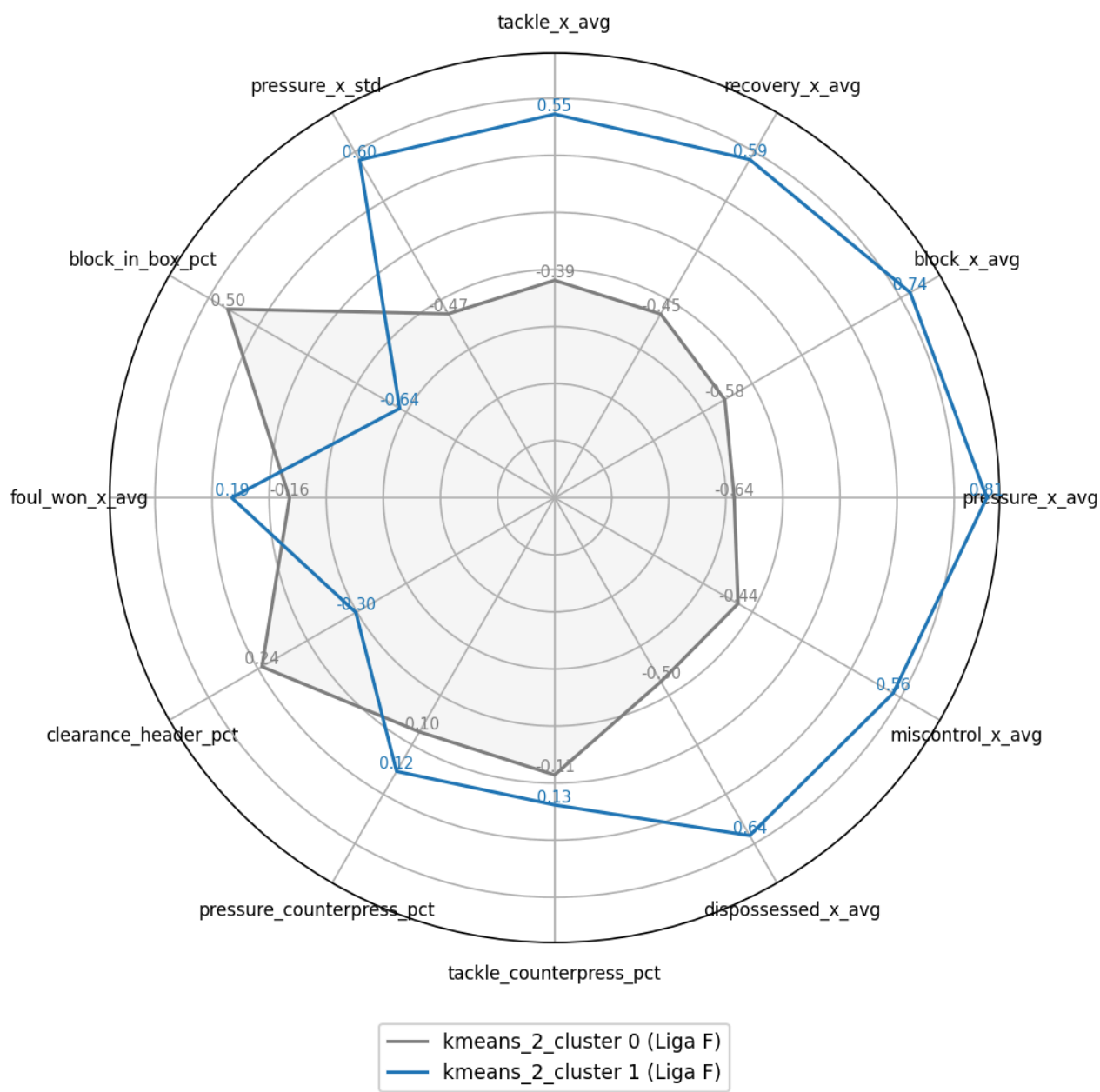


Figure 24: Spain Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

Serie A Women

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Passing — kmeans_2_cluster comparison (Serie A Women, raw)

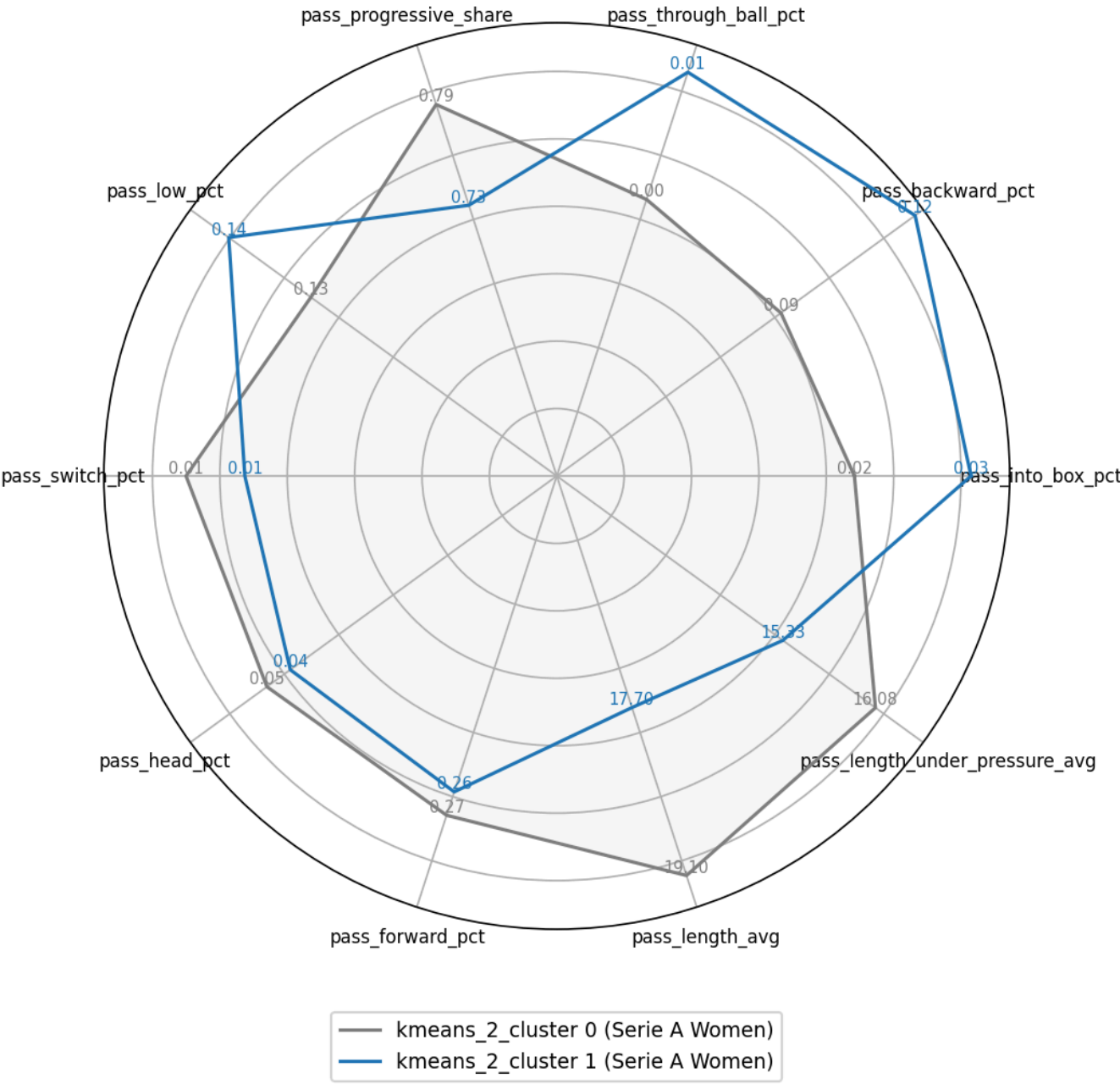


Figure 25: Italy Center Midfields Passing — Raw Statistics Comparison, 2 Clusters

Passing — kmeans_2_cluster comparison (Serie A Women, adj)

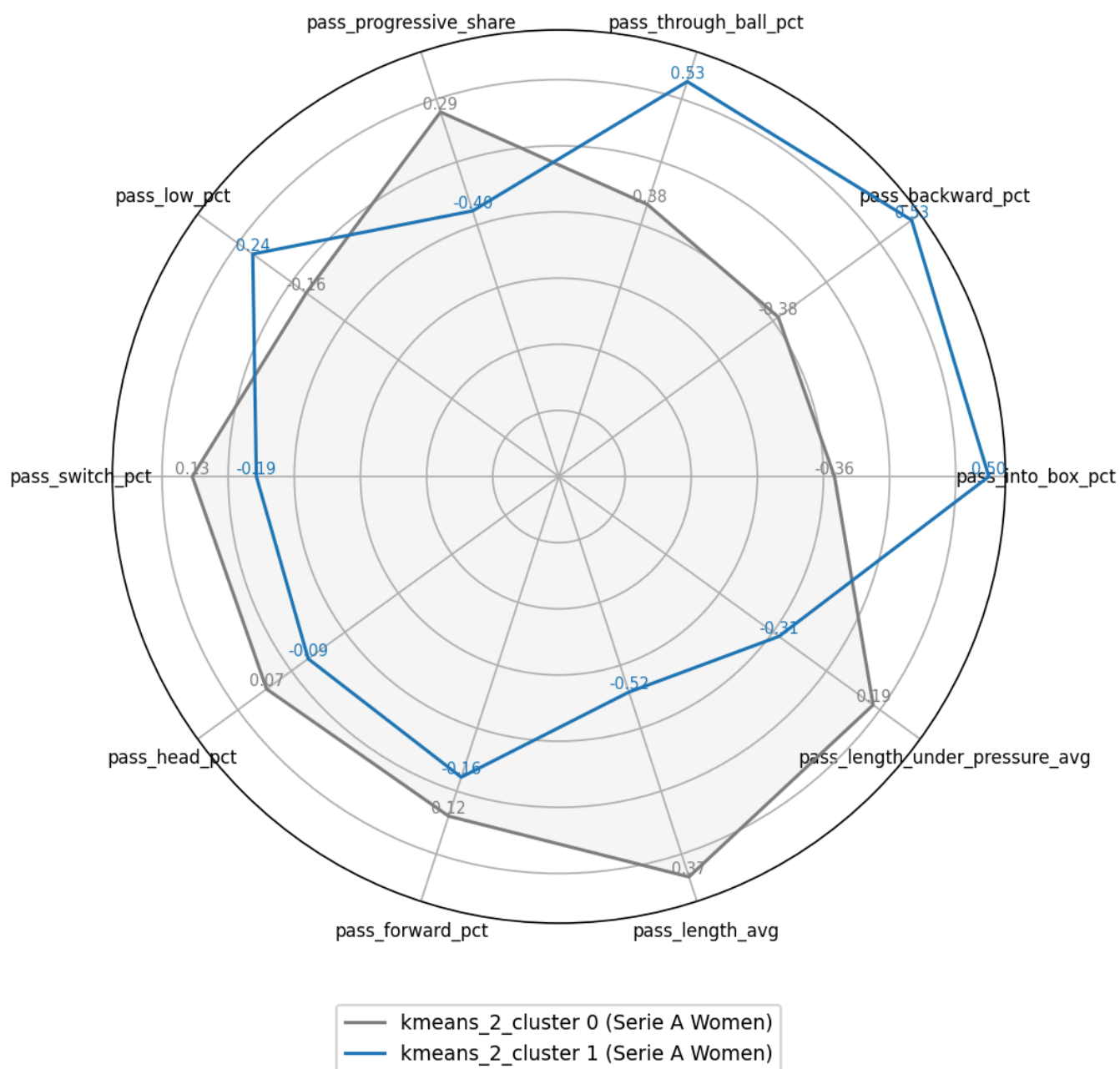


Figure 26: Italy Center Midfields Passing — Adjusted Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (Serie A Women, raw)

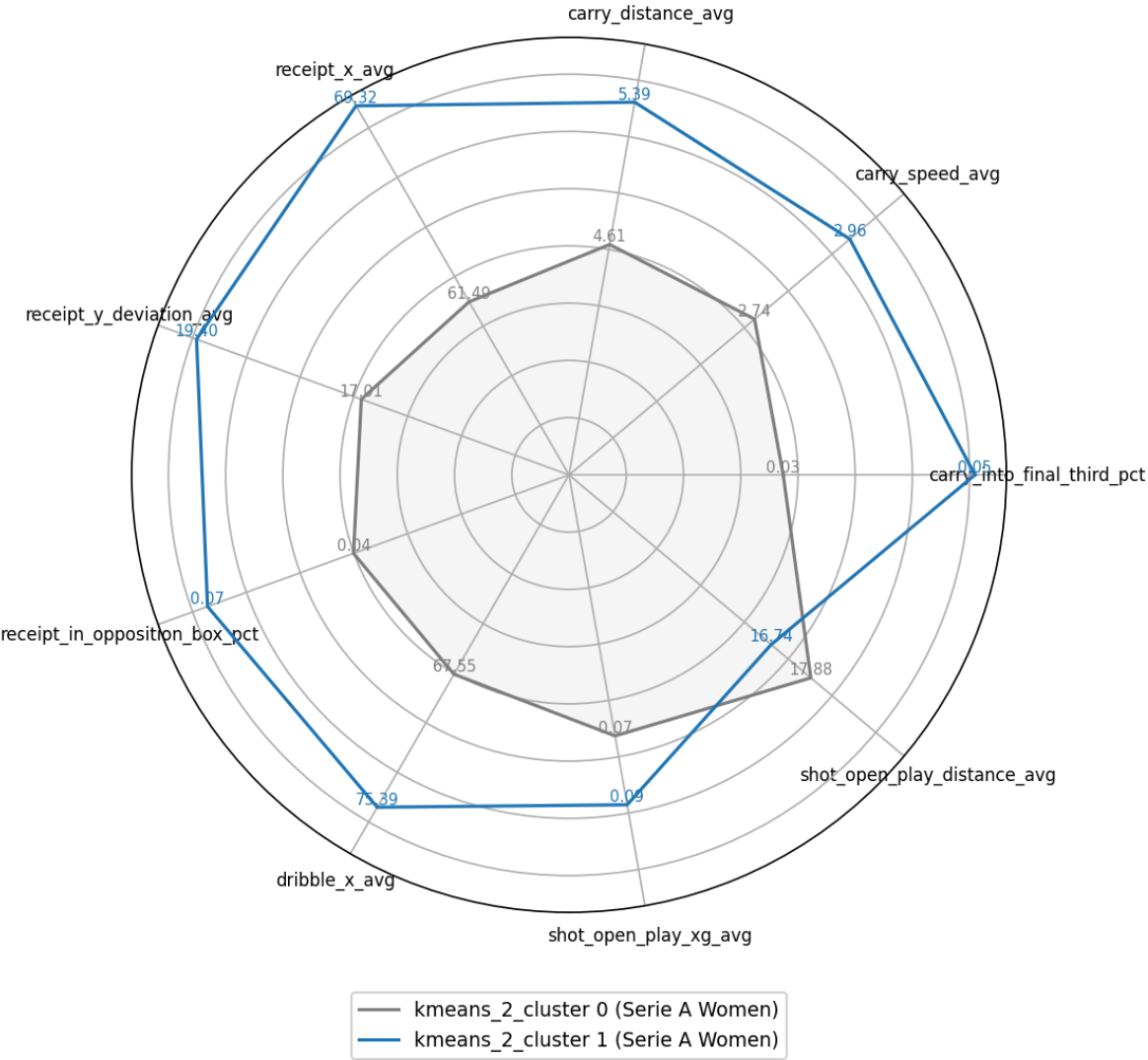


Figure 27: Italy Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 2 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_2_cluster comparison (Serie A Women, adj)

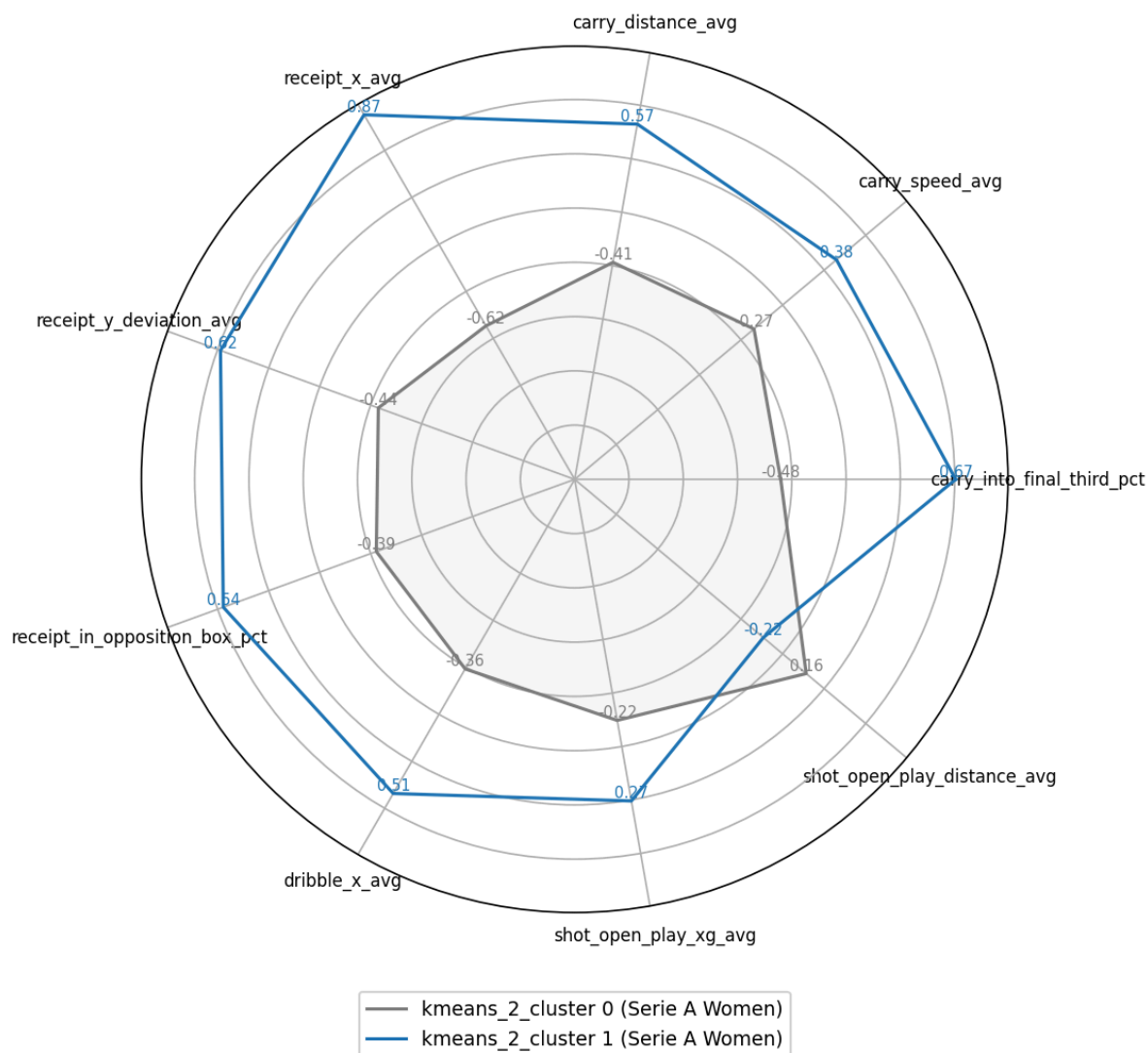


Figure 28: Italy Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (Serie A Women, raw)

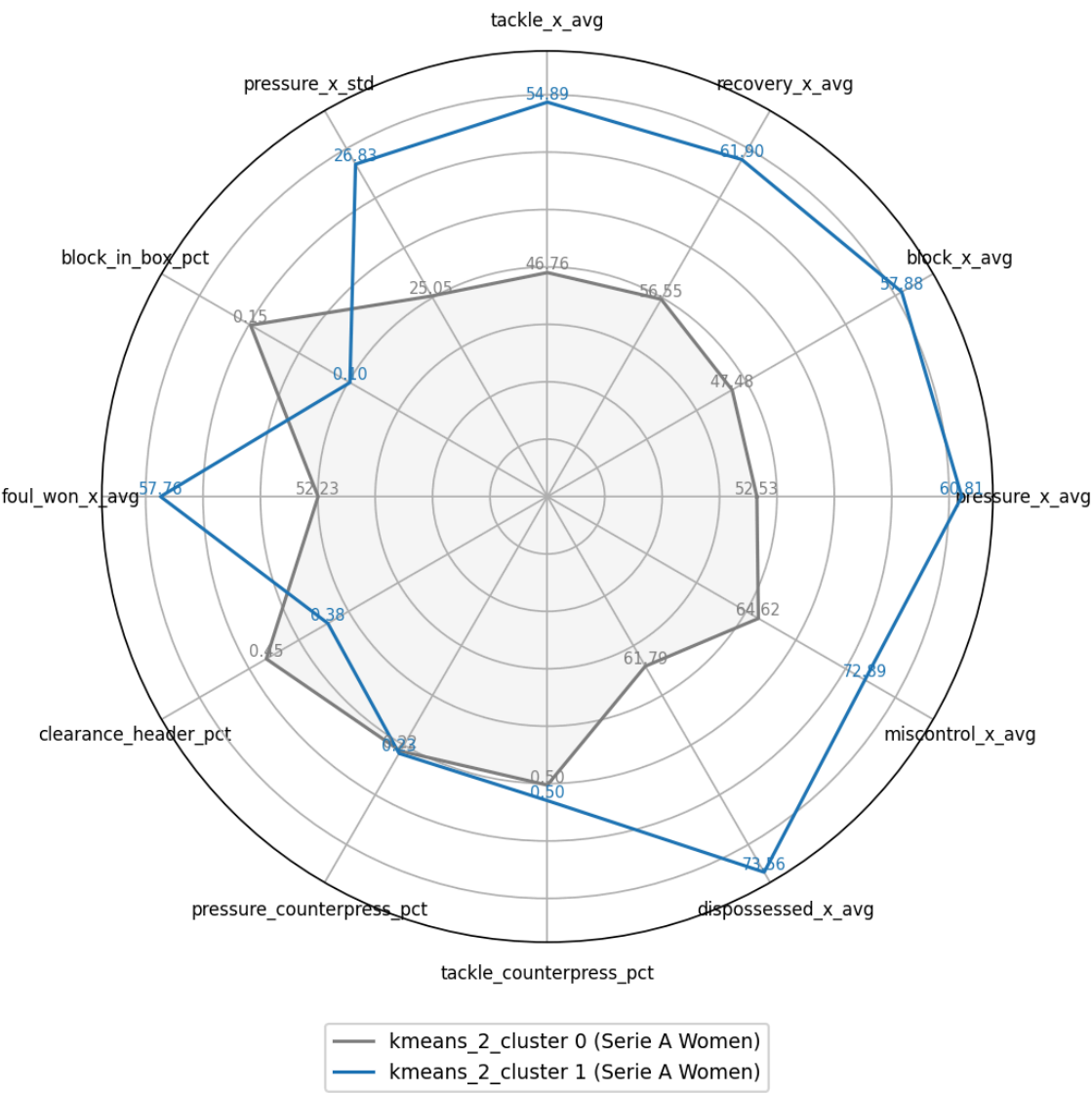


Figure 29: Italy Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_2_cluster comparison (Serie A Women, adj)

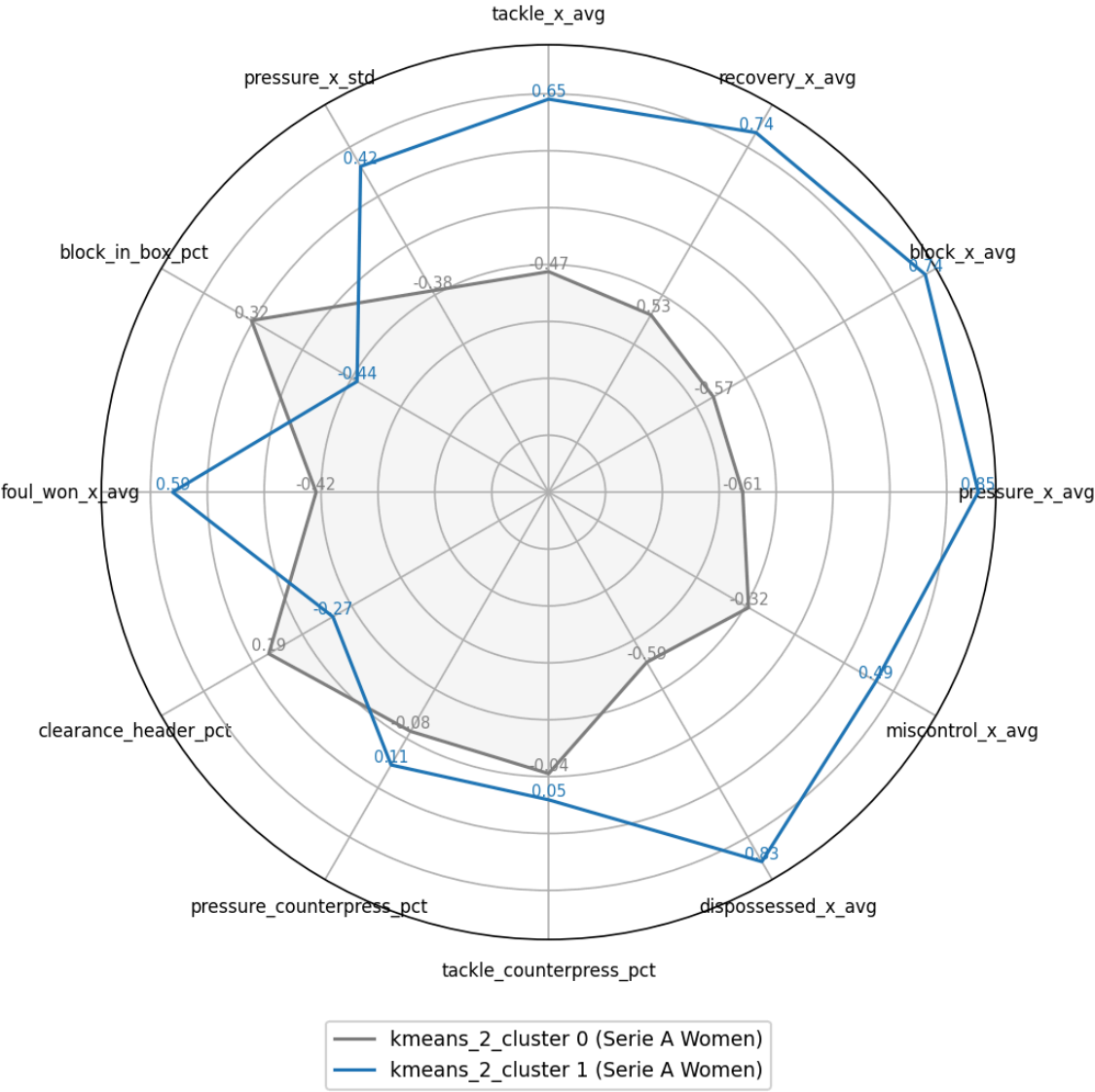


Figure 30: Italy Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

Radar plots for 3 clusters center midfielders' averages

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Passing — kmeans_3_cluster comparison (raw)

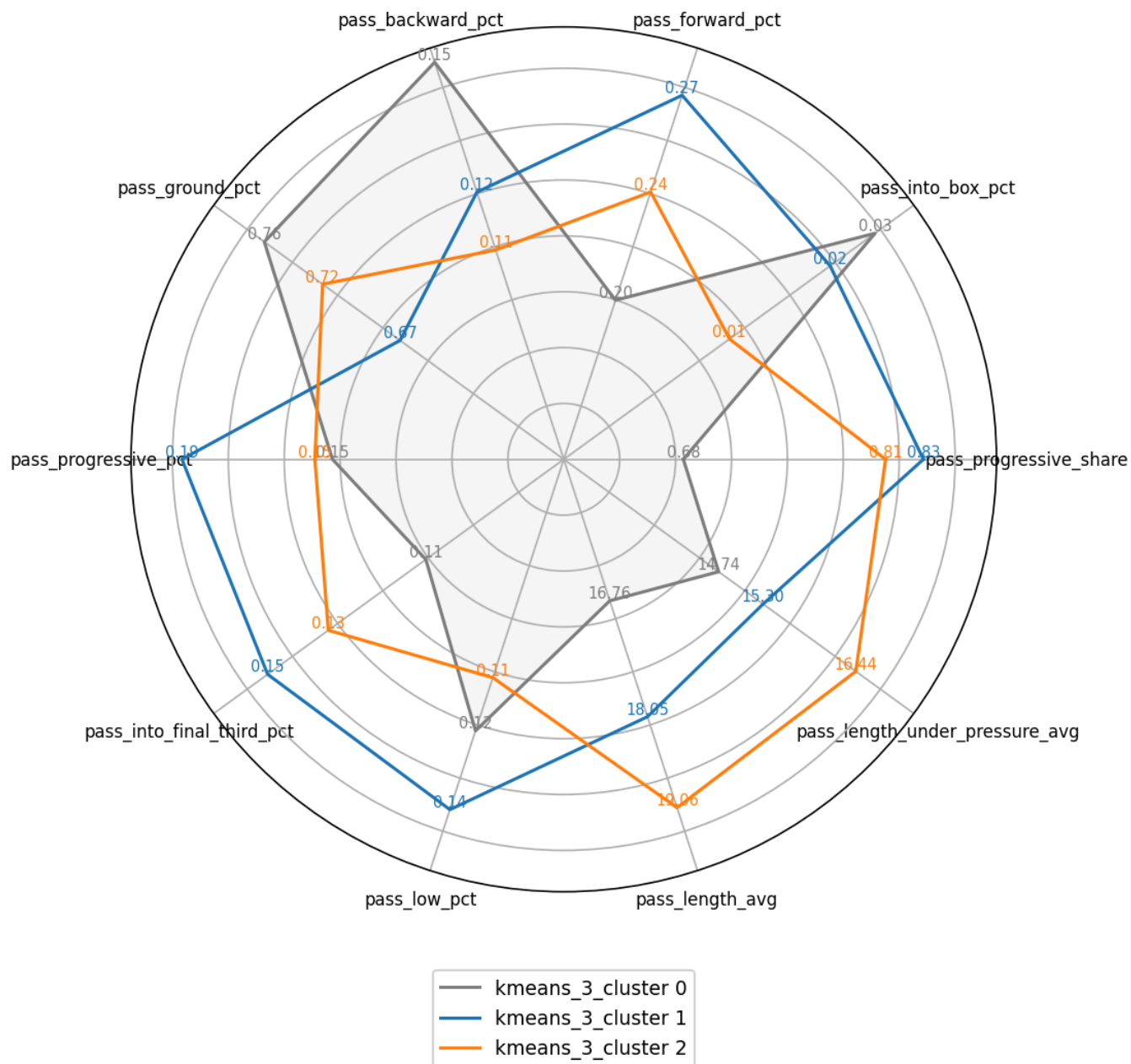


Figure 31: Center Midfields Passing — Raw Statistics Comparison, 3 Clusters

Passing — kmeans_3_cluster comparison (adj)

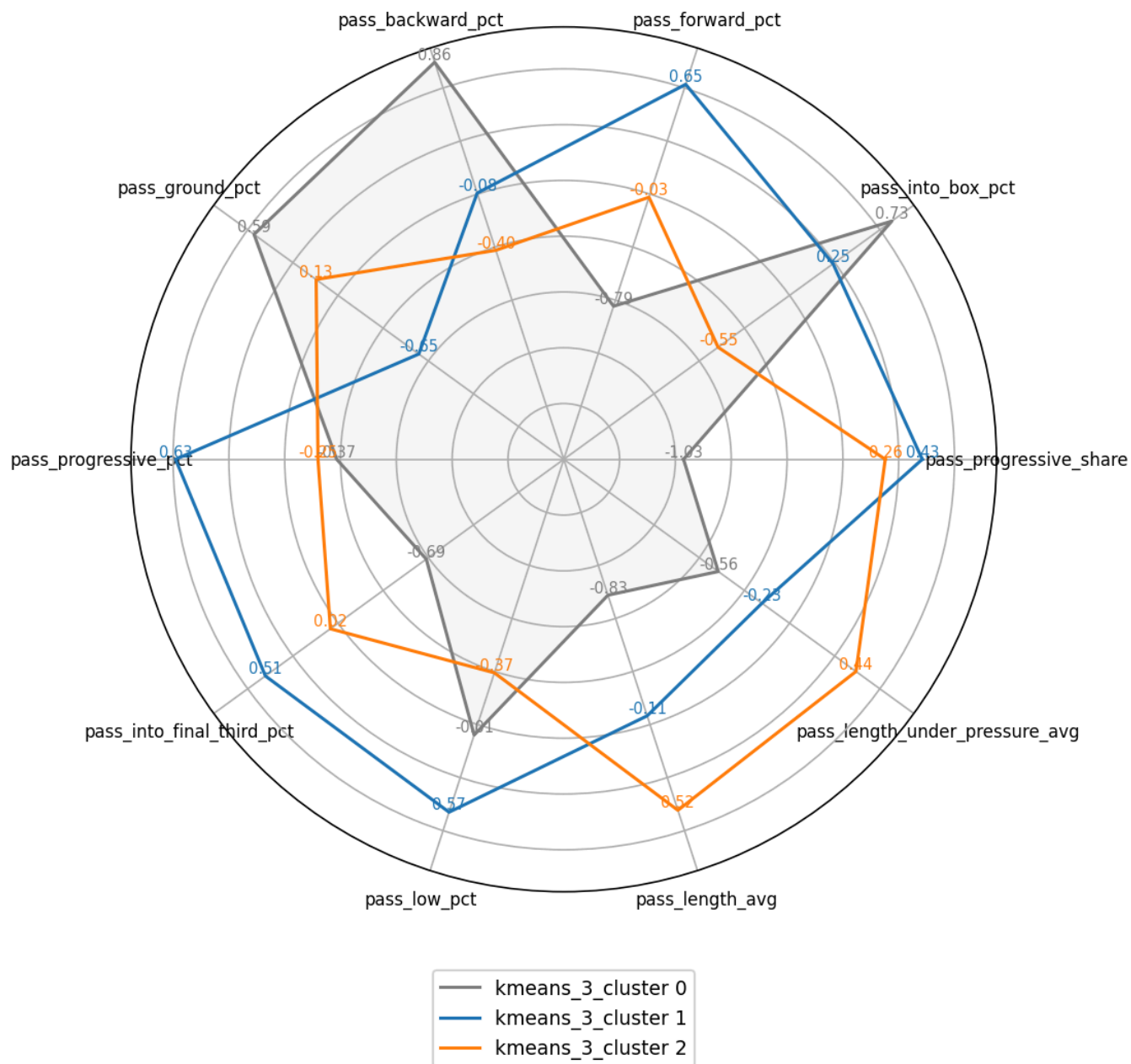


Figure 32: Center Midfields Passing — Adjusted Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (raw)

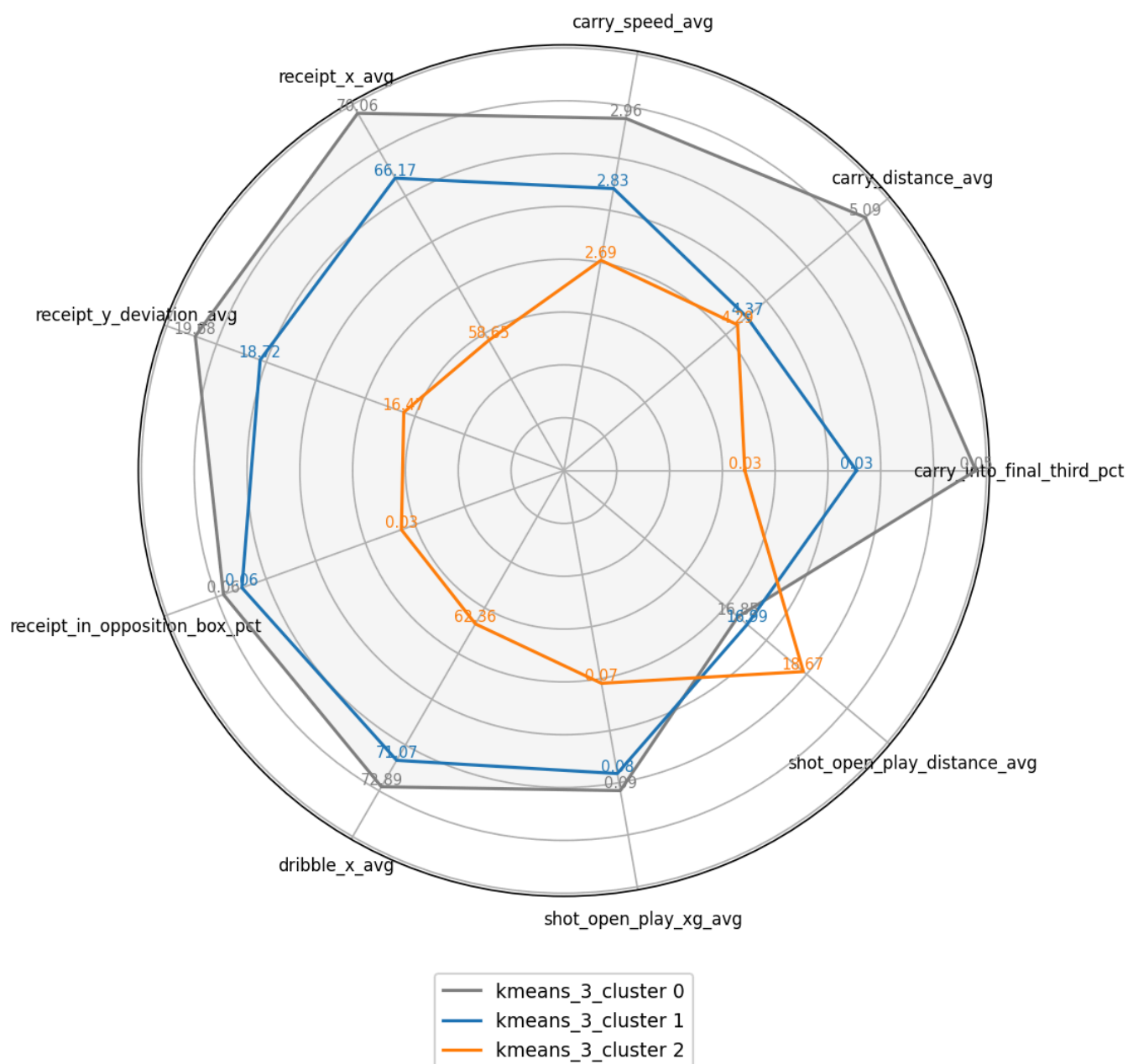


Figure 33: Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (adj)

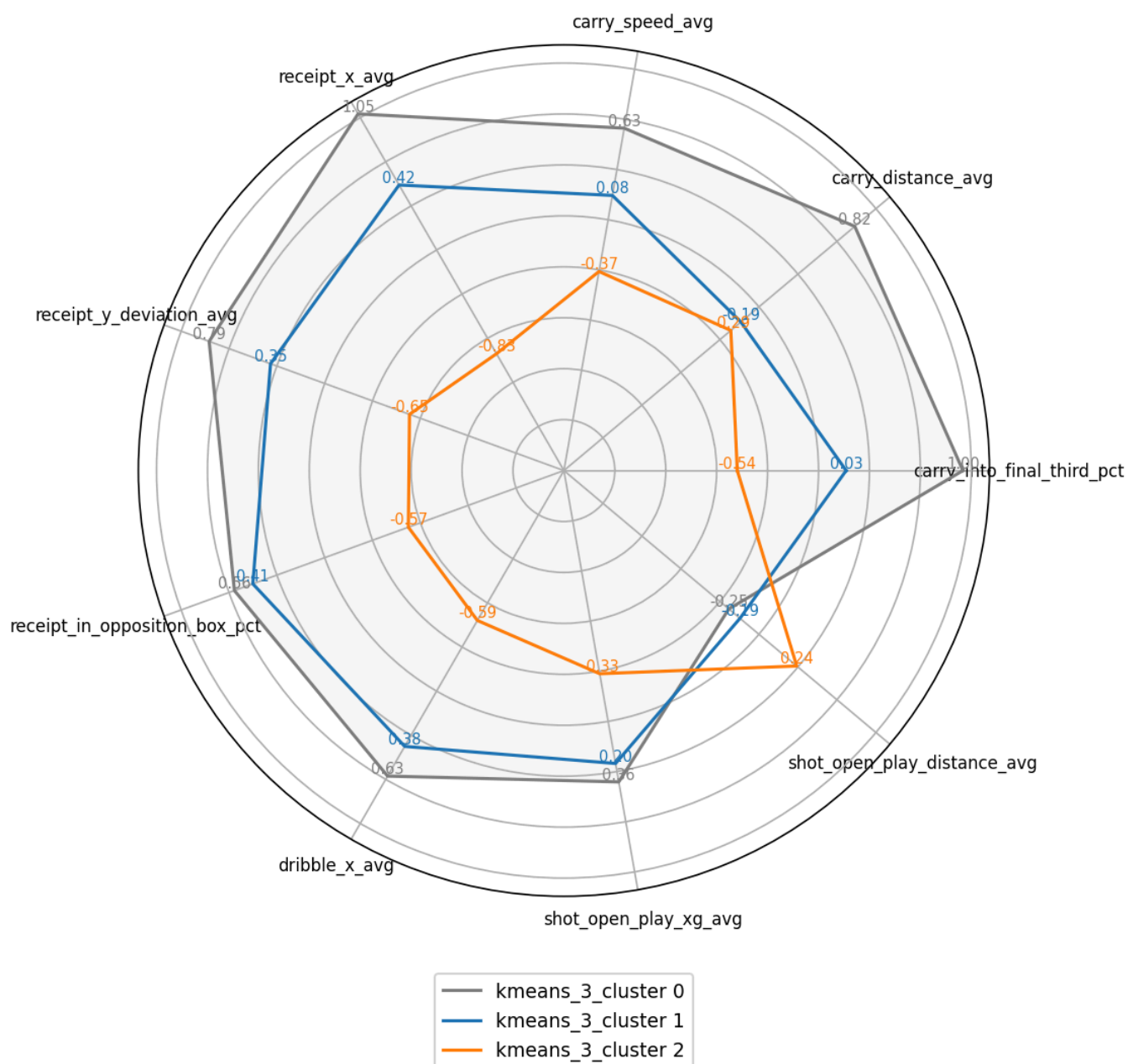


Figure 34: Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (raw)

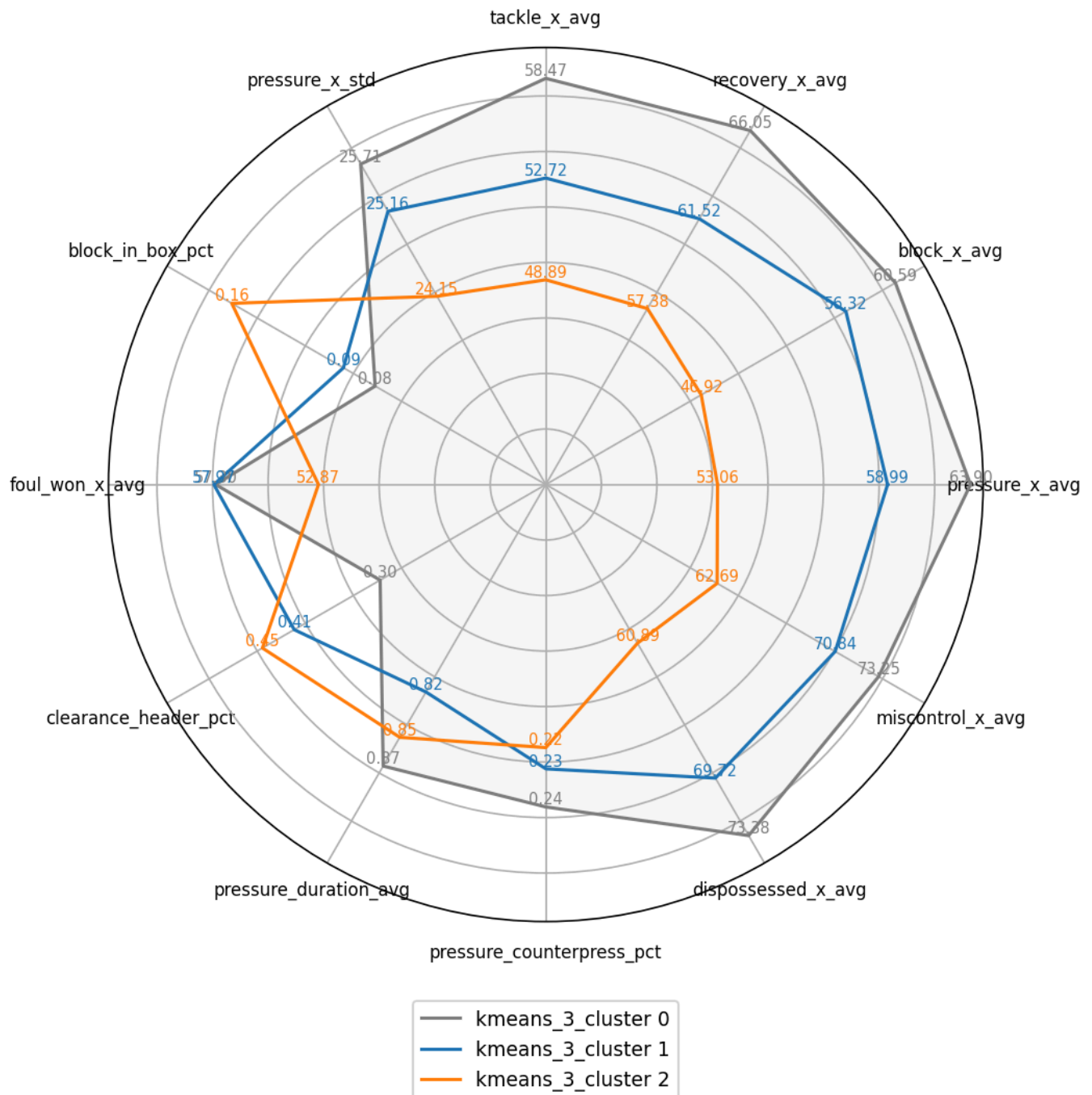


Figure 35: Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (adj)

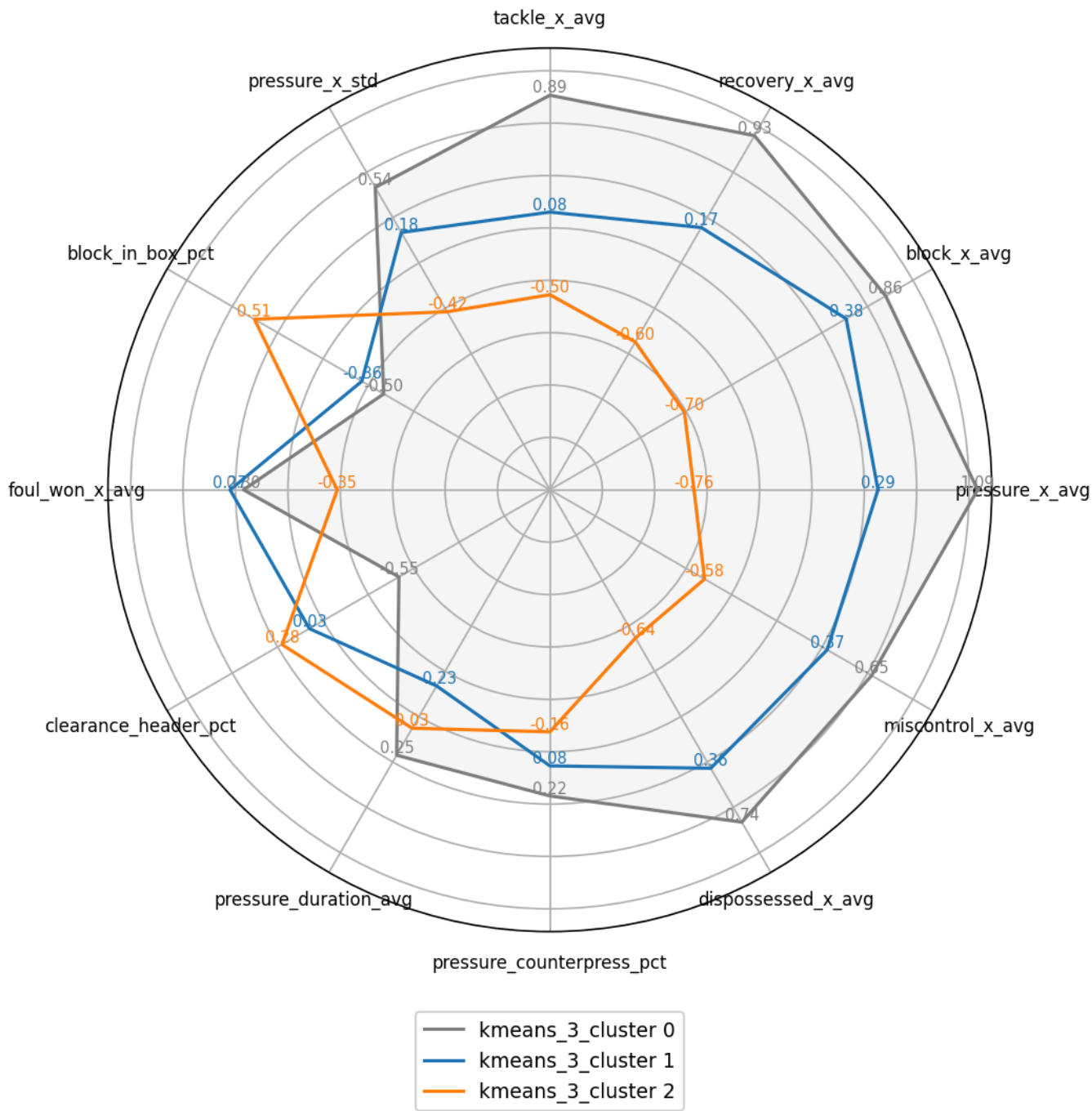


Figure 36: Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters

FA Women’s Super League

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Passing — kmeans_3_cluster comparison (FA Women's Super League, raw)

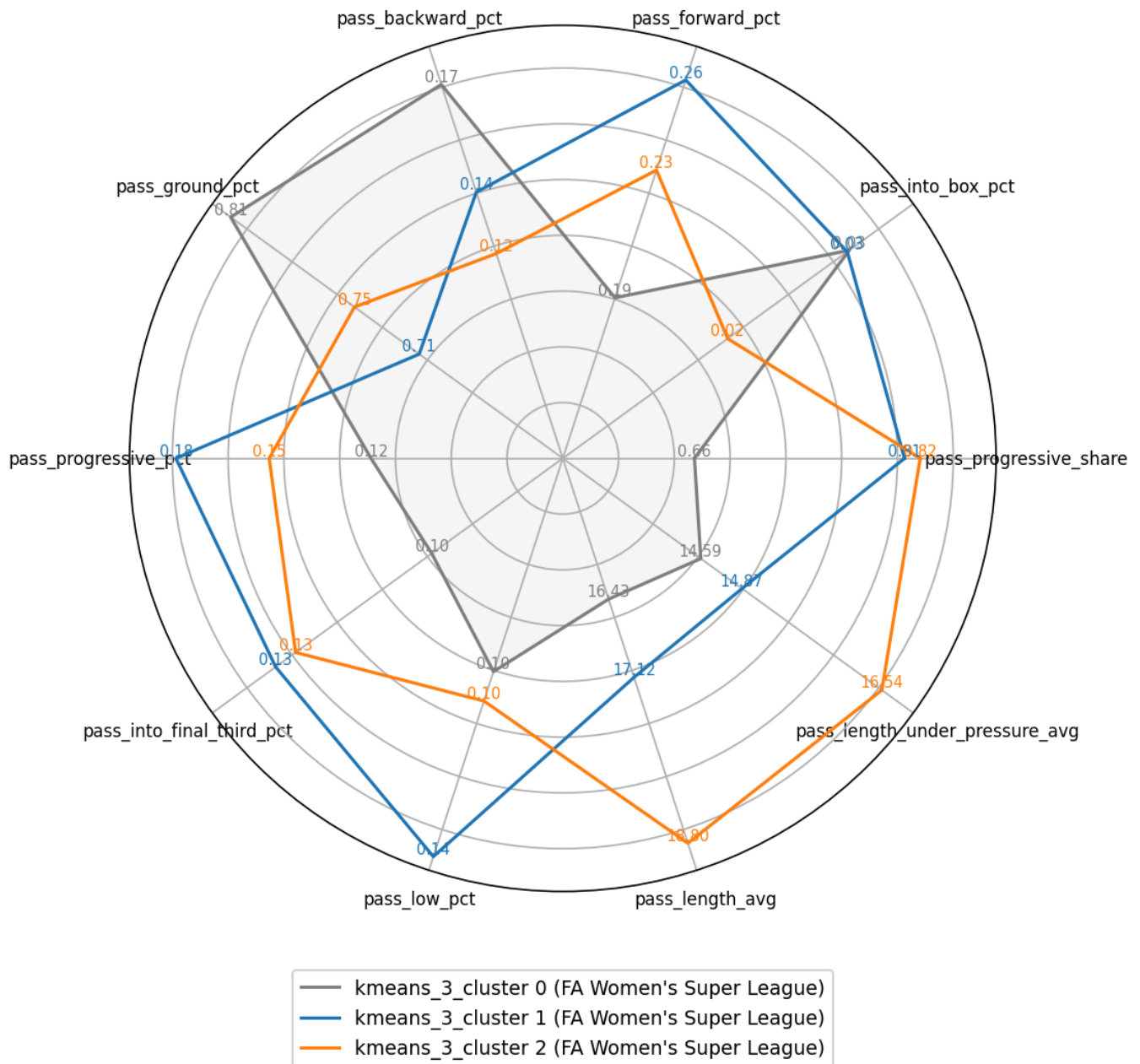


Figure 37: England Center Midfields Passing — Raw Statistics Comparison, 3 Clusters

Passing — kmeans_3_cluster comparison (FA Women's Super League, adj)

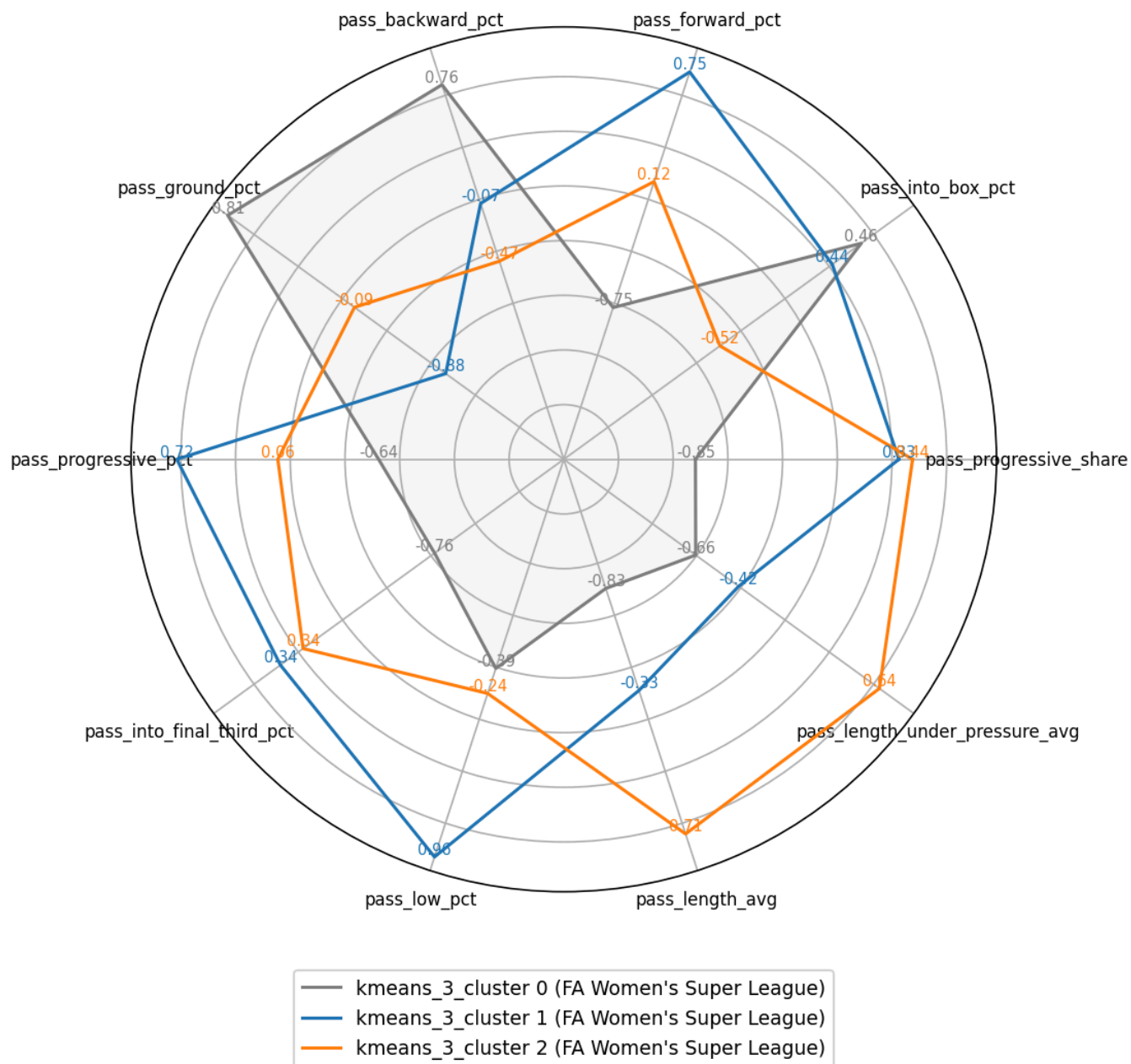


Figure 38: England Center Midfields Passing — Adjusted Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (FA Women's Super League, raw)

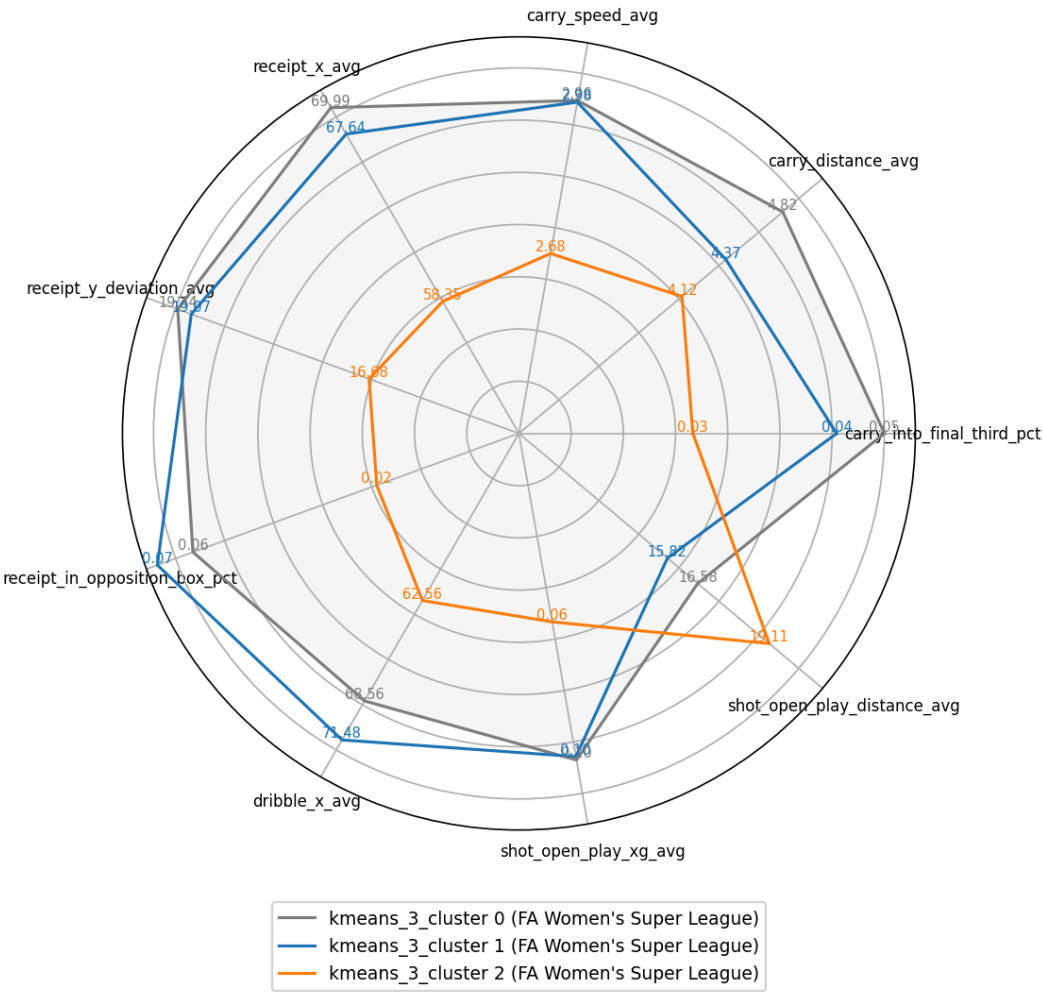


Figure 39: England Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (FA Women's Super League, adj)

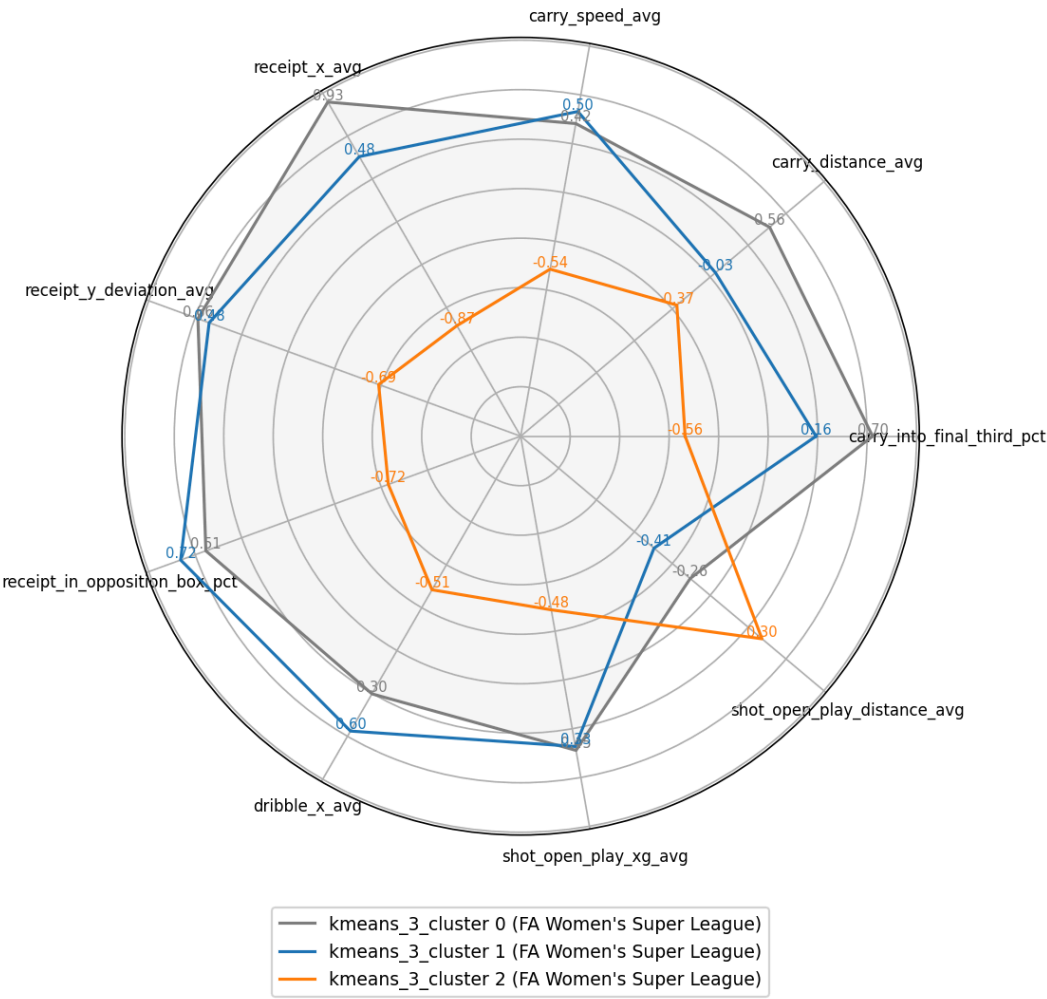


Figure 40: England Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (FA Women's Super League, raw)

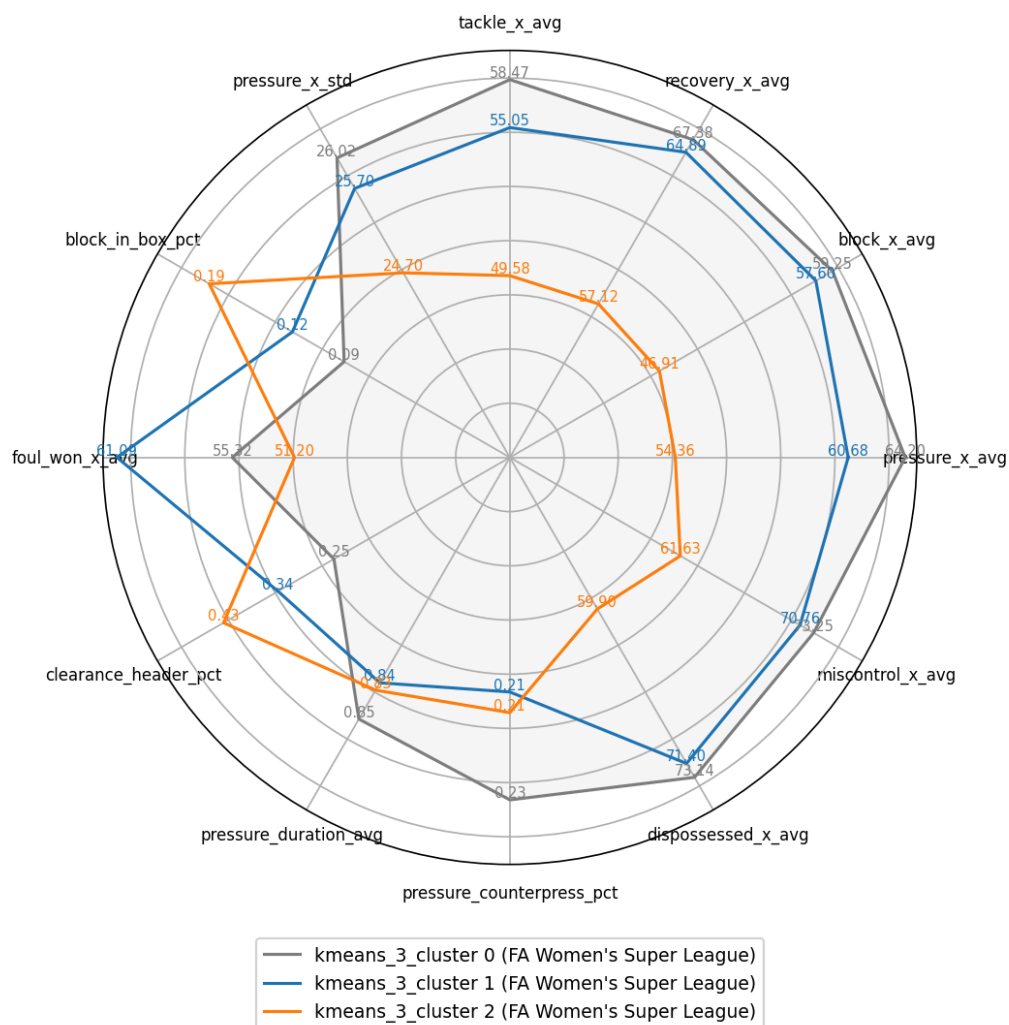


Figure 41: England Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (FA Women's Super League, adj)

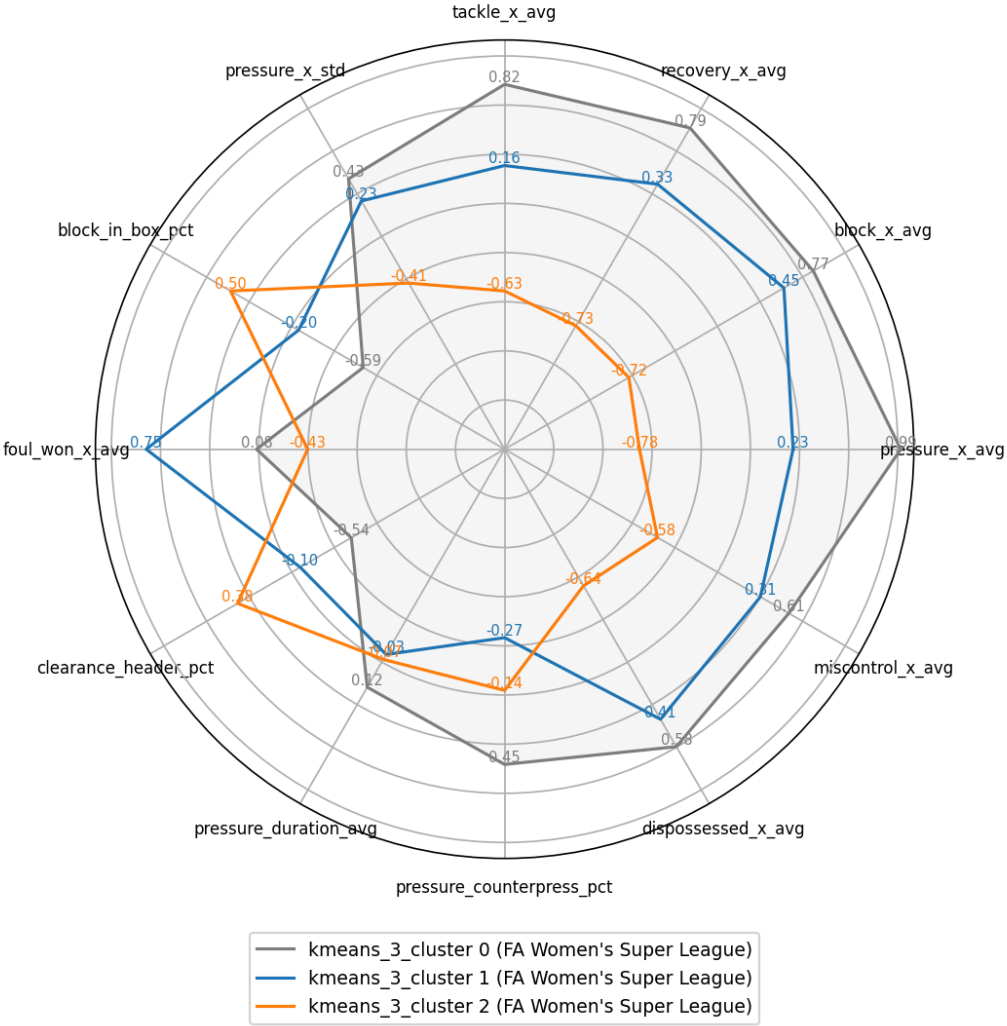


Figure 42: England Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters

Frauen Bundesliga

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Passing — kmeans_3_cluster comparison (Frauen Bundesliga, raw)

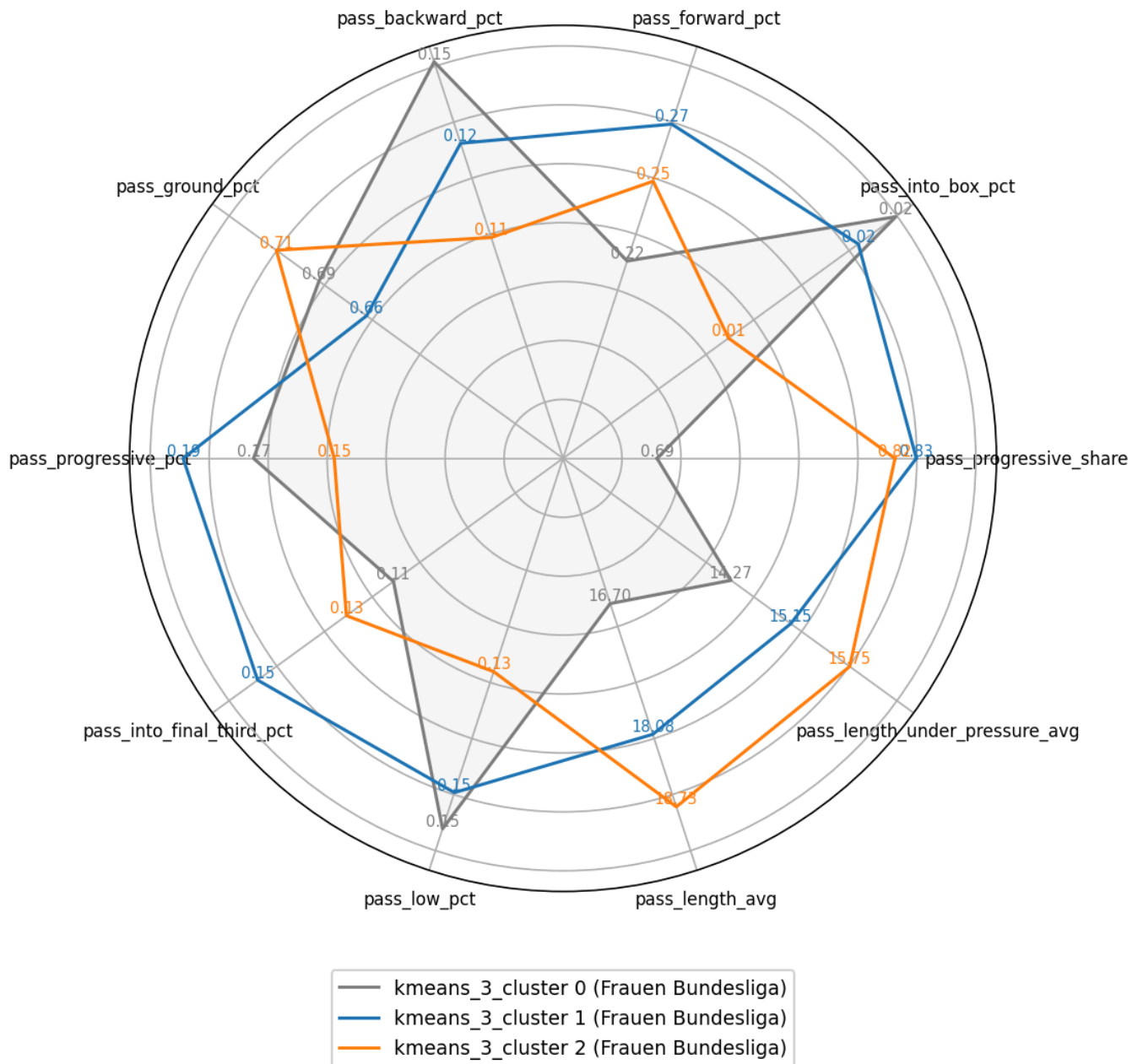


Figure 43: Germany Center Midfields Passing — Raw Statistics Comparison, 3 Clusters

Passing — kmeans_3_cluster comparison (Frauen Bundesliga, adj)

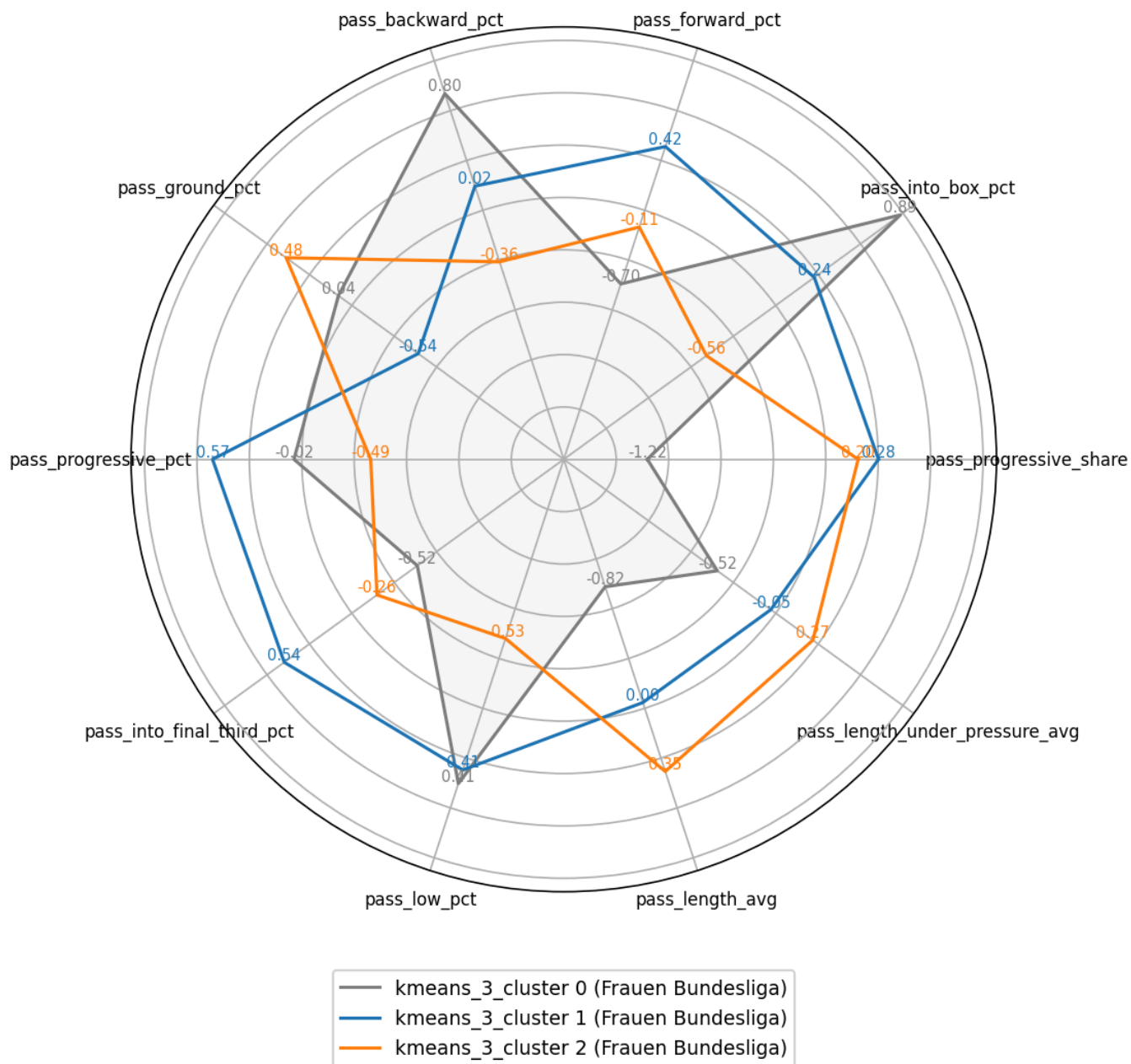


Figure 44: Germany Center Midfields Passing — Adjusted Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (Frauen Bundesliga, raw)

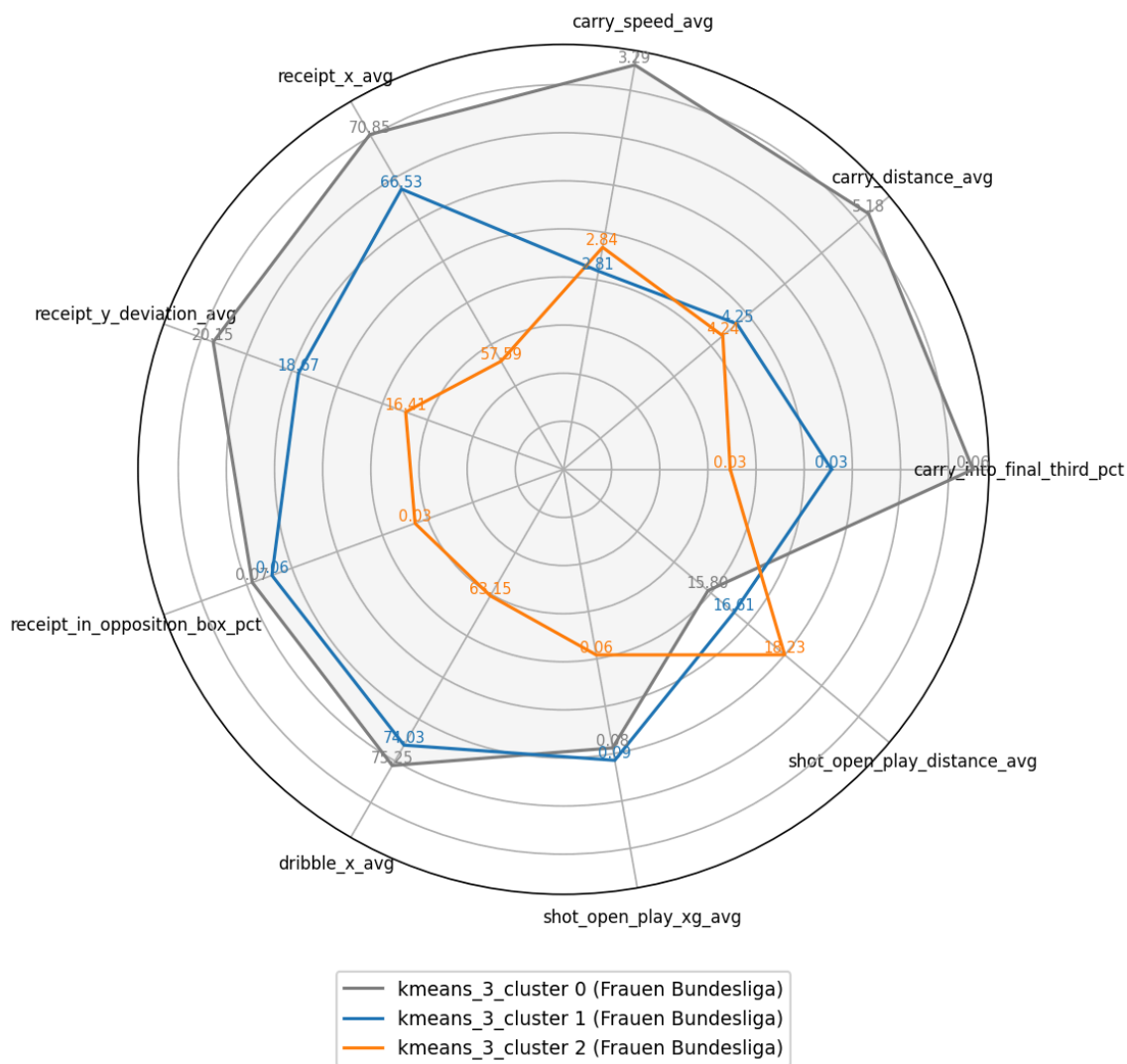


Figure 45: Germany Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (Frauen Bundesliga, adj)

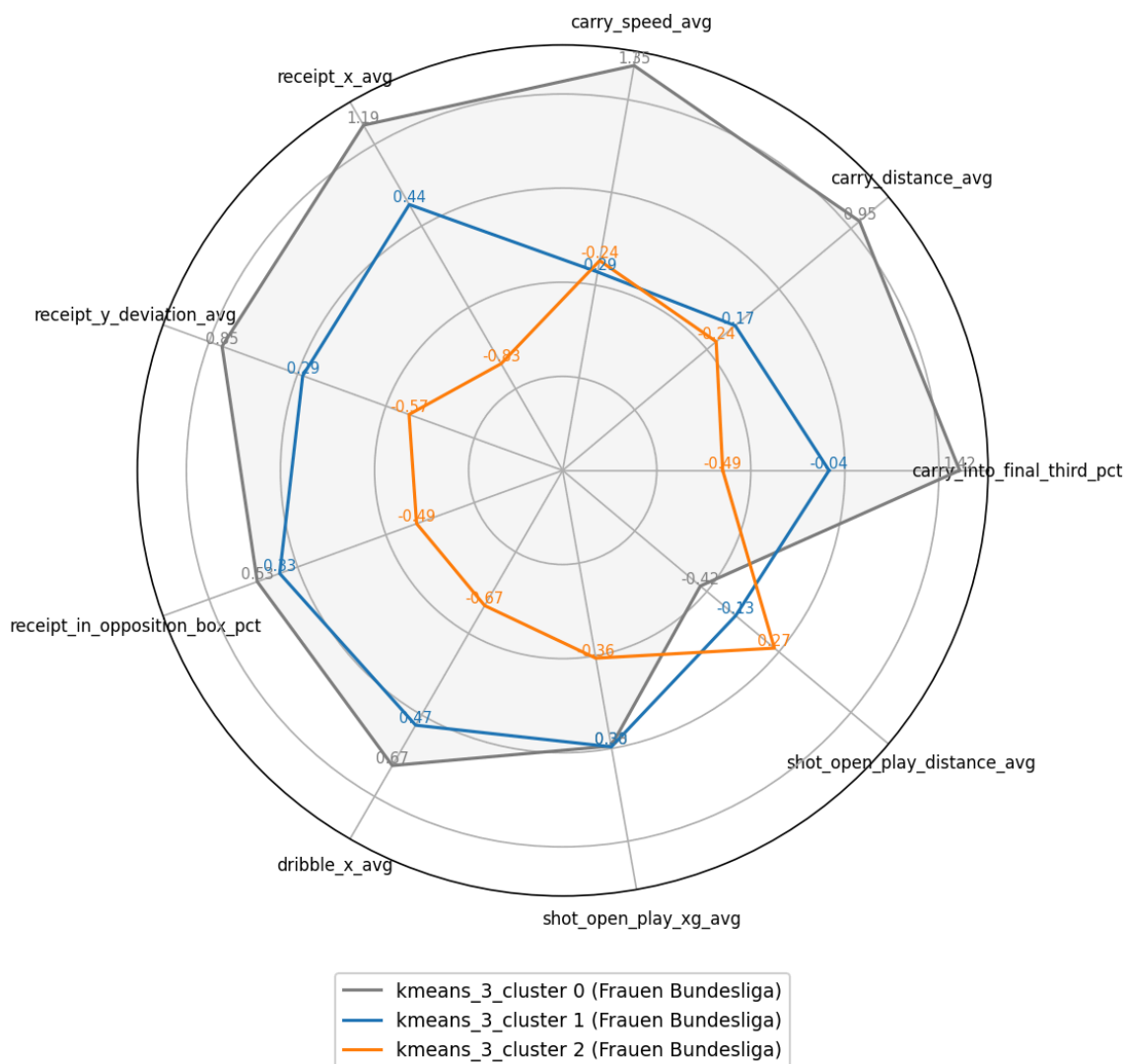


Figure 46: Germany Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (Frauen Bundesliga, raw)

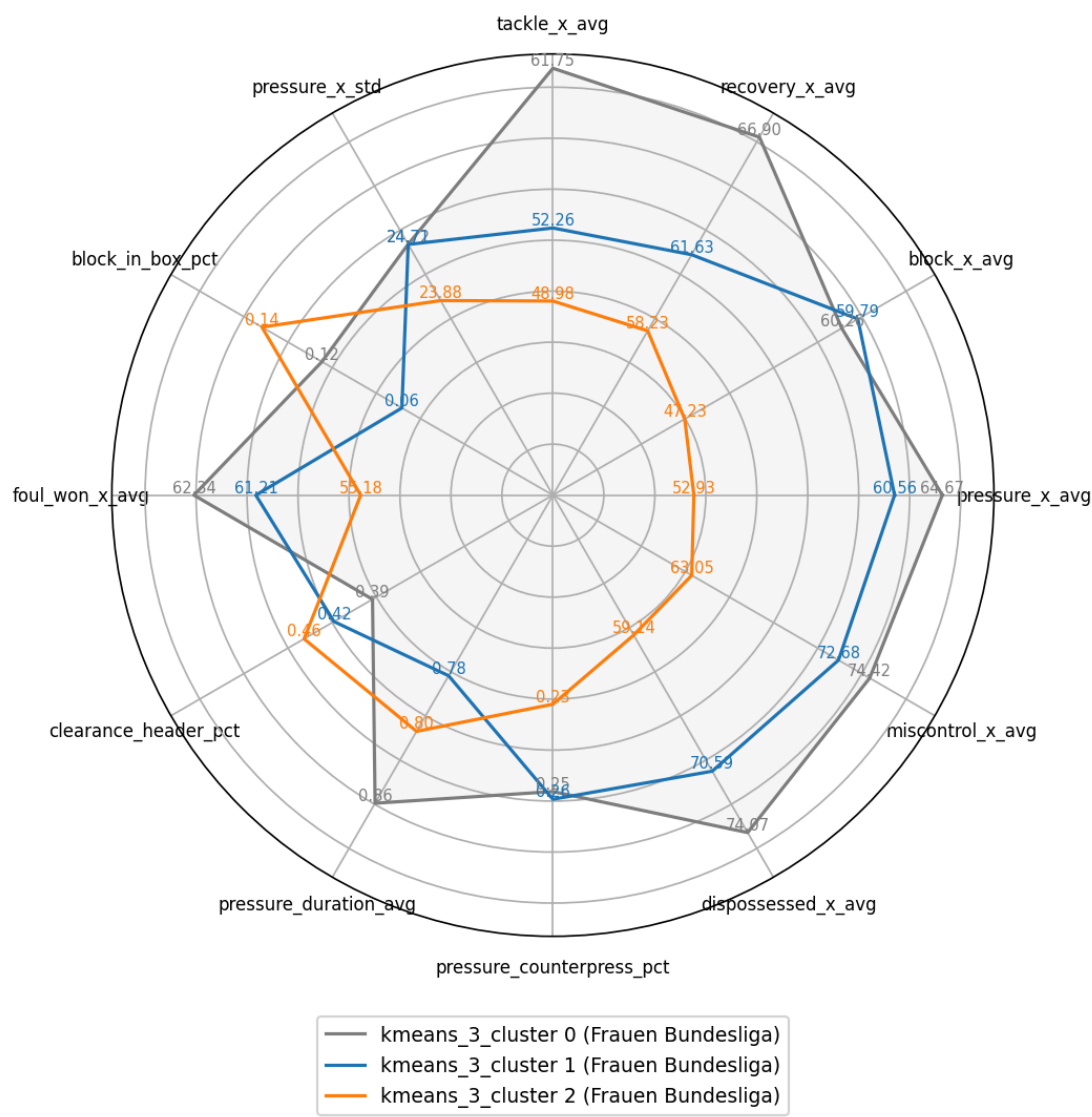


Figure 47: Germany Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (Frauen Bundesliga, adj)

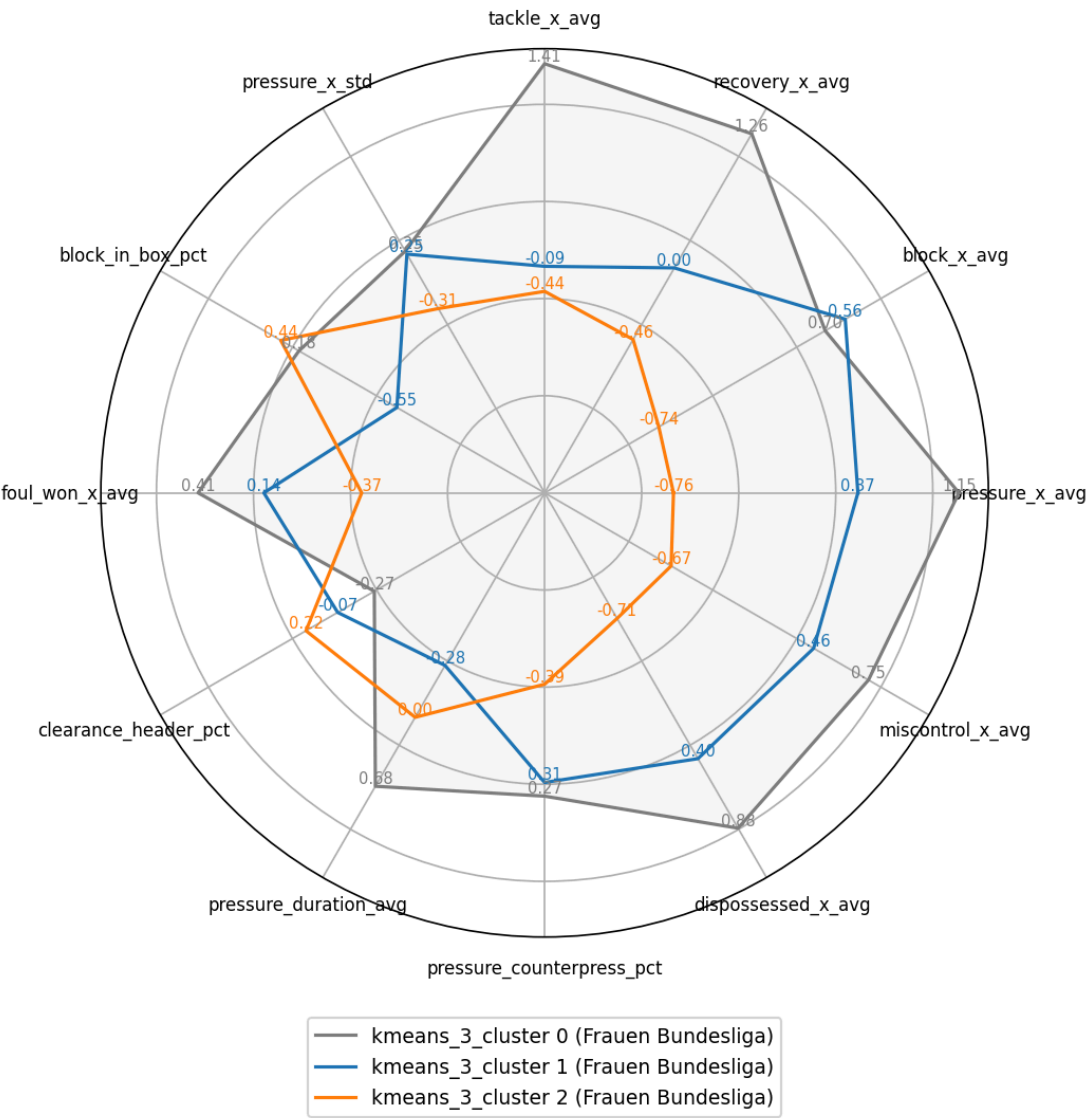


Figure 48: Germany Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters

Liga F

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Passing — kmeans_3_cluster comparison (Liga F, raw)

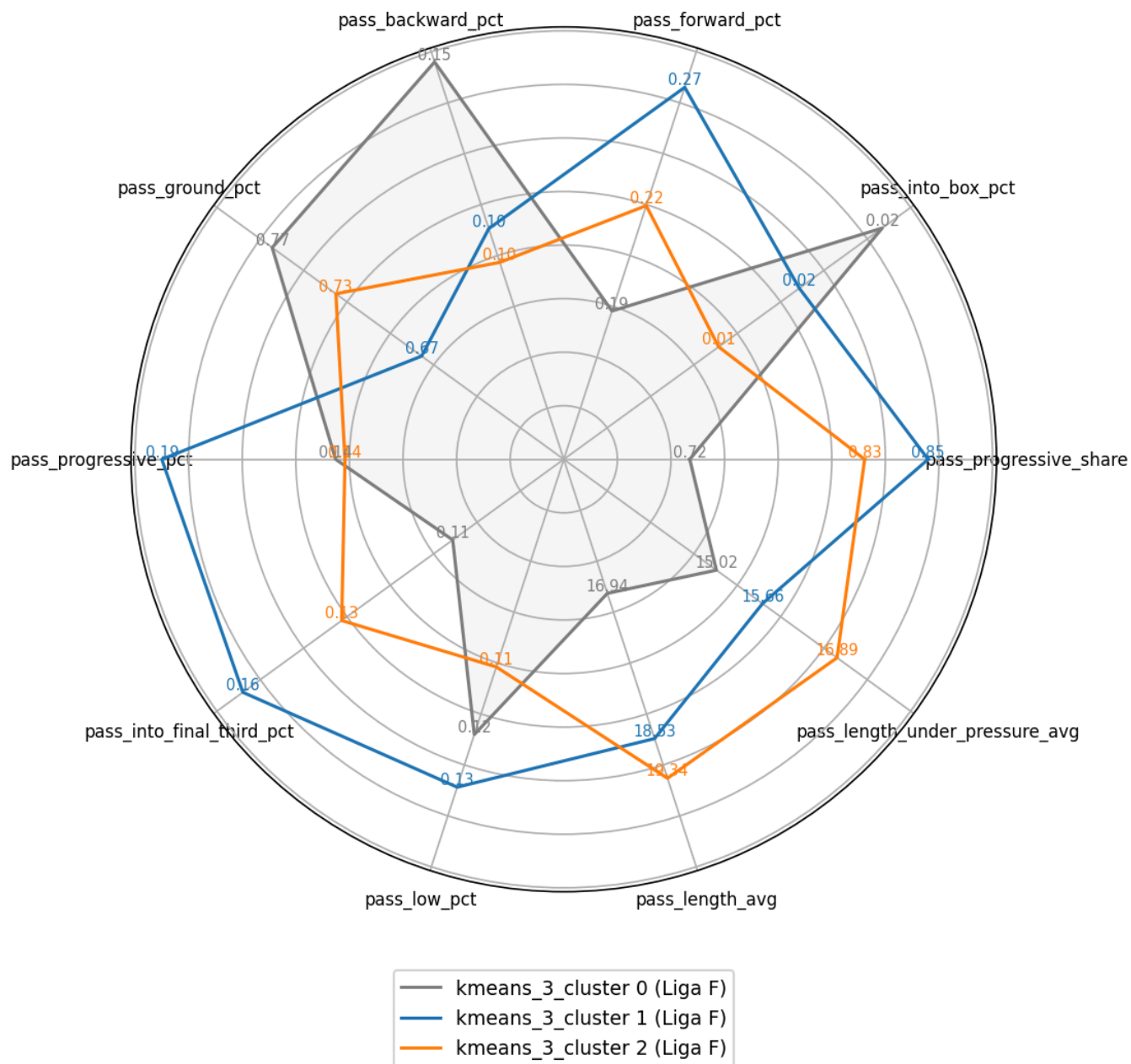


Figure 49: Spain Center Midfields Passing — Raw Statistics Comparison, 3 Clusters

Passing — kmeans_3_cluster comparison (Liga F, adj)

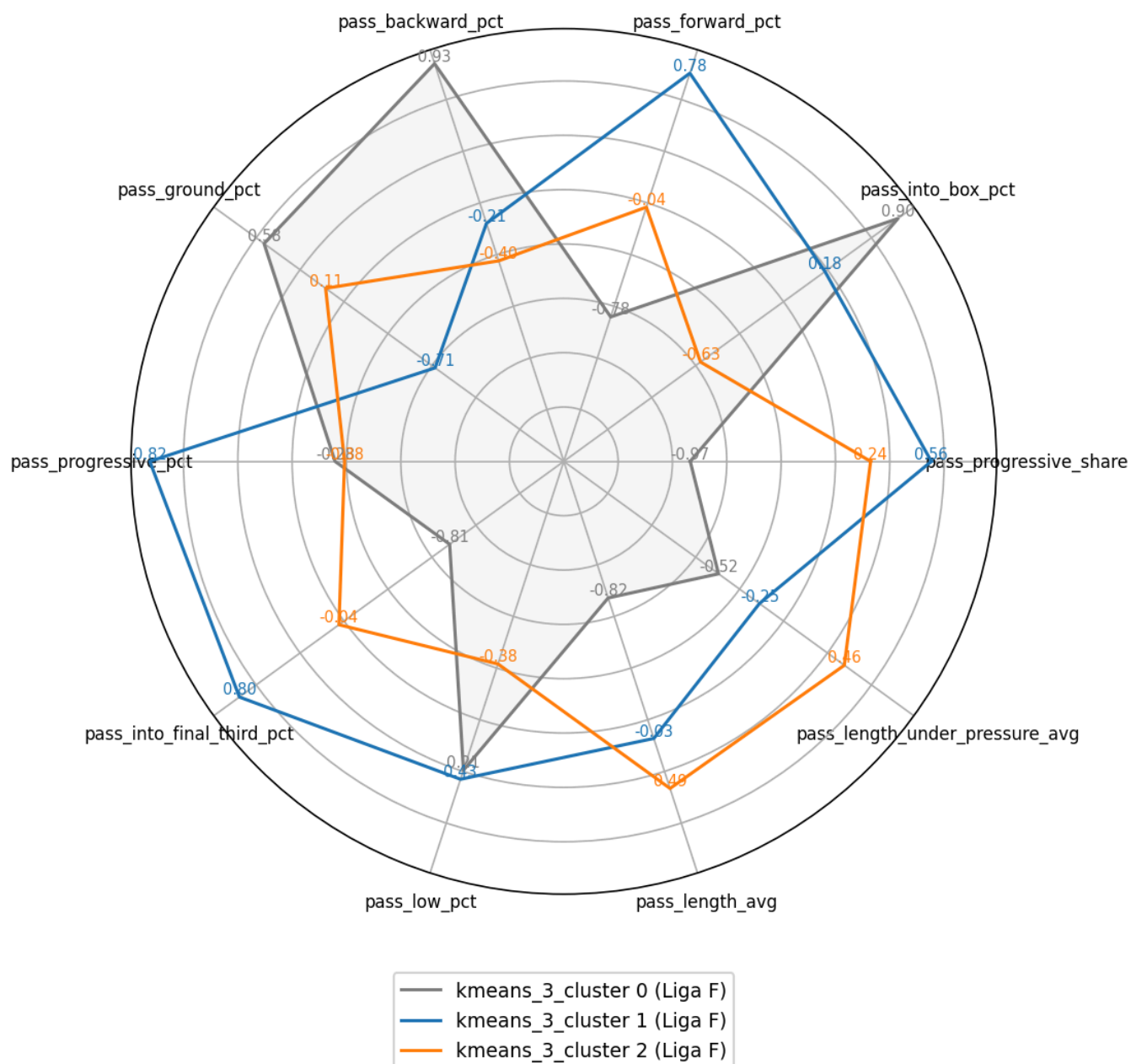


Figure 50: Spain Center Midfields Passing — Adjusted Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (Liga F, raw)

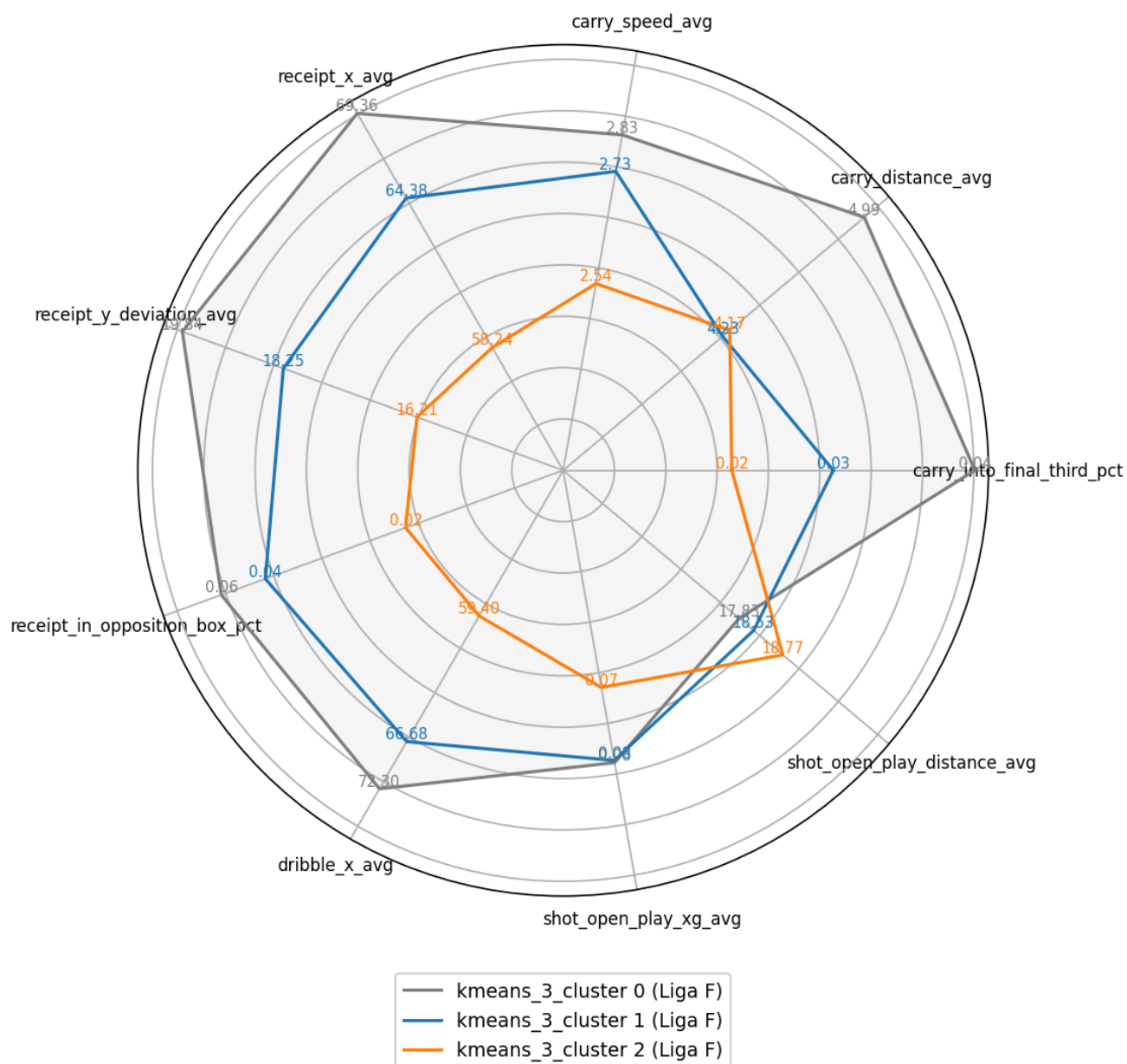


Figure 51: Spain Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (Liga F, adj)

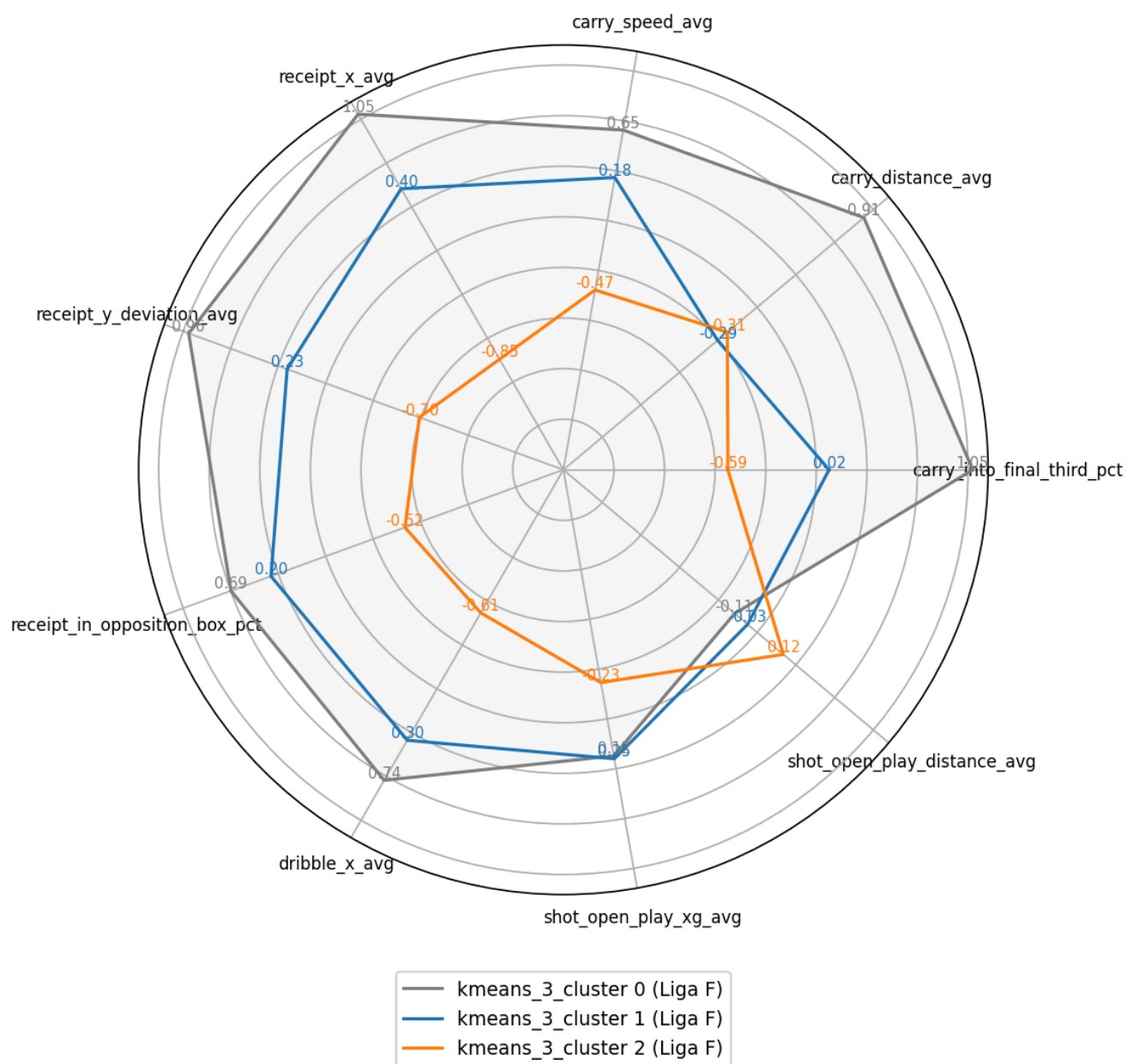


Figure 52: Spain Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (Liga F, raw)

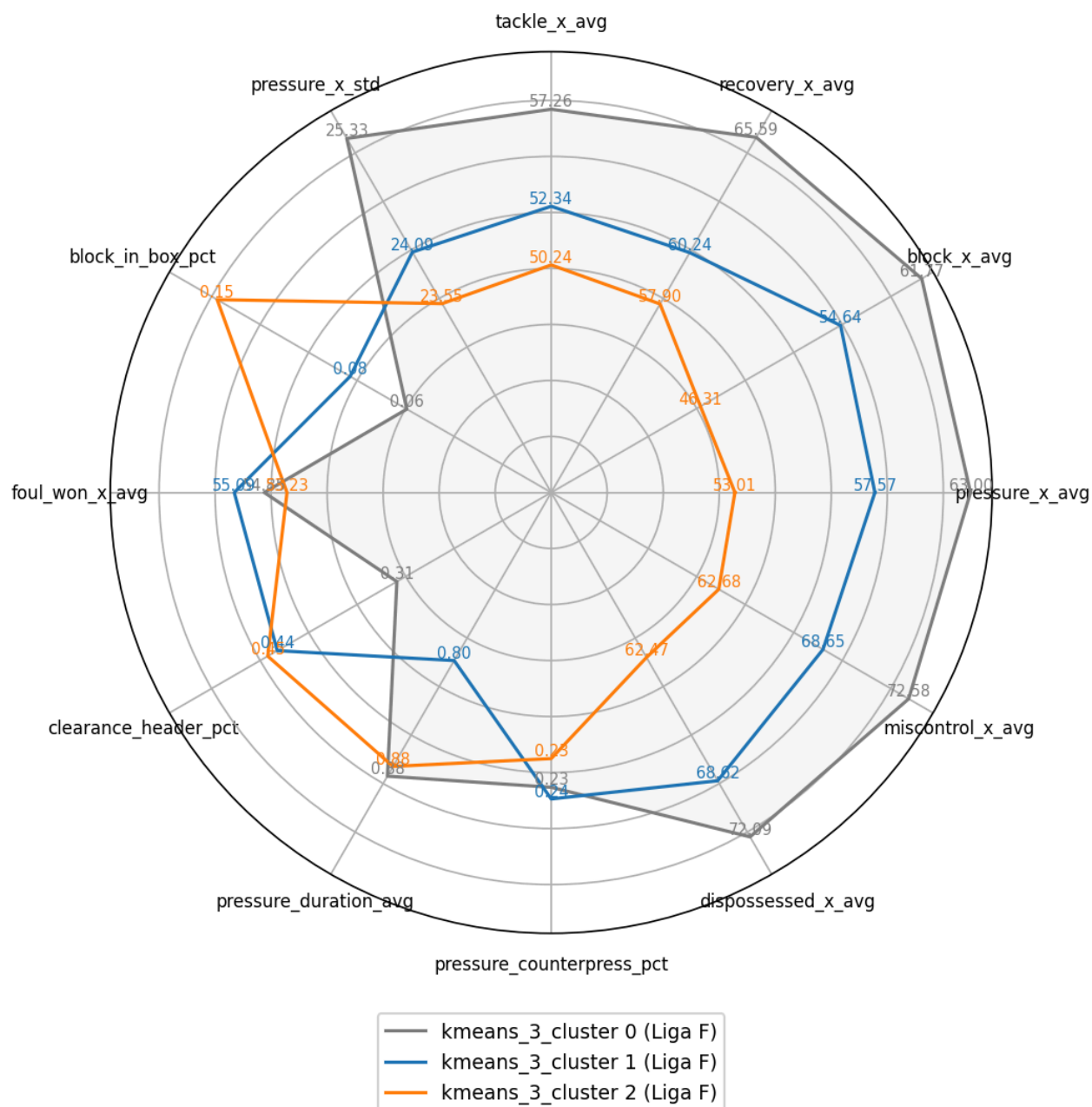


Figure 53: Spain Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (Liga F, adj)

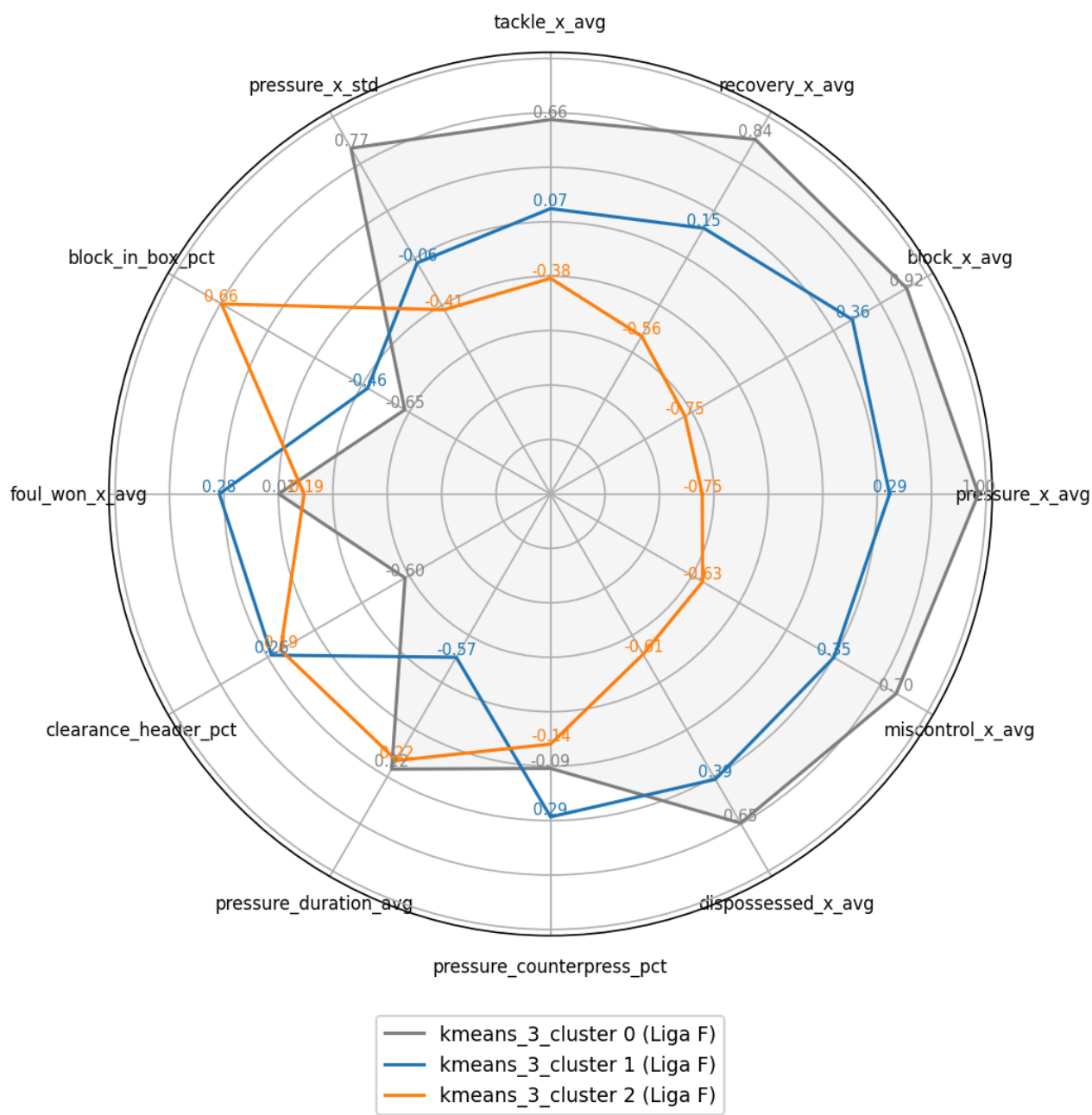


Figure 54: Spain Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters

Serie A Women

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Passing — kmeans_3_cluster comparison (Serie A Women, raw)

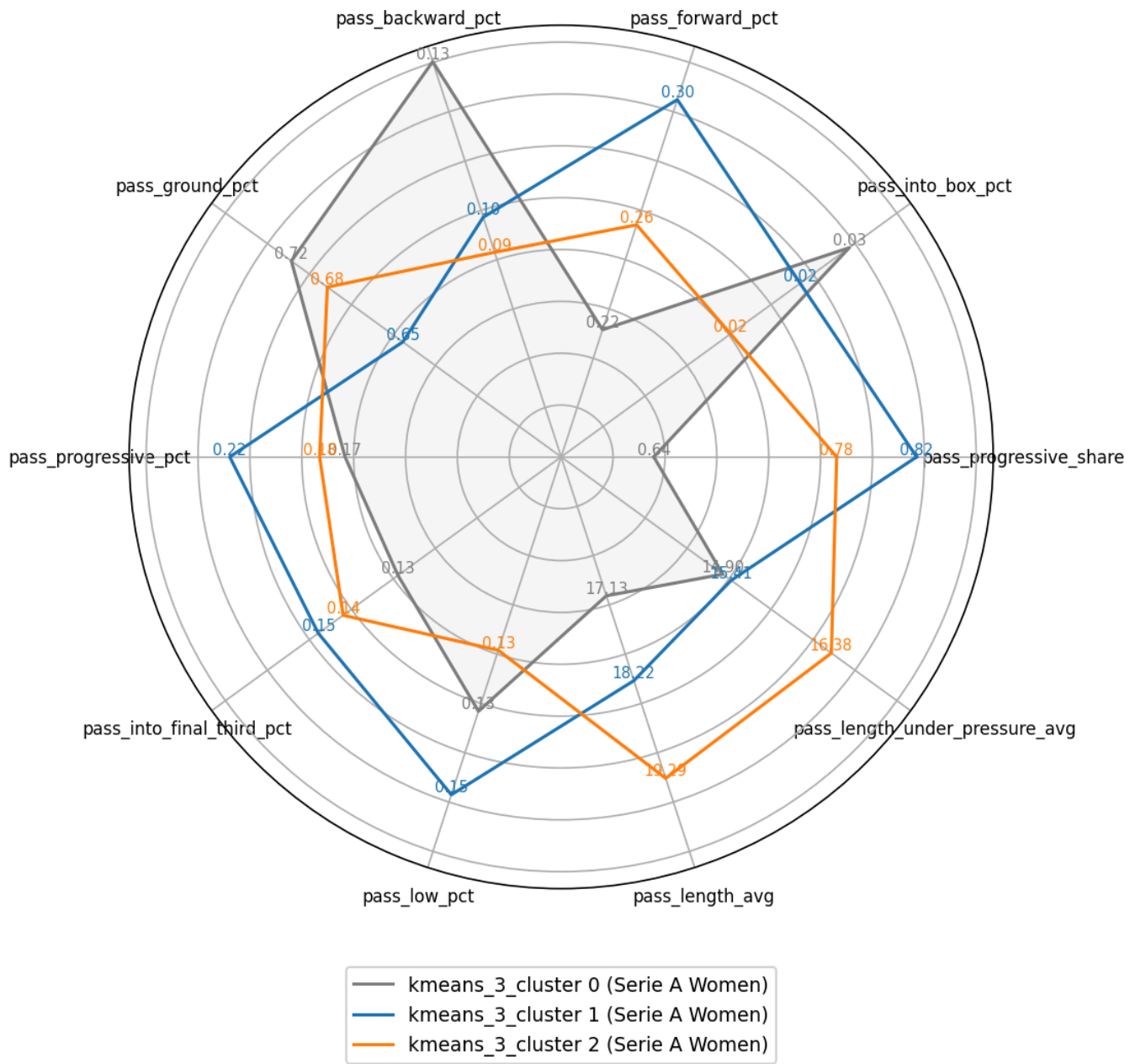


Figure 55: Italy Center Midfields Passing — Raw Statistics Comparison, 3 Clusters

Passing — kmeans_3_cluster comparison (Serie A Women, adj)

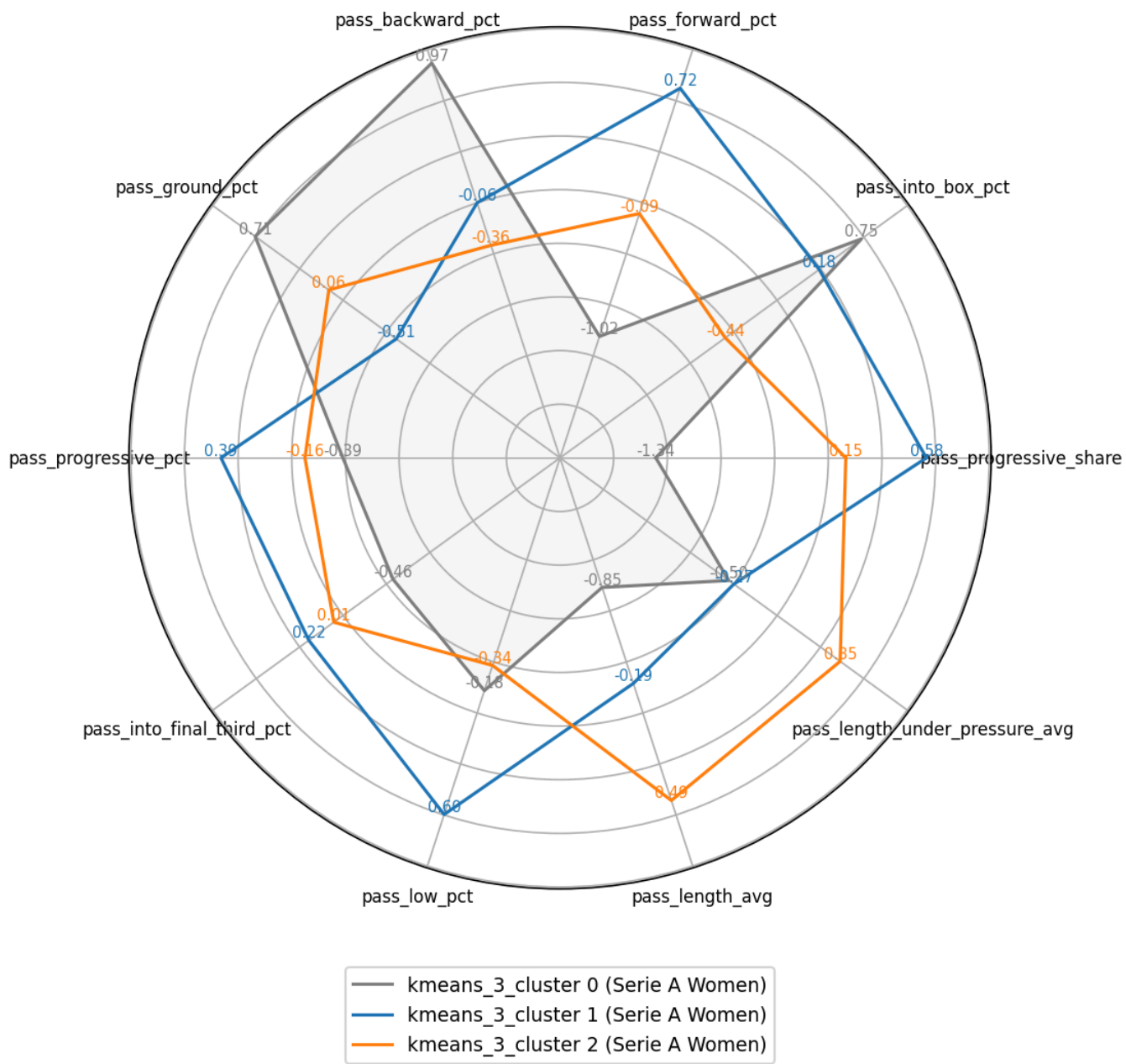


Figure 56: Italy Center Midfields Passing — Adjusted Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (Serie A Women, raw)

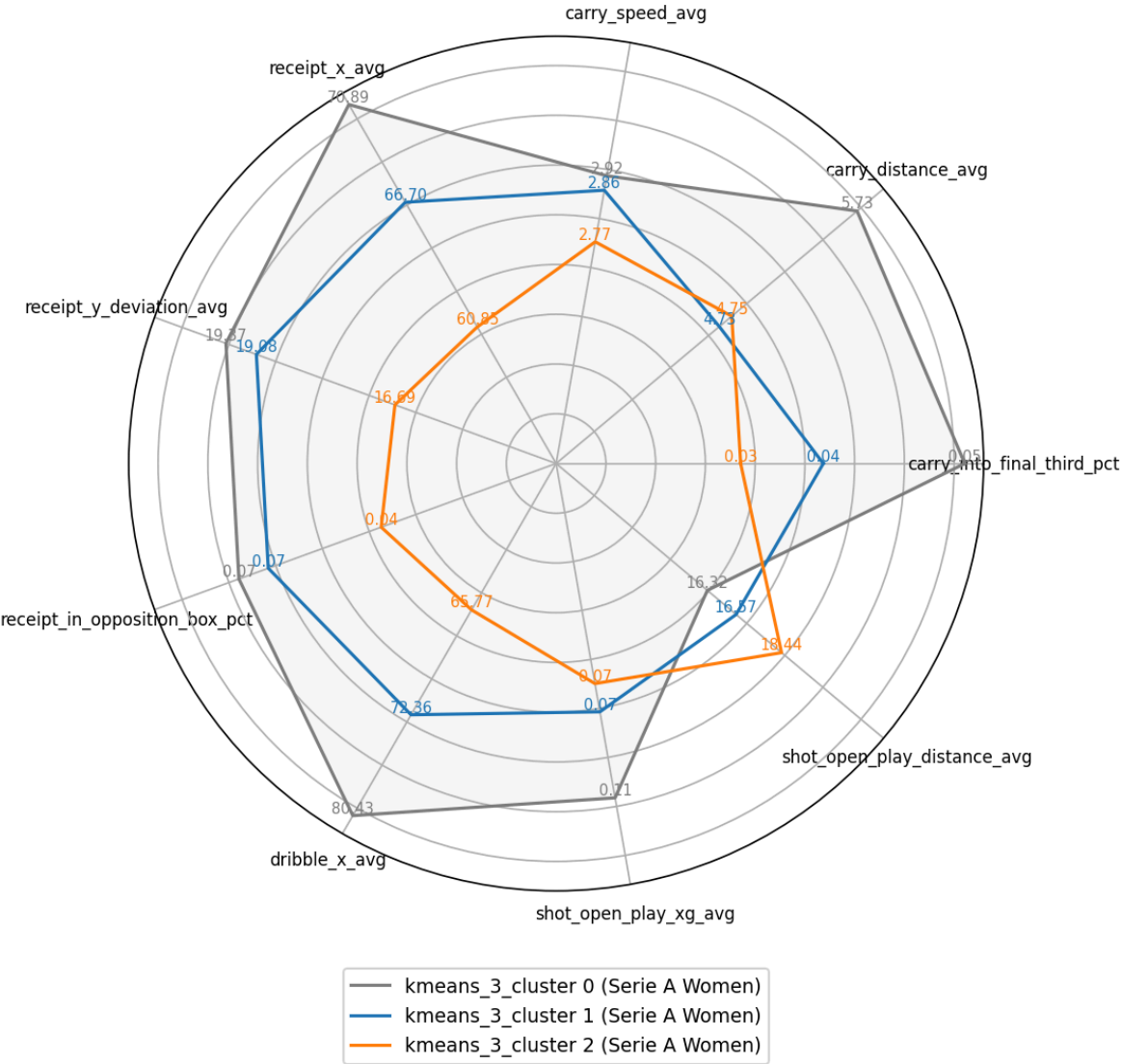


Figure 57: Italy Center Midfields Carrying Receipt Dribbling Shooting — Raw Statistics Comparison, 3 Clusters

Carrying_Receipt_Dribbling_Shooting — kmeans_3_cluster comparison (Serie A Women, adj)

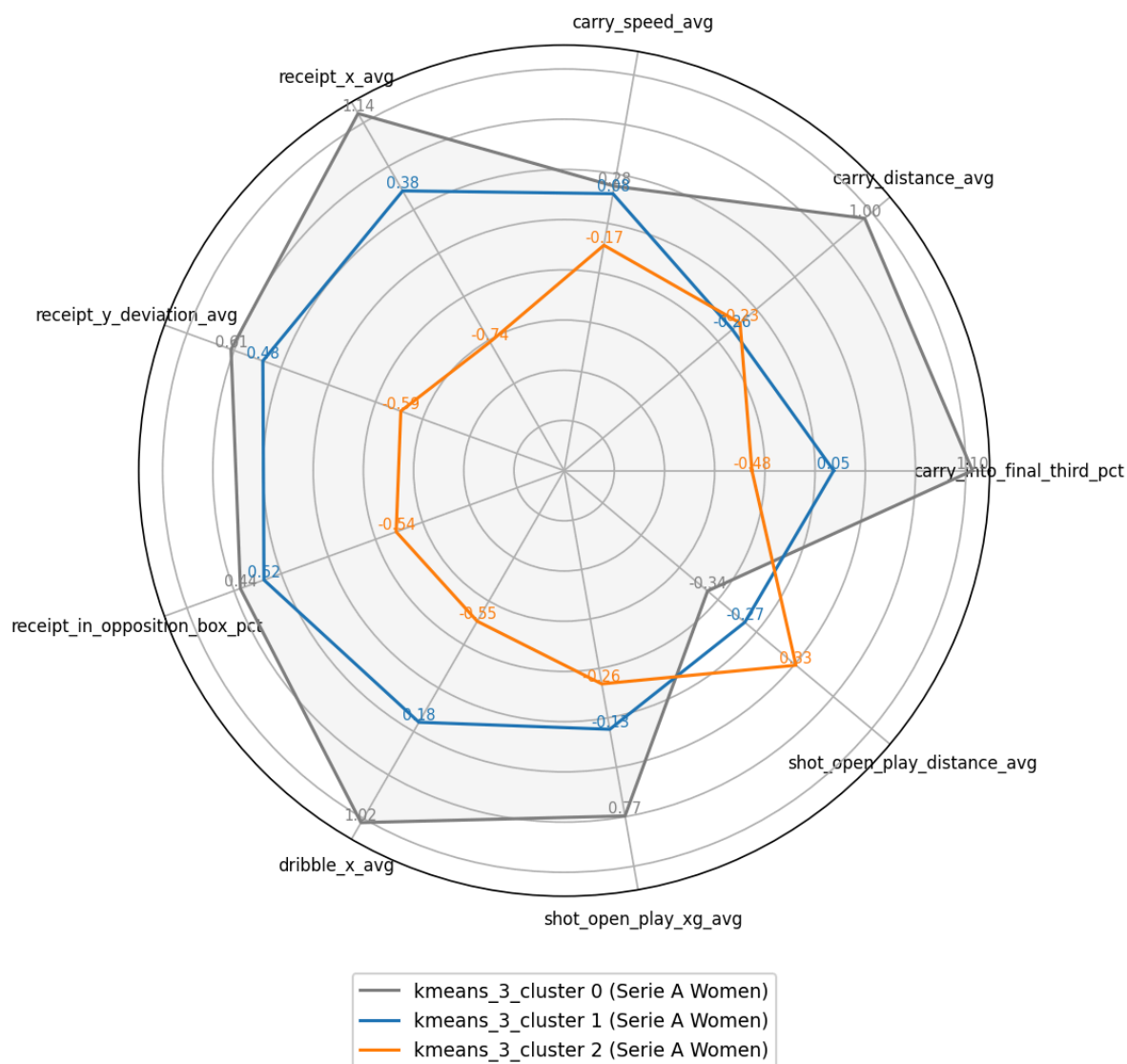


Figure 58: Italy Center Midfields Carrying Receipt Dribbling Shooting — Adjusted Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (Serie A Women, raw)

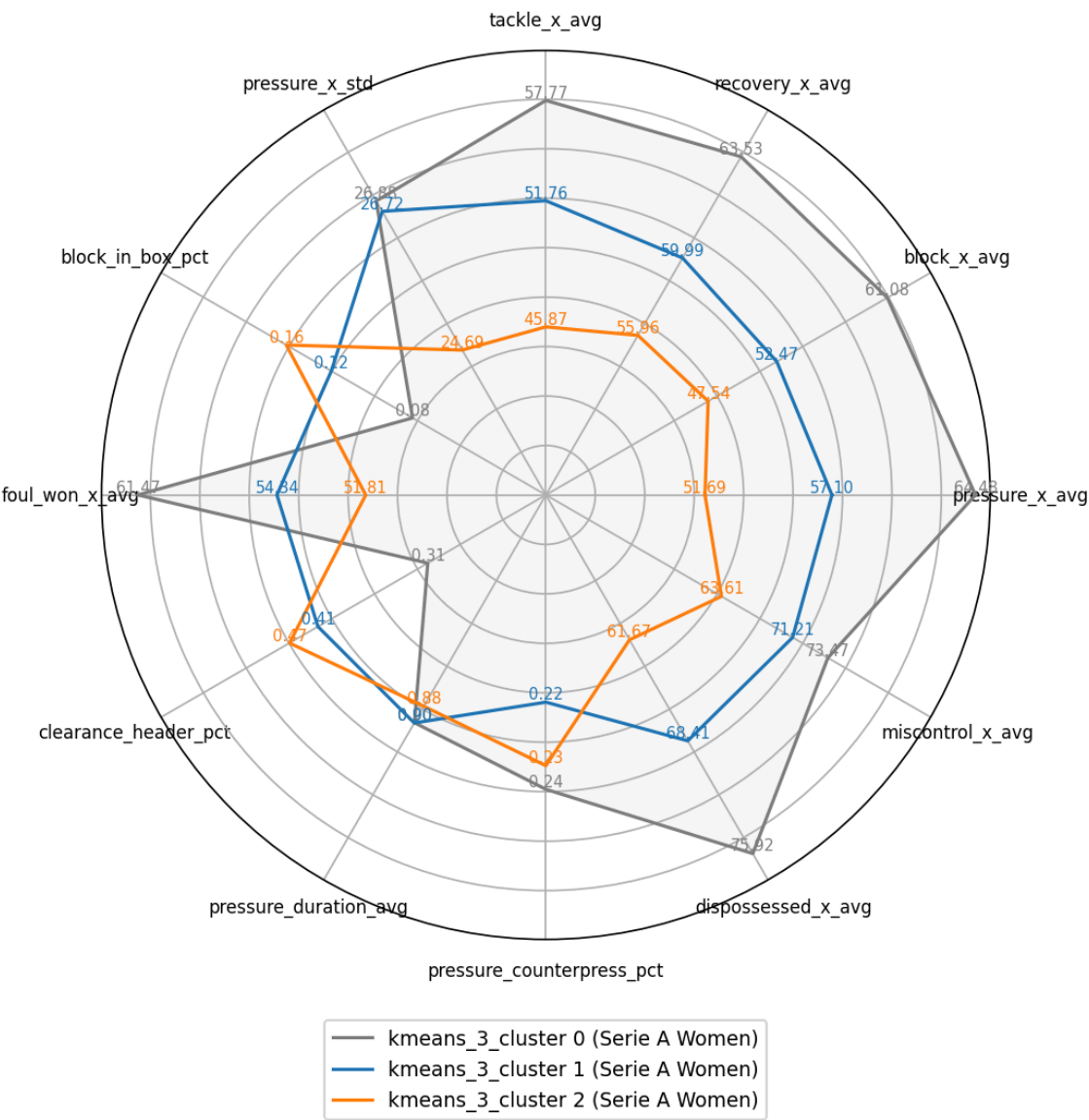


Figure 59: Italy Center Midfields Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

Defending_Vulnerability_Fifty_Fifty — kmeans_3_cluster comparison (Serie A Women, adj)

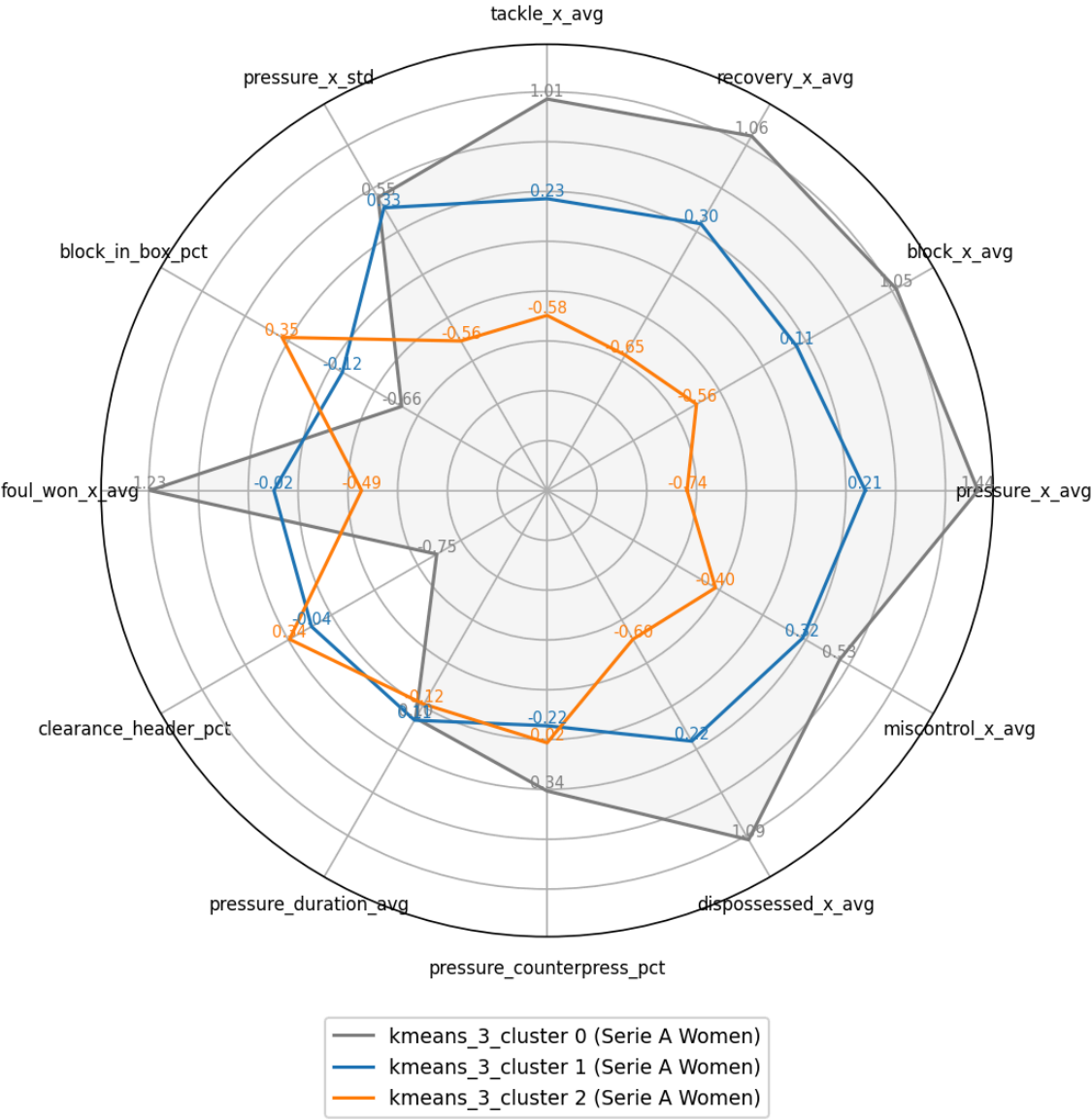


Figure 60: Italy Center Midfields Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters